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Part I

Quantum light

Chapter 1

Field quantization

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1.1 Electrodinamics review

1.1.1 Plane waves and algebra

1.1.2 Helmholtz and potentials

1.1.3 Wave equations

1.2 Quantization of a single-mode field

1.2.1 Fields in a cavity

Lets consider the following one-dimensional problem, where a cavity of length L is oriented along the z-axis.

A linear polarized E-field is assumed, the medium is free space, perfect conducting walls and there is no free charges nor free current. The scheme is shown in figure 1.1.

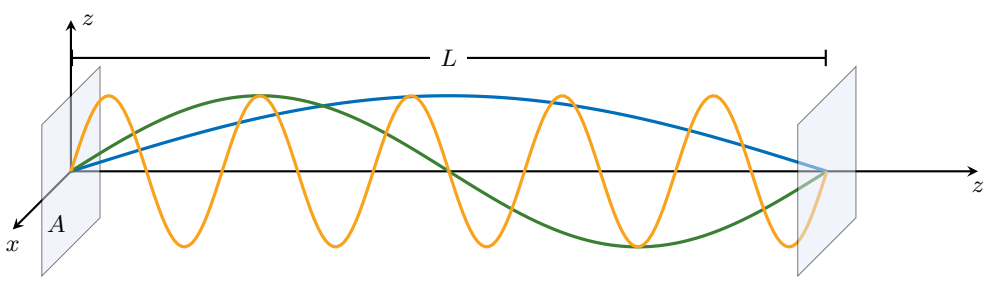


Figure 1.1 One-dimensional cavity problem. Perfect conducting walls.

Our goal is to find the E- and B-field inside the cavity. Maxwell's equations in this case are:

$$\text{Maxwell's equations with free sources} \quad \begin{cases} \nabla \cdot \boldsymbol{E} = 0 \\ \nabla \cdot \boldsymbol{B} = 0 \\ \nabla \times \boldsymbol{E} = -\partial_t \boldsymbol{B} \\ \nabla \times \boldsymbol{B} = \frac{1}{c^2}\partial_t \boldsymbol{E} \end{cases} \tag{1.1}$$

The E-field will be assumed to be $\boldsymbol{E}(z,t) = \boldsymbol{e}E(z,t)$, where $\boldsymbol{e}$ is the polarization vector. Because fields depends only on z, $\nabla = \hat{\boldsymbol{z}}\partial_z$. First Maxwell equation yields:

$$\nabla \cdot \boldsymbol{E} = \partial_z(\hat{\boldsymbol{z}} \cdot \boldsymbol{E}) = \partial_z(\hat{\boldsymbol{z}} \cdot \boldsymbol{e}E) = 0 \Longrightarrow \boldsymbol{e} \cdot \hat{\boldsymbol{z}} = 0.$$

This implies that the polarization vector must be unitary in the transverse plane:

$$\boldsymbol{e} = \cos\phi\hat{\boldsymbol{x}} + \sin\phi\hat{\boldsymbol{y}}.$$

Third Maxwell's equation yields

$$\nabla \times \boldsymbol{E} = (\hat{\boldsymbol{z}}\partial_z) \times (\boldsymbol{e}E) = (\hat{\boldsymbol{z}} \times \boldsymbol{e})\partial_z E = -\partial_t \boldsymbol{B}.$$

Taking the curl of this equation, using Fourth Maxwell's equation and vector identities:

$$\begin{aligned} \nabla \times (\nabla \times \boldsymbol{E}) &= -\partial_t(\nabla \times \boldsymbol{B}) \\ \nabla(\cancelto{0}{\nabla \cdot \boldsymbol{E}}) - \nabla^2 \boldsymbol{E} &= -\frac{1}{c^2}\partial_t^2 \boldsymbol{E} \\ -\boldsymbol{e}\partial_z^2 E &= -\boldsymbol{e}\frac{1}{c^2}\partial_t^2 E. \end{aligned}$$

From here, we have the E-field wave equation for this particular problem.

$$\partial_z^2 E - \frac{1}{c^2}\partial_t^2 E = 0. \tag{1.2}$$

Before solving this equation, we need the boundary condition set by the PEC condition. We need to $\hat{n} \times \boldsymbol{E} = 0$ on the surface. Because the normal surface unit vector is $\hat{n} = \pm\hat{\boldsymbol{z}}$, we have

$$\text{Boundary condition} \qquad \hat{n} \times \boldsymbol{E} = \hat{\boldsymbol{z}} \times (\boldsymbol{e}E) = 0 \Longrightarrow E(z=0,t) = E(z=L,t) = 0.$$

In order to solve the PDE, we assum a product form $E(z,t) = Z(z)q(t)$. Then, by replacing in it in (1.2):

$$\begin{aligned}
\partial_z^2[Z(z)q(t)] - \frac{1}{c^2}\partial_t^2[Z(z)q(t)] &= 0 \\
Z''(z)q(t) - \frac{1}{c^2}Z(z)\ddot{q}(t) &= \Big/ [Z(z)q(t)]^{-1} \\
\frac{Z''(z)}{Z(z)} &= \frac{1}{c^2}\frac{\ddot{q}(z)}{q(z)}.
\end{aligned}$$

Left side depends only on z, while the right side only on y. The only way this can be true is if both are a constant, say, $-k^2$. Then,

$$\text{Spatial and temporal differential equations} \qquad \begin{cases} \dfrac{Z''}{Z} = -k^2 \longrightarrow Z'' + k^2 Z = 0 \\ \dfrac{1}{c^2}\dfrac{\ddot{T}}{T} = -k^2 \longrightarrow \ddot{q} + \omega^2 q(t) = 0 \end{cases}.$$

For the spatial ODE, we assume a solution of the form

$$Z(z) = A\sin(kz) + B\cos(kz), \quad Z(0) = Z(L) = 0.$$

Setting the boundaries:

$$\begin{aligned}
Z(0) &= A\sin(0) + B\cos(0) = 0 \Longrightarrow B = 0 \\
Z(L) &= A\sin(kL) = 0 \Longrightarrow k_m = \frac{m\pi}{L}, \quad m \in \mathbb{N}.
\end{aligned}$$

We left the temporal ODE unsolved. Finally, putting all together yields the initial E-field:

$$\boldsymbol{E}_{\boldsymbol{k},\lambda}(z,t) = \boldsymbol{e}_\lambda \sqrt{\frac{2\omega^2}{V\varepsilon_0}} q_{\boldsymbol{k},\lambda}(t)\sin(kz), \quad k_m = \frac{m\pi}{L}, \quad \omega = ck.$$

We have includnig the subscript λ and $\boldsymbol{k}$ to consider multiple mode $\boldsymbol{k}$ varied with m and λ. Also, the coefficient in red is for better results in the future. Using Faraday's law:

$$\boldsymbol{B}_{\boldsymbol{k},\lambda}(z,t) = (\boldsymbol{k} \times \boldsymbol{e}_\lambda)\frac{1}{kc^2}\sqrt{\frac{2\omega^2}{V\varepsilon_0}}\dot{q}_{\boldsymbol{k},\lambda}(t)\cos(kz).$$

The term $\dot{q}(t)$ will play the role of a canonical momentum for a particle of unit mass, $p(t) = \dot{q}(t)$.

1.2.2 Single-mode Hamiltonian

The classical field energy, or Hamiltonian H, of the single-mode field is given by

$$\begin{aligned}
H_{\boldsymbol{k},\lambda} &= \frac{1}{2}\int dV\left[\varepsilon_0 \boldsymbol{E}^2_{\boldsymbol{k},\lambda} + \frac{1}{\mu_0}\boldsymbol{B}^2_{\boldsymbol{k},\lambda}\right] \\
&= \frac{1}{2}A\int_0^L dz\left[\frac{2\omega^2}{V\varepsilon_0}q^2_{\boldsymbol{k},\lambda}(t)\sin^2(kz) + \frac{1}{\mu_0}\frac{1}{k^2c^4}\frac{2\omega^2}{V\varepsilon_0}p^2_{\boldsymbol{k},\lambda}(t)\cos^2(kz)\right] \\
&= \frac{A}{V}\int_0^L dz\left[\omega^2 q^2_{\boldsymbol{k},\lambda}(t)\sin^2(kz) + p^2_{\boldsymbol{k},\lambda}(t)\cos^2(kz)\right] \\
&= \frac{1}{L}\left[\omega^2 q^2_{\boldsymbol{k},\lambda}(t)\frac{L}{2} + p^2_{\boldsymbol{k},\lambda}(t)\frac{L}{2}\right] \\
H_{\boldsymbol{k},\lambda} &= \frac{1}{2}\left[\omega^2 q^2_{\boldsymbol{k},\lambda}(t) + p^2_{\boldsymbol{k},\lambda}(t)\right].
\end{aligned}$$

It is apparent that a single-mode field if formally equivalent to a harmonic quantim oscillator of unit mass, where the E- and B-fields play the roles of canonical position and momentum.

$$H_{\text{harmonic oscillator}} = \frac{1}{2}m\omega^2 x^2 + \frac{p^2}{2m}.$$

To begin the quantization, we make q, p operators $\hat{q}, \hat{p}$, which needs to satisfy the canonical commutation relations

$$[\hat{q}_{\boldsymbol{k},\lambda}, \hat{p}_{\boldsymbol{k},\lambda}] = i\hbar. \tag{1.3}$$

The EM fields with the operators are:

Quantized single-mode EM fields

$$\begin{aligned}
\hat{\boldsymbol{E}}_{\boldsymbol{k},\lambda}(z,t) &= \boldsymbol{e}_{\boldsymbol{k},\lambda}\sqrt{\frac{2\omega^2}{V\varepsilon_0}}\hat{q}(t)\sin(kz) \\
\hat{\boldsymbol{B}}_{\boldsymbol{k},\lambda}(z,t) &= (\boldsymbol{k}\times\boldsymbol{e}_{\boldsymbol{k},\lambda})\frac{1}{kc^2}\sqrt{\frac{2\omega^2}{V\varepsilon_0}}\hat{p}(t)\cos(kz)
\end{aligned}. \tag{1.4}$$

The operators $\hat{q}, \hat{p}$ are Hermitian and therefore correspond to obserable quantites. It is convenient to introduce the ladder operators to simplify the expresison:

Ladder operator

$$\begin{aligned}
\hat{a}_{\boldsymbol{k},\lambda} &= \sqrt{\tfrac{1}{2\hbar\omega}}(\omega\hat{q}_{\boldsymbol{k},\lambda} + i\hat{p}_{\boldsymbol{k},\lambda}) \\
\hat{a}^\dagger_{\boldsymbol{k},\lambda} &= \sqrt{\tfrac{1}{2\hbar\omega}}(\omega\hat{q}_{\boldsymbol{k},\lambda} - i\hat{p}_{\boldsymbol{k},\lambda})
\end{aligned} \longrightarrow \begin{aligned}
\hat{q}_{\boldsymbol{k},\lambda} &= \sqrt{\tfrac{\hbar}{2\omega}}(\hat{a}_{\boldsymbol{k},\lambda} + \hat{a}^\dagger_{\boldsymbol{k},\lambda}) \\
\hat{p}_{\boldsymbol{k},\lambda} &= -i\sqrt{\tfrac{\hbar\omega}{2}}(\hat{a}_{\boldsymbol{k},\lambda} - \hat{a}^\dagger_{\boldsymbol{k},\lambda})
\end{aligned}.$$

Using these defintion, the Hamiltonian yields:

$$\begin{aligned}
\hat{H}_{\boldsymbol{k},\lambda} &= -\frac{1}{2}\frac{\hbar\omega}{2}(\hat{a}_{\boldsymbol{k},\lambda} - \hat{a}^\dagger_{\boldsymbol{k},\lambda})(\hat{a}_{\boldsymbol{k},\lambda} - \hat{a}^\dagger_{\boldsymbol{k},\lambda}) + \frac{1}{2}\frac{\hbar\omega^2}{2\omega}(\hat{a}_{\boldsymbol{k},\lambda} + \hat{a}^\dagger_{\boldsymbol{k},\lambda})(\hat{a}_{\boldsymbol{k},\lambda} + \hat{a}^\dagger_{\boldsymbol{k},\lambda}) \\
&= \frac{\hbar\omega}{2}[\hat{a}^\dagger_{\boldsymbol{k},\lambda}\hat{a}_{\boldsymbol{k},\lambda} + \hat{a}_{\boldsymbol{k},\lambda}\hat{a}^\dagger_{\boldsymbol{k},\lambda}] = \hbar\omega\left[\hat{a}^\dagger_{\boldsymbol{k},\lambda}\hat{a}_{\boldsymbol{k},\lambda} + \frac{1}{2}\right] = \hbar\omega\left[n_{\boldsymbol{k},\lambda} + \frac{1}{2}\right].
\end{aligned}$$

The time dependence of the ladder operator in Heisenberg equation is:

$$\frac{d\hat{a}_{\boldsymbol{k},\lambda}}{dt} = \frac{i}{\hbar}[\hat{H}_{\boldsymbol{k},\lambda}, \hat{a}_{\boldsymbol{k},\lambda}] = \frac{i}{\hbar}\left[\hbar\omega\left(\hat{a}^\dagger_{\boldsymbol{k},\lambda}\hat{a}_{\boldsymbol{k},\lambda} + \frac{1}{2}\right), \hat{a}_{\boldsymbol{k},\lambda}\right] = -i\omega\hat{a}_{\boldsymbol{k},\lambda} \longrightarrow \begin{aligned}
\hat{a}_{\boldsymbol{k},\lambda}(t) &= \hat{a}_{\boldsymbol{k},\lambda}(0)e^{-i\omega t} \\
\hat{a}^\dagger_{\boldsymbol{k},\lambda}(t) &= \hat{a}^\dagger_{\boldsymbol{k},\lambda}(0)e^{i\omega t}
\end{aligned}.$$

For example, we have the following applications:

$$\hat{a}^\dagger_{\boldsymbol{k},\lambda}\left|0\right\rangle = \left|1\right\rangle_{\boldsymbol{k},\lambda}, \quad \hat{a}_{\boldsymbol{k},\lambda}\left|n\right\rangle_{\boldsymbol{k},\lambda} = \sqrt{n}\left|n-1\right\rangle_{\boldsymbol{k},\lambda}, \quad \hat{a}^\dagger_{\boldsymbol{k}',\lambda'}\left|n\right\rangle_{\boldsymbol{k},\lambda} = \left|n\right\rangle_{\boldsymbol{k},\lambda}\left|1\right\rangle_{\boldsymbol{k}',\lambda'}, \quad \hat{a}_{\boldsymbol{k},\lambda}\left|n\right\rangle_{\boldsymbol{k}',\lambda'}\left|0\right\rangle_{\boldsymbol{k},\omega} = 0.$$

The operator product $\hat{a}^\dagger\hat{a}$ has an important significance and is called the number operator $\hat{n}$, whose eigenequation is

$$\hat{n}\left|n\right\rangle = n\left|n\right\rangle.$$

The state $\left|n\right\rangle$ is the energy eigenstte of the single mode field with energy eiganvlaue E_n:

$$\hat{H}\left|n\right\rangle = \hbar\omega(\hat{a}^\dagger\hat{a} + \frac{1}{2})\left|n\right\rangle = E_n\left|n\right\rangle$$

The relation between the number state and the raising operator $\hat{a}^\dagger$ is:

$$\left|n\right\rangle = \frac{(\hat{a}^\dagger)^n}{\sqrt{n}}\left|0\right\rangle.$$

1.2.3 Quantizing the multimode field

We now consider all the modes in the optical cavity:

Quantized multi-mode EM fields
$$\begin{aligned}\boldsymbol{E}(z,t) &= \sum_{m,\lambda}\hat{\boldsymbol{e}}_\lambda\sqrt{\frac{2\omega_m^2}{\varepsilon_0 V}}\hat{q}_{m,\lambda}(t)\sin(k_m z)\\ \boldsymbol{B}(z,t) &= \sum_{m,\lambda}(\hat{\boldsymbol{z}}\times\hat{\boldsymbol{e}}_\lambda)\frac{1}{k_m c^2}\sqrt{\frac{2\omega_m^2}{\varepsilon_0 V}}\dot{\hat{q}}_{m,\lambda}(t)\cos(k_m z)\end{aligned} \tag{1.5}$$

where $kL = m\pi$ and $\hat{\boldsymbol{e}}_\lambda = \{\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}}\}$. The process is anaglogous; we compute the Hamiltonian:

$$\begin{aligned}H &= \int dV\left[\frac{1}{2}\varepsilon_0\boldsymbol{E}^2(z,t) + \frac{1}{2\mu_0}\boldsymbol{B}(z,t)\right]\\ &= \int dV\left[\frac{1}{2}\varepsilon_0\left\{\sum_{m,\alpha}\hat{\boldsymbol{e}}_\alpha\sqrt{\frac{2\omega_m^2}{\varepsilon_0 V}}q_{m,\alpha}(t)\sin(k_m z)\right\}^2 + \frac{1}{2\mu_0}\varepsilon_0\left\{\sum_{m,\alpha}(\hat{\boldsymbol{z}}\times\hat{\boldsymbol{e}}_\alpha)\frac{\mu_0}{\varepsilon_0}k_m\sqrt{\frac{2\omega_m^2}{\varepsilon_0 V}}\dot{q}_{m,\alpha}(t)\cos(k_m z)\right\}^2\right]\\ &\vdots\\ H &= \frac{1}{2}\sum_{m,\lambda}(\omega_m^2 q_{m,\lambda}^2 + \dot{q}_{m,\lambda}^2).\end{aligned}$$

That is,

Each mode of the EM field is an independent harmonic oscillator.

We can quantize the multimode EM field in a similar way as the sige mode field to obtain the Hamiltonian:

Hamiltoninan multimode EM field
$$\hat{H} = \sum_{m,\lambda}\hbar\omega_m\left[\hat{a}^\dagger_{m,\lambda}a_{m,\lambda} + \frac{1}{2}\right] = \sum_{m,\lambda}\hbar\omega_m\left[\hat{n}_{m,\lambda} + \frac{1}{2}\right],$$

where $\hat{n}_{m,\lambda}$ is the number of excitations in mode m, λ.

1.2.4 Quantizing EM field in free space

We define the scalar and vector potential

$$\boldsymbol{B} = \nabla \times \boldsymbol{A} \quad \text{and} \quad \boldsymbol{E} = -\partial_t \boldsymbol{A} - \nabla\phi.$$

Using these automatically satisfy the Gauss magnetic equation and Ampere law. In QO, we assume **Coulomb gauge**: $\nabla \cdot \boldsymbol{A} = 0$. With this, the E-field is

$$\boldsymbol{E} = -\partial_t \boldsymbol{A} - \nabla\phi = \boldsymbol{E}^{\perp} + \boldsymbol{E}^{\parallel}, \quad \text{where} \begin{cases} \nabla \cdot \boldsymbol{E}^{\perp} = 0, & \boldsymbol{A} \text{ corresponds to radiation (all transverse)} \\ \boldsymbol{E} = -\nabla\phi, & \boldsymbol{A} \text{ corresponds to the field of sources (all longitudinal)} \end{cases}.$$

In Coulomb gauge, in absence of charge, we have $\phi = 0$ such that the vector potential-wave equation is

$$\nabla^2 \boldsymbol{A} - \frac{1}{c^2}\partial_t^2 \boldsymbol{A} = 0.$$

It can be solved with the following plane wave

$$\boldsymbol{A}(\boldsymbol{r}, t) = \sum_{\boldsymbol{k},\lambda} \hat{\boldsymbol{e}}_{\boldsymbol{k},\lambda} \left[A_{\boldsymbol{k},\lambda} e^{i(\boldsymbol{k}\cdot\boldsymbol{r} - \omega t)} + A^*_{\boldsymbol{k},\lambda} e^{-i(\boldsymbol{k}\cdot\boldsymbol{r} - \omega t)} \right], \quad \nabla \cdot \boldsymbol{A} = 0 \Longrightarrow \boldsymbol{k} \cdot \hat{\boldsymbol{e}}_{\boldsymbol{k},\lambda} = 0.$$

The vector potential in free space is expressed as a sueprposition of plane waves such that linear dispersion $\omega = |\boldsymbol{k}|c$ is satisfied with $\hat{\boldsymbol{e}}_{\boldsymbol{k},\lambda} \cdot \hat{\boldsymbol{e}}_{\boldsymbol{k},\lambda'} = \delta_{\lambda\lambda'}$.

The quantization volume we quantize the field is considered to have finite volume of V and we impose periodic boundary such that each side is equal.

The E- and B-field are then:

$$\begin{aligned} \boldsymbol{E}(\boldsymbol{r}, t) &= -\partial_t \boldsymbol{A}(\boldsymbol{r}, t) = i \sum_{\boldsymbol{k},\lambda} \hat{\boldsymbol{e}}_{\boldsymbol{k},\lambda} \omega \left[A_{\boldsymbol{k},\lambda} e^{i(\boldsymbol{k}\cdot\boldsymbol{r} - \omega t)} - A^*_{\boldsymbol{k},\lambda} e^{-i(\boldsymbol{k}\cdot\boldsymbol{r} - \omega t)} \right], \\ \boldsymbol{B}(\boldsymbol{r}, t) &= \nabla \times \boldsymbol{A}(\boldsymbol{r}, t) = i \sum_{\boldsymbol{k},\lambda} (\boldsymbol{k} \times \hat{\boldsymbol{e}}_{\boldsymbol{k},\lambda}) \left[A_{\boldsymbol{k},\lambda} e^{i(\boldsymbol{k}\cdot\boldsymbol{r} - \omega t)} - A^*_{\boldsymbol{k},\lambda} e^{-i(\boldsymbol{k}\cdot\boldsymbol{r} - \omega t)} \right]. \end{aligned}$$

The energy (Hamiltonian) associated with the fields is:

$$\begin{aligned} H &= \int dV \left[\frac{1}{2}\varepsilon_0 \boldsymbol{E}^2(\boldsymbol{r}, t) + \frac{1}{2\mu_0} \boldsymbol{B}(\boldsymbol{r}, t) \right] \\ &= \text{Square becomes a dot product of itself for each field} \\ &\vdots \\ H &= 2\varepsilon_0 V \sum_{\boldsymbol{k},\lambda} \omega^2 |A_{\boldsymbol{k},\lambda}|^2. \end{aligned}$$

We have used the following relation:

$$\int dV e^{\pm i(\boldsymbol{k} - \boldsymbol{k}')\cdot\boldsymbol{r}} = \delta_{\boldsymbol{k}\boldsymbol{k}'} V.$$

Defining

$$A_{\boldsymbol{k},\lambda} = \frac{1}{2\omega\sqrt{\varepsilon_0 V}}(\omega q_{\boldsymbol{k},\lambda} + i p_{\boldsymbol{k},\lambda}), \quad A^*_{\boldsymbol{k},\lambda} = \frac{1}{2\omega\sqrt{\varepsilon_0 V}}(\omega q_{\boldsymbol{k},\lambda} - i p_{\boldsymbol{k},\lambda}).$$

The Hamiltonian is:

$$H = \frac{1}{2}\sum_{\boldsymbol{k},\lambda}(\omega^2 q_{\boldsymbol{k},\lambda}^2 + p_{\boldsymbol{k},\lambda}^2) \xrightarrow{\text{quantizing}} \hat{H} = \sum_{\boldsymbol{k},\lambda}\hbar\omega\left(\hat{a}_{\boldsymbol{k},\lambda}^\dagger \hat{a}_{\boldsymbol{k},\lambda} + \frac{1}{2}\right),$$

where we have defined the ladder operators:

$$\hat{a} = \frac{1}{\sqrt{2\hbar\omega}}(\omega\hat{q} + i\hat{p}), \quad \hat{a}^\dagger = \frac{1}{\sqrt{2\hbar\omega}}(\omega\hat{q} - i\hat{p}). \tag{1.6}$$

So, the coefficients are expressed in term of the ladder:

$$A_{\boldsymbol{k},\lambda} = \sqrt{\frac{\hbar}{2\omega\varepsilon_0 V}}\hat{a}_{\boldsymbol{k},\lambda}, \quad A_{\boldsymbol{k},\lambda}^* = \sqrt{\frac{\hbar}{2\omega\varepsilon_0 V}}\hat{a}_{\boldsymbol{k},\lambda}^\dagger.$$

Thus the fields are:

$$\begin{aligned}
\boldsymbol{A}(\boldsymbol{r},t) &= \textstyle\sum_{\boldsymbol{k},\lambda}\sqrt{\frac{\hbar}{2\omega\varepsilon_0 V}}\hat{\boldsymbol{e}}_{\boldsymbol{k},\lambda}\left[\hat{a}_{\boldsymbol{k},\lambda}e^{i(\boldsymbol{k}\cdot\boldsymbol{r}-\omega t)} + \hat{a}_{\boldsymbol{k},\lambda}^\dagger e^{-i(\boldsymbol{k}\cdot\boldsymbol{r}-\omega t)}\right],\\
\boldsymbol{E}(\boldsymbol{r},t) &= i\textstyle\sum_{\boldsymbol{k},\lambda}\sqrt{\frac{\hbar\omega}{2\varepsilon_0 V}}\hat{\boldsymbol{e}}_{\boldsymbol{k},\lambda}\left[\hat{a}_{\boldsymbol{k},\lambda}e^{i(\boldsymbol{k}\cdot\boldsymbol{r}-\omega t)} - \hat{a}_{\boldsymbol{k},\lambda}^\dagger e^{-i(\boldsymbol{k}\cdot\boldsymbol{r}-\omega t)}\right],\\
\boldsymbol{B}(\boldsymbol{r},t) &= \tfrac{i}{c}\textstyle\sum_{\boldsymbol{k},\lambda}\sqrt{\frac{\hbar\omega}{2\varepsilon_0 V}}(\boldsymbol{k}\times\hat{\boldsymbol{e}}_{\boldsymbol{k},\lambda})\left[\hat{a}_{\boldsymbol{k},\lambda}e^{i(\boldsymbol{k}\cdot\boldsymbol{r}-\omega t)} - \hat{a}_{\boldsymbol{k},\lambda}^\dagger e^{-i(\boldsymbol{k}\cdot\boldsymbol{r}-\omega t)}\right].
\end{aligned} \tag{1.7}$$

1.2.5 Electric field expectation and uncertainty

Lets conside one mode of the E-field in a number state: $|n\rangle_{\boldsymbol{k},\lambda}|\{0\}\rangle$. Lets add the time dependence to the ladder operators:

$$\hat{a}_{\boldsymbol{k},\lambda}e^{-i\omega t} \to \hat{a}_{\boldsymbol{k},\lambda}(t), \quad \hat{a}_{\boldsymbol{k},\lambda}^\dagger e^{i\omega t} \to \hat{a}_{\boldsymbol{k},\lambda}^\dagger(t).$$

The E-field is:

$$\hat{\boldsymbol{E}}(\boldsymbol{r},t) = i\sum_{\boldsymbol{k},\lambda} e_{\boldsymbol{k},\lambda}\sqrt{\frac{\hbar\omega}{2\varepsilon_0 V}}\left[\hat{a}_{\boldsymbol{k},\lambda}(t)e^{i\boldsymbol{k}\cdot\boldsymbol{r}} - \hat{a}_{\boldsymbol{k},\lambda}^\dagger(t)e^{-i\boldsymbol{k}\cdot\boldsymbol{r}}\right].$$

Expectation value of the E-field is:

$$\begin{aligned}
{}_{\boldsymbol{k}_0,\lambda_0}\langle n|\hat{\boldsymbol{E}}(\boldsymbol{r},t)|n\rangle_{\boldsymbol{k}_0,\lambda_0} &= i_{\boldsymbol{k}_0,\lambda_0}\langle n|\sum_{\boldsymbol{k},\lambda}\hat{\boldsymbol{e}}_{\boldsymbol{k},\lambda}\sqrt{\frac{\hbar\omega}{2\varepsilon_0 V}}\left[\hat{a}_{\boldsymbol{k},\lambda}(t)e^{i\boldsymbol{k}\cdot\boldsymbol{r}} - \hat{a}_{\boldsymbol{k},\lambda}^\dagger(t)e^{-i\boldsymbol{k}\cdot\boldsymbol{r}}\right]|n\rangle_{\boldsymbol{k}_0,\lambda_0}\\
&= i_{\boldsymbol{k}_0,\lambda_0}\langle n|\hat{\boldsymbol{e}}_{\boldsymbol{k}_0,\lambda_0}\sqrt{\frac{\hbar\omega_0}{2\varepsilon_0 V}}\left[\sqrt{n}\,|n-1\rangle_{\boldsymbol{k}_0,\lambda_0}e^{i\boldsymbol{k}_0\cdot\boldsymbol{r}} - \sqrt{n+1}\,|n+1\rangle_{\boldsymbol{k}_0,\lambda_0}e^{-i\boldsymbol{k}_0\cdot\boldsymbol{r}}\right]\\
&= 0.
\end{aligned}$$

Thus, the average E-field is zero.

Lets consider the uncertainty of the E-field:

$$\begin{aligned}
\langle\hat{\boldsymbol{E}}\rangle =& {}_{\boldsymbol{k}_0,\lambda_0}\langle n|\sum_{\boldsymbol{k},\lambda}\sum_{\boldsymbol{k}',\lambda'}\sqrt{\frac{\hbar\omega}{2\varepsilon_0 V}}\sqrt{\frac{\hbar\omega'}{2\varepsilon_0 V}}\hat{\boldsymbol{e}}_{\boldsymbol{k},\lambda}\cdot\hat{\boldsymbol{e}}_{\boldsymbol{k}',\lambda'}\left[\hat{a}_{\boldsymbol{k},\lambda}e^{i\boldsymbol{k}\cdot\boldsymbol{r}} - \hat{a}_{\boldsymbol{k},\lambda}^\dagger e^{-i\boldsymbol{k}\cdot\boldsymbol{r}}\right]\left[\hat{a}_{\boldsymbol{k}',\lambda'}e^{i\boldsymbol{k}'\cdot\boldsymbol{r}} - \hat{a}_{\boldsymbol{k}',\lambda'}^\dagger e^{-i\boldsymbol{k}'\cdot\boldsymbol{r}}\right]|n\rangle_{\boldsymbol{k}_0,\lambda_0}\\
=& \sum_{\boldsymbol{k},\lambda}\sum_{\boldsymbol{k}',\lambda'}\sqrt{\frac{\hbar\omega}{2\varepsilon_0 V}}\sqrt{\frac{\hbar\omega'}{2\varepsilon_V}}{}_{\boldsymbol{k},\lambda}\langle 1|1\rangle_{\boldsymbol{k}',\lambda'}e^{i(\boldsymbol{k}-\boldsymbol{k}')\cdot\boldsymbol{r}} + \frac{\hbar\omega_0}{2\varepsilon_0 V}[{}_{\boldsymbol{k}_0,\lambda_0}\langle n+1|\sqrt{n+1} + {}_{\boldsymbol{k}_0,\lambda_0}\langle n-1|\sqrt{n}][\sqrt{n+1}\,|n+1\rangle_{\boldsymbol{k}_0,\lambda_0} +\\
=& \sum_{\{\boldsymbol{k},\lambda\}\neq\{\boldsymbol{k}_0,\lambda_0\}}\frac{\hbar\omega}{2\varepsilon_0 V} + \frac{\hbar\omega_0}{2\varepsilon_0 V}(2n+1).
\end{aligned}$$

So, the uncertainty is:

$$\text{E-field uncertainty} \qquad \langle \Delta E^2 \rangle = \underbrace{\sum_{\{\boldsymbol{k},\lambda\}\neq\{\boldsymbol{k}_0,\lambda_0\}} \frac{\hbar\omega}{2\varepsilon_0 V}}_{\text{All modes in vaccum state}} + \underbrace{\frac{\hbar\omega_0}{2\varepsilon_0 V}(2n+1)}_{\text{Mode } (\boldsymbol{k}_0,\lambda_0)}.$$

Thus, there is an uncertainty in the E-field due to the *quantum fluctuations*, even in vacuum.

1.3 Coherent states

The coherent state is defined as the eigenstate of the annihilator operator:

$$\text{Coherent state} \qquad \hat{a}\,|\alpha\rangle = \alpha\,|\alpha\rangle\,, \quad [\hat{a},\hat{a}^\dagger] = 1, \quad \alpha \in \mathbb{C}.$$

The expected number of photons in the coherent state is:

$$\langle\alpha|\hat{N}|\alpha\rangle = \langle\alpha|\hat{a}^\dagger\hat{a}|\alpha\rangle = |\alpha|^2.$$

Coherent states are the most classical states of the quantized EM field.

It has the following results:

$$\langle\alpha|\hat{a}^2|\alpha\rangle = \alpha^2, \quad \langle\alpha|\hat{a}^{\dagger 2}|\alpha\rangle = \alpha^{*2}, \quad \langle\alpha|\hat{a}^\dagger\hat{a}|\alpha\rangle = |\alpha|^2, \quad \langle\alpha|\hat{a}\hat{a}^\dagger|\alpha\rangle = |\alpha|^2 + 1.$$

For example, computing the expectation of the E-field (1.7) is:

$$\begin{aligned}
\langle\boldsymbol{E}\rangle &= \langle\alpha|i\sum_{\boldsymbol{k},\lambda}\sqrt{\frac{\hbar\omega}{2\varepsilon_0 V}}\hat{\boldsymbol{e}}_{\boldsymbol{k},\lambda}[\hat{a}_{\boldsymbol{k},\lambda}e^{i(\boldsymbol{k}\cdot\boldsymbol{r}-\omega t)} - \hat{a}^\dagger_{\boldsymbol{k},\lambda}e^{-i(\boldsymbol{k}\cdot\boldsymbol{r}-\omega t)}]|\alpha\rangle \\
&= i\sqrt{\frac{\hbar\omega}{2\varepsilon_0 V}}\hat{\boldsymbol{e}}_{\boldsymbol{k},\lambda}[\alpha e^{i(\boldsymbol{k}\cdot\boldsymbol{r}-\omega t)} - \alpha^* e^{-i(\boldsymbol{k}\cdot\boldsymbol{r}-\omega t)}] \\
\langle\boldsymbol{E}\rangle &= \boldsymbol{\mathcal{E}}e^{i(\boldsymbol{k}\cdot\boldsymbol{r}-\omega t)} + \boldsymbol{\mathcal{E}}^* e^{-i(\boldsymbol{k}\cdot\boldsymbol{r}-\omega t)} \longrightarrow \boldsymbol{\mathcal{E}} = i\sqrt{\frac{\hbar\omega}{2\varepsilon_0 V}}\alpha\hat{\boldsymbol{e}}_{\boldsymbol{k},\lambda}.
\end{aligned}$$

The expectation value of the E-field for a coherent state corresponds to a classical E-field.

Let us now consider the uncertainty of the E-field for $|\alpha\rangle$:

$$\begin{aligned}
\langle\Delta E^2\rangle &= \langle\boldsymbol{E}^2\rangle - \langle\boldsymbol{E}\rangle^2 \\
&= \langle\alpha|i\hat{\boldsymbol{e}}_{\boldsymbol{k},\lambda}\sqrt{\frac{\hbar\omega}{2\varepsilon_0 V}}[\hat{a}e^{i(\boldsymbol{k}\cdot\boldsymbol{r}-\omega t)} - \hat{a}^\dagger e^{-i(\boldsymbol{k}\cdot\boldsymbol{r}-\omega t)}]\cdot i\hat{\boldsymbol{e}}_{\boldsymbol{k},\lambda}\sqrt{\frac{\hbar\omega}{2\varepsilon_0 V}}[\hat{a}e^{i(\boldsymbol{k}\cdot\boldsymbol{r}-\omega t)} - \hat{a}^\dagger e^{-i(\boldsymbol{k}\cdot\boldsymbol{r}-\omega t)}]|\alpha\rangle - \langle\boldsymbol{E}\rangle^2 \\
&= -\frac{\hbar\omega}{2\varepsilon_0 V}[\alpha^2 e^{i2(\boldsymbol{k}\cdot\boldsymbol{r}-\omega t)} + \alpha^{*2}e^{-i2(\boldsymbol{k}\cdot\boldsymbol{r}-\omega t)} - |\alpha|^2 - (|\alpha|^2+1)] - \boldsymbol{\mathcal{E}}^2 e^{i2(\boldsymbol{k}\cdot\boldsymbol{r}-\omega t)} - \boldsymbol{\mathcal{E}}^{*2}e^{-i2(\boldsymbol{k}\cdot\boldsymbol{r}-\omega t)} - 2|\boldsymbol{\mathcal{E}}|^2 \\
&= \boldsymbol{\mathcal{E}}^2 e^{i2(\boldsymbol{k}\cdot\boldsymbol{r}-\omega t)} + \boldsymbol{\mathcal{E}}^{*2}e^{-i2(\boldsymbol{k}\cdot\boldsymbol{r}-\omega t)} + 2|\boldsymbol{\mathcal{E}}|^2 + \frac{\hbar\omega}{2\varepsilon_0 V} - \boldsymbol{\mathcal{E}}^2 e^{i2(\boldsymbol{k}\cdot\boldsymbol{r}-\omega t)} - \boldsymbol{\mathcal{E}}^{*2}e^{-2i(\boldsymbol{k}\cdot\boldsymbol{r}-\omega t)} - 2|\boldsymbol{\mathcal{E}}|^2 \\
\langle\Delta E^2\rangle &= \frac{\hbar\omega}{2\varepsilon_0 V}.
\end{aligned}$$

This is the uncertainty in E-field limited by quantum noise.

Also, we know that the eigenstates of $|\alpha\rangle$ do not form an orthonormal basis. However, we know that the number state $|n\rangle$ form an orthonormal basis and we can expand the coherent state in terms of the number states, as follows,

$$|\alpha\rangle = e^{-|\alpha|^2/2} \sum_n \frac{\alpha^n}{\sqrt{n!}} |n\rangle \,.$$

The probability of finding n photons in a coherent state is:

$$P_n = |\langle n|\alpha\rangle|^2 = \left| \langle n| e^{-|\alpha|^2/2} \sum_m \frac{\alpha^m}{\sqrt{m!}} |m\rangle \right|^2 = \left| e^{-|\alpha|^2/2} \frac{\alpha^n}{\sqrt{n!}} \right|^2 = e^{-|\alpha|^2} \frac{|\alpha|^{2n}}{n!}.$$

Recalling that $\langle \hat{n} \rangle = |\alpha|^2 = \overline{n}$:

Probability of finding n photons in a coherent state $$\boxed{P_n = \frac{e^{-\overline{n}} \overline{n}^n}{n!}}.$$

1.4 Quadrature operators

Recall the E-field operator for a singel mode is:

$$\begin{aligned}
\hat{\boldsymbol{E}}(\boldsymbol{r}, t) &= i\sqrt{\frac{\hbar\omega}{2\varepsilon_0 V}} \hat{\boldsymbol{e}}_{\boldsymbol{k},\lambda} [\hat{a} e^{i(\boldsymbol{k}\cdot\boldsymbol{r} - \omega t)} - \hat{a}^\dagger e^{-i(\boldsymbol{k}\cdot\boldsymbol{r} - \omega t)}] \\
&= i\sqrt{\frac{\hbar\omega}{2\varepsilon_0 V}} \hat{\boldsymbol{e}}_{\boldsymbol{k},\lambda} [\hat{a} e^{-i\omega t} e^{i\boldsymbol{k}\cdot\boldsymbol{r}} - \hat{a}^\dagger e^{i\omega t} e^{-i\boldsymbol{k}\cdot\boldsymbol{r}}] \\
&= i\sqrt{\frac{\hbar\omega}{2\varepsilon_0 V}} \hat{\boldsymbol{e}}_{\boldsymbol{k},\lambda} [\hat{a}(t) e^{i\boldsymbol{k}\cdot\boldsymbol{r}} - \hat{a}^\dagger(t) e^{-i\boldsymbol{k}\cdot\boldsymbol{r}}] \\
\hat{\boldsymbol{E}}(\boldsymbol{r}, t) &= i\sqrt{\frac{\hbar\omega}{2\varepsilon_0 V}} \hat{\boldsymbol{e}}_{\boldsymbol{k},\lambda} [\hat{a}(t)\cos(\boldsymbol{k}\cdot\boldsymbol{r}) + i\hat{a}(t)\sin(\boldsymbol{k}\cdot\boldsymbol{r}) - \hat{a}^\dagger(t)\cos(\boldsymbol{k}\cdot\boldsymbol{r}) + i\hat{a}^\dagger(t)\sin(\boldsymbol{k}\cdot\boldsymbol{r})].
\end{aligned}$$

Lets define the quadrature operators:

Quadrature operators $$\boxed{\begin{aligned} \hat{X}_1 &= \frac{1}{2}(\hat{a} + \hat{a}^\dagger) \\ \hat{X}_2 &= \frac{1}{2i}(\hat{a} - \hat{a}^\dagger) \end{aligned}, \quad [\hat{X}_1, \hat{X}_2] = \frac{i}{2}}, \tag{1.8}$$

In the E-field, if we collect the cosine and sine, we get

$$\hat{\boldsymbol{E}}(\boldsymbol{r}, t) = -\sqrt{\frac{2\hbar\omega}{\varepsilon_0 V}} \hat{\boldsymbol{e}}_{\boldsymbol{k},\lambda} [\hat{X}_1(t)\sin(\boldsymbol{k}\cdot\boldsymbol{r}) + \hat{X}_2(t)\cos(\boldsymbol{k}\cdot\boldsymbol{r})].$$

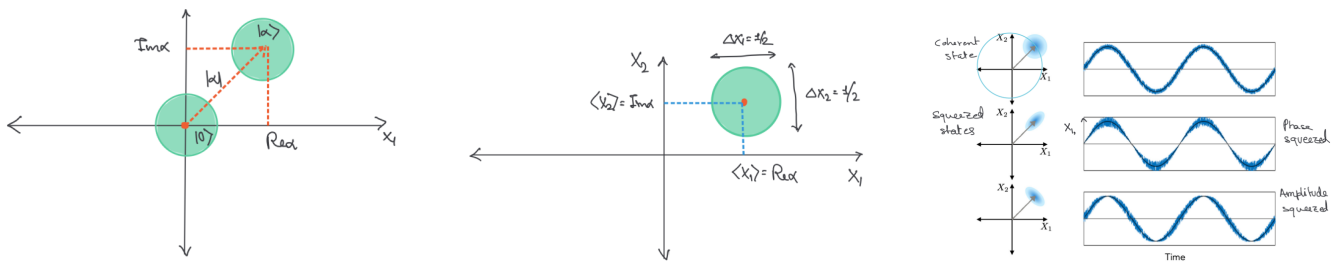
The expectation value of these operators is:

$$\begin{aligned}
&\langle \hat{X}_1 \rangle = \langle \alpha | \hat{X}_1(0) | \alpha \rangle = \langle \alpha | \frac{\hat{a} + \hat{a}^\dagger}{2} | \alpha \rangle = \mathrm{Re}\,(\alpha)\,, \quad \langle \hat{X}_2 \rangle = \langle \alpha | \hat{X}_2(0) | \alpha \rangle = \langle \alpha | \frac{\hat{a} - \hat{a}^\dagger}{2i} | \alpha \rangle = \mathrm{Im}\,(\alpha)\,, \\
&\langle \Delta \hat{X}_1^2 \rangle = \langle \alpha | \frac{1}{4}(\hat{a} + \hat{a}^\dagger)^2 | \alpha \rangle - \frac{1}{4}(\alpha + \alpha^*)^2 = \frac{1}{4}, \quad \langle \Delta \hat{X}_2^2 \rangle = \frac{1}{4}.
\end{aligned}$$

That is, there is a minimum uncertainty that a state can possible have in the two quadrature $\hat{X}_1, \hat{X}_2$, such a state is refferred as *minimum uncertainty state*. When $\alpha = 0$ we have the vaccum state and therefore $\langle \hat{X}_1(0) \rangle = \langle \hat{X}_2(0) \rangle = 0$, $\langle \hat{X}_1^2(0) \rangle = \langle \hat{X}_2^2(0) \rangle = 1/4$.

Coherent states are also referred to as displaced vaccum states:

$$|\alpha\rangle = \hat{D}(\alpha)\,|0\rangle\,, \quad \hat{D}(\alpha) = e^{\alpha\hat{a}^\dagger - \alpha^*\hat{a}}.$$



(a) Displacement of vacuum **(b)** Coherent state **(c)** Evolution of coherent state

The phase space representation of states allows one to access information aabout the noise/variance associated with any state when a specific E-field quadrature is measured.

1.4.1 Time dependence of quadrature operators

Using (1.8) and the inclusion of time-dependence yields

Time-dependent Quadrature operators

$$\boxed{\begin{aligned} \hat{X}_1(t) &= \frac{\hat{a}e^{-i\omega t} + \hat{a}^\dagger e^{i\omega t}}{2} \\ \hat{X}_2(t) &= \frac{\hat{a}e^{-i\omega t} - \hat{a}^\dagger e^{i\omega t}}{2i} \end{aligned}} \,. \tag{1.9}$$

We can use them to express them in terms of $\hat{X}_1(0)$ and $\hat{X}_2(0)$.

$$\hat{a} + \hat{a}^\dagger = 2\hat{X}_1(0), \tag{1.10}$$

$$\hat{a} - \hat{a}^\dagger = 2i\hat{X}_2(0). \tag{1.11}$$

$$\begin{aligned} (1.10) + (1.11): \quad & 2\hat{a} = 2[\hat{X}_1(0) + i\hat{X}_2(0)] \longrightarrow \hat{a} = \hat{X}_1(0) + i\hat{X}_2(0) \\ (1.10) - (1.11): \quad & 2\hat{a}^\dagger = 2[\hat{X}_1(0) - i\hat{X}_2(0)] \longrightarrow \hat{a}^\dagger = \hat{X}_1(0) - i\hat{X}_2(0) \end{aligned}$$

Using these in the time-evolved quadrature operators:

$$\begin{aligned} \hat{X}_1(t) &= \frac{\hat{a}e^{-i\omega t} + \hat{a}^\dagger e^{i\omega t}}{2} = \frac{[\hat{X}_1(0) + i\hat{X}_2(0)]e^{-i\omega t} + [\hat{X}_1(0) - i\hat{X}_2(0)]e^{i\omega t}}{2} \\ \hat{X}_1(t) &= \frac{e^{i\omega t} + e^{-i\omega t}}{2}\hat{X}_1(0) + \frac{(e^{i\omega t} - e^{-i\omega t})}{2i}\hat{X}_2(0). \end{aligned}$$

Likewise,

$$\begin{aligned} \hat{X}_2(t) &= \frac{\hat{a}e^{-i\omega t} - \hat{a}^\dagger e^{i\omega t}}{2i} = \frac{[\hat{X}_1(0) + i\hat{X}_2(0)]e^{-i\omega t} - [\hat{X}_1(0) - i\hat{X}_2(0)]e^{i\omega t}}{2i} \\ \hat{X}_2(t) &= -\frac{e^{i\omega t} - e^{-i\omega t}}{2i}\hat{X}_1(0) + i\frac{(e^{i\omega t} + e^{-i\omega t})}{2i}\hat{X}_2(0). \end{aligned}$$

Then, we have

$$\begin{aligned}\hat{X}_1(t) &= \cos(\omega t)\hat{X}_1(0) + \sin(\omega t)\hat{X}_2(0),\\ \hat{X}_2(t) &= -\sin(\omega t)\hat{X}_1(0) + \cos(\omega t)\hat{X}_2(0).\end{aligned} \tag{1.12}$$

1.4.2 Taking the continuum limit

In free space the sum over all $\boldsymbol{k}$ becomes an integral:

$$\sum_{\boldsymbol{k}} \to N \int d^2\boldsymbol{k} = N \int_0^\infty dk k^2 \int_0^\pi d\theta \sin\theta \int_0^{2\pi} d\phi,$$

where N is the normalization equal to volume/mode. We consider a volume of quantization that is the region in free space where we are quantizing the field $V = L_x L_y L_z$. Allowed $\boldsymbol{k}$ vectors are $k_x = (2\pi/L_x)m_x$, $k_y = (2\pi/L_y)m_y$ y $k_z = (2\pi/L_z)m_z$, that is, we ensure that the lenths fit an integer number of wavelengths.

The number of modes in between $\boldsymbol{k}$ and $\boldsymbol{k} + \Delta\boldsymbol{k}$ is:

$$\Delta m_x \Delta m_y \Delta m_z = \frac{L_x L_y L_z}{(2\pi)^2}\Delta\boldsymbol{k}_x\Delta\boldsymbol{k}_y\Delta\boldsymbol{k}_z = \frac{V}{(2\pi)^3}\Delta\boldsymbol{k}_x\Delta\boldsymbol{k}_y\Delta\boldsymbol{k}_z.$$

Using the above normalization:

$$\sum_{\boldsymbol{k},\lambda} \to \sum_{\lambda} \int dk k^2 \int_0^\pi d\theta \sin\theta \int_0^{2\pi} d\phi \frac{V}{8\pi^2}.$$

Thus, any continuum limit involves

$$\text{Continuum limit} \qquad \sum_{\boldsymbol{k}} \to \frac{V}{8\pi^2}\int d^3\boldsymbol{k}.$$

1.5 Squeezed states

Lets define the squeeze operator

$$\text{Squeezing state} \qquad |\xi, 0\rangle = \hat{S}(\xi)\,|0\rangle\,, \quad \text{with} \quad \hat{S}(\xi) = e^{\frac{1}{2}(\xi^*\hat{a}^2 - \xi\hat{a}^{\dagger 2})}, \quad \xi \in \mathbb{C}.$$

Lets use it to compute the expectation value for the quadrature operators.

$$\langle\xi, 0|\hat{X}_1|\xi, 0\rangle = \langle 0|\hat{S}^\dagger(\xi)\left(\frac{\hat{a} + \hat{a}^\dagger}{2}\right)\hat{S}(\xi)|0\rangle = \frac{1}{2}\langle 0|(\hat{S}^\dagger\hat{a}\hat{S} + \hat{S}^\dagger\hat{a}^\dagger\hat{S})|0\rangle.$$

We define displaced ladder operators as

$$\tilde{a} = \hat{S}^\dagger(\xi)\hat{a}\hat{S}(\xi), \quad \xi = re^{i\phi}.$$

Using the BCH formula

$$e^L A e^{-L} = A + [L, A] + \frac{1}{2!}[L, [L, A]] + \cdots$$

and $A = \hat{a}$ with $L = -\frac{1}{2}(\xi^* \hat{a}^2 - \xi \hat{a}^{\dagger 2})$ allow us to express the squeezing ladder operator

Squeezing ladder operators
$$\begin{aligned} \tilde{a} &= \hat{S}^\dagger(\xi)\hat{a}\hat{S}(\xi) = \hat{a}\cosh(r) - \hat{a}^\dagger e^{i\phi}\sinh(r) \\ \tilde{a}^\dagger &= S(\xi)^\dagger \hat{a}^\dagger \hat{S}(\xi) = \hat{a}^\dagger \cosh(r) - \hat{a} e^{i\phi}\sinh(r). \end{aligned} \tag{1.13}$$

We have used several orders. Its derivative wrt r is:

$$\frac{d\hat{a}}{dr} = \hat{a}\sinh(r) - \hat{a}^\dagger e^{i\phi}\cosh(r) = -e^{i\phi}\tilde{a}^\dagger, \quad \frac{d\tilde{a}^\dagger}{dr} = -e^{-i\phi}\tilde{a}.$$

Taking another derivative yields:

$$\frac{d^2\tilde{a}}{dr^2} = \tilde{a}.$$

Now we can continue the expectation of the quadrature:

$$\begin{aligned} \langle \hat{X}_1 \rangle &= \frac{1}{2}\langle 0|(\hat{S}^\dagger \hat{a}\hat{S} + \hat{S}^\dagger \hat{a}^\dagger \hat{S})|0\rangle = \frac{1}{2}\langle 0| \left[\tilde{a} + \tilde{a}^\dagger\right]|0\rangle \\ &= \frac{1}{2}\langle 0|[\hat{a}\cosh(r) - e^{i\phi}\hat{a}^\dagger \sinh(r) + \hat{a}^\dagger \cosh(r) - e^{-i\phi}\sinh(r)]|0\rangle = 0. \\ \langle \hat{X}_2 \rangle &= \text{similarly} = 0. \end{aligned}$$

We can do the same for the operator square:

$$\begin{aligned} \langle \hat{X}_1^2 \rangle &= \frac{1}{4}\langle 0|\hat{S}^\dagger(\xi)[(\hat{a} + \hat{a}^\dagger)(\hat{a} + \hat{a}^\dagger)]\hat{S}(\xi)|0\rangle = \frac{1}{4}\langle 0|\hat{S}^\dagger(\hat{a} + \hat{a}^\dagger)\hat{S}\hat{S}^\dagger(\hat{a} + \hat{a}^\dagger)\hat{S}|0\rangle \\ &= \frac{1}{4}\langle 0|(\tilde{a} + \tilde{a}^\dagger)(\tilde{a} + \tilde{a}^\dagger)|0\rangle = \frac{1}{4}\langle 0|e^{-2r}(\hat{a} + \hat{a}^\dagger)^2|0\rangle = \frac{e^{-2r}}{4}. \end{aligned}$$

Similarly we can obtain the respective quantities for $\hat{X}_2$ and the uncertainties are:

$$\langle \hat{X}_1 \rangle = \langle \hat{X}_2 \rangle = 0 \Longrightarrow \Delta\hat{X}_1 = \frac{e^{-r}}{2}, \quad \Delta\hat{X}_2 = \frac{e^r}{2}.$$

1.5.1 Displaced squeezing state

Now,

Displaced squeezing state
$$|\xi, \alpha\rangle = \hat{D}(\alpha)\hat{S}(\xi)\,|0\rangle\,.$$

Now, the operators are:

Displaced squeezing operator
$$\begin{aligned} \tilde{a} &= \hat{S}^\dagger(\xi)\hat{D}^\dagger(\alpha)\hat{a}\hat{D}(\alpha)\hat{S}(\xi) = \cosh(r)\hat{a} - e^{i\phi}\sinh(r)\hat{a}^\dagger + \alpha, \\ \tilde{a}^\dagger &= \hat{S}^\dagger(\xi)\hat{D}^\dagger(\alpha)\hat{a}^\dagger\hat{D}(\alpha)\hat{S}(\xi) = \cosh(r)\hat{a}^\dagger - e^{-i\phi}\sinh(r)\hat{a} + \alpha^*. \end{aligned} \tag{1.14}$$

The expected number of photons in a displaced squeezed state is:

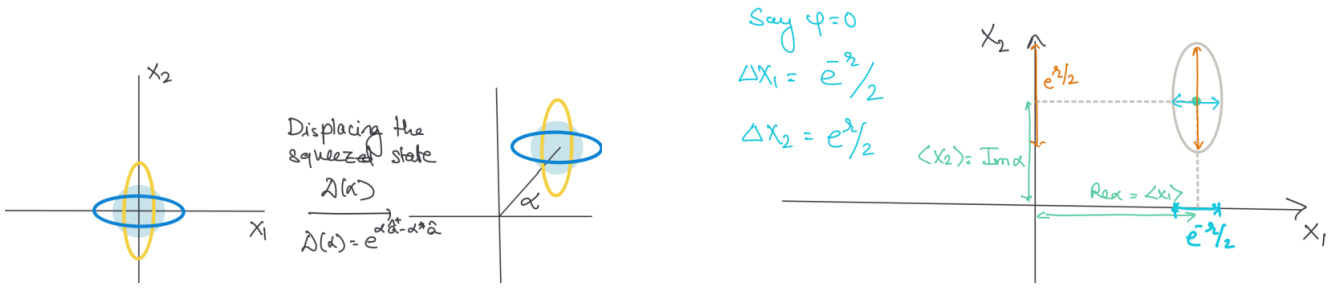
$$\langle 0|\tilde{a}^\dagger\tilde{a}|0\rangle = \langle 0|[\cosh(r)\hat{a}^\dagger - e^{-i\phi}\sinh(r)\hat{a} + \alpha^*][\cosh(r)\hat{a} - e^{i\phi}\sinh(r)\hat{a}^\dagger + \alpha]|0\rangle = \sinh^2|\xi| + |\alpha|^2.$$

The expected value in the quadrature value is:

$$\begin{aligned} \langle \xi, \alpha|\hat{X}_1|\xi, \alpha\rangle &= \langle 0|\hat{S}^\dagger\hat{D}^\dagger\hat{X}_1\hat{D}\hat{S}|0\rangle = \langle 0|\hat{S}^\dagger\hat{D}^\dagger(\frac{\hat{a} + \hat{a}^\dagger}{2})\hat{D}\hat{S}|0\rangle = \frac{1}{2}\langle 0|(\tilde{a} + \tilde{a}^\dagger)|0\rangle \\ &= \frac{1}{2}\langle 0|[\cosh(r)\hat{a} - e^{i\phi}\sinh(r)\hat{a}^\dagger + \alpha + \cosh(r)\hat{a}^\dagger - e^{-i\phi}\sinh(r)\hat{a} + \alpha^*]|0\rangle = \mathrm{Re}\,(\alpha)\,. \\ \langle \xi, \alpha|\hat{X}_2|\xi, \alpha\rangle &= \mathrm{Im}\,(\alpha)\,. \end{aligned}$$

On the other hand, the quadrature uncertainties are:

$$
\begin{aligned}
\Delta\hat{X}_1^2 &= \langle\xi,\alpha|\hat{X}_1^2|\xi,\alpha\rangle - \langle\xi,\alpha|\hat{X}_1|\xi,\alpha\rangle^2 = \langle 0|\hat{S}^\dagger\hat{D}^\dagger\hat{X}_1\hat{D}\hat{S}\hat{S}^\dagger\hat{D}^\dagger\hat{X}_1\hat{D}\hat{S}|0\rangle - \mathrm{Re}\,(\alpha)^2 \\
&= \langle 0|\left(\frac{\tilde{a}+\tilde{a}^\dagger}{2}\right)\left(\frac{\tilde{a}+\tilde{a}^\dagger}{2}\right)|0\rangle - \mathrm{Re}\,(\alpha)^2 \\
&= \frac{1}{4}\langle 0|[\cosh(r)\hat{a} - e^{i\phi}\sinh(r)\hat{a}^\dagger + \alpha + \cosh(r)\hat{a}^\dagger - e^{-i\phi}\sinh(r)\hat{a} + \alpha^*]^2|0\rangle - \mathrm{Re}\,(\alpha)^2 \\
&= \frac{1}{4}(\cosh^2(r) + \sinh^2(r) - 2\cosh(r)\sinh(r)\cos\phi) + \mathrm{Re}\,(\alpha)^2 - \mathrm{Re}\,(\alpha)^2 \\
\Delta\hat{X}_1^2 &= \frac{1}{4}[\cosh^2(r) + \sinh^2(r) - 2\cosh(r)\sinh(r)\cos(\phi)]. \\
\Delta\hat{X}_2^2 &= \frac{1}{4}[\cosh^2(r) + \sinh^2(r) + 2\cosh(r)\sinh(r)\cos(\phi)].
\end{aligned}
$$



(a) **(b)**

Figure 1.1

1.5.2 Displaced squeezed temporal state

This is the most general case of the two above, because it considers the squeezing displaced states and the temporal quadrature operators.

The expected value of $\hat{X}_1(t)$ is

$$
\begin{aligned}
\langle\hat{X}_1(t)\rangle &= \langle 0|\hat{S}^\dagger\hat{D}^\dagger\frac{\hat{a}e^{-i\omega t} + \hat{a}^\dagger e^{i\omega t}}{2}\hat{D}\hat{S}|0\rangle = \frac{1}{2}\langle 0|\tilde{a}e^{-i\omega t} + \tilde{a}^\dagger e^{i\omega t}|0\rangle \\
&= \frac{1}{2}\langle 0|\left[\cosh(r)\hat{a}e^{-i\omega t} - e^{i\phi}\sinh(r)\hat{a}^\dagger e^{-i\omega t} + \alpha e^{-i\omega t} + \cosh(r)\hat{a}^\dagger e^{i\omega t} - e^{-i\phi}\sinh(r)\hat{a}e^{i\omega t} + \alpha^* e^{i\omega t}\right]|0\rangle \\
&= \frac{1}{2}[\alpha e^{-i\omega t} + \alpha^* e^{i\omega t}] \\
\langle\hat{X}_1(t)\rangle &= \mathrm{Re}\left(\alpha e^{-i\omega t}\right).
\end{aligned}
$$

Similarly,

$$
\begin{aligned}
\langle\hat{X}_2(t)\rangle &= \langle 0|\hat{S}^\dagger\hat{D}^\dagger\frac{\hat{a}e^{-i\omega t} - \hat{a}^\dagger e^{i\omega t}}{2i}\hat{D}\hat{S}|0\rangle = \frac{1}{2i}\langle 0|\tilde{a}e^{-i\omega t} + \tilde{a}^\dagger e^{i\omega t}|0\rangle \\
&= \frac{1}{2i}\langle 0|\left[\cosh(r)\hat{a}e^{-i\omega t} - e^{i\phi}\sinh(r)\hat{a}^\dagger e^{-i\omega t} + \alpha e^{-i\omega t} - \cosh(r)\hat{a}^\dagger e^{i\omega t} + e^{-i\phi}\sinh(r)\hat{a}e^{i\omega t} - \alpha^* e^{i\omega t}\right]|0\rangle \\
&= \frac{1}{2}[\alpha e^{-i\omega t} - \alpha^* e^{i\omega t}] \\
\langle\hat{X}_2(t)\rangle &= \mathrm{Im}\left(\alpha e^{-i\omega t}\right).
\end{aligned}
$$

Doing the same for the operators squared:

$$\begin{aligned}
\langle \hat{X}_1^2(t)\rangle &= \langle 0|\hat{S}^\dagger \hat{D}^\dagger \left(\frac{\hat{a}e^{-i\omega t}+\hat{a}^\dagger e^{i\omega t}}{2}\right)^2 \hat{D}\hat{S}|0\rangle \\
&= \frac{1}{4}\langle 0|\hat{S}^\dagger \hat{D}^\dagger \left(\hat{a}e^{-i\omega t}+\hat{a}^\dagger e^{i\omega t}\right)\hat{D}\hat{S}\hat{S}^\dagger \hat{D}^\dagger \left(\hat{a}e^{-i\omega t}+\hat{a}^\dagger e^{i\omega t}\right)\hat{D}\hat{S}|0\rangle \\
&= \frac{1}{4}\langle 0|(\tilde{a}e^{-i\omega t}+\tilde{a}^\dagger e^{i\omega t})^2|0\rangle \\
&= \frac{1}{4}\left[e^{-i2\omega t}\langle 0|\tilde{a}^2|0\rangle + e^{i2\omega t}\langle 0|\tilde{a}^{\dagger 2}|0\rangle + \langle 0|\tilde{a}\tilde{a}^\dagger|0\rangle + \langle 0|\tilde{a}^\dagger\tilde{a}|0\rangle\right].
\end{aligned}$$

We compute each term separately:

$$\begin{aligned}
\langle 0|\tilde{a}^2|0\rangle =& \langle 0|(\cosh(r)\hat{a} - e^{i\phi}\sinh(r)\hat{a}^\dagger + \alpha)(\cosh(r)\hat{a} - e^{i\phi}\sinh(r)\hat{a}^\dagger + \alpha)|0\rangle \\
=& \cosh^2(r)\langle \hat{a}^2\rangle - e^{i\phi}\sinh(r)\cosh(r)\langle \hat{a}\hat{a}^\dagger\rangle + \alpha\cosh(r)\langle \hat{a}\rangle - e^{i\phi}\sinh(r)\cosh(r)\langle \hat{a}^\dagger\hat{a}\rangle \\
&+ e^{i2\phi}\sinh^2(r)\langle \hat{a}^{\dagger 2}\rangle - \alpha e^{i\phi}\sinh(r)\langle \hat{a}^\dagger\rangle + \alpha\cosh(r)\langle \hat{a}\rangle - \alpha e^{i\phi}\sinh(r)\langle \hat{a}^\dagger\rangle + \alpha^2 \\
&= \alpha^2 - e^{i\phi}\sinh(r)\cosh(r) \\
\langle 0|\tilde{a}^2|0\rangle =& \alpha^2 - \frac{1}{2}e^{i\phi}\sinh(2r).
\end{aligned}$$

$$\begin{aligned}
\langle 0|\tilde{a}^{\dagger 2}|0\rangle =& \langle 0|(\cosh(r)\hat{a}^\dagger - e^{-i\phi}\sinh(r)\hat{a} + \alpha^*)(\cosh(r)\hat{a}^\dagger - e^{-i\phi}\sinh(r)\hat{a} + \alpha^*)|0\rangle \\
=& \cosh^2(r)\langle \hat{a}^{\dagger 2}\rangle - e^{-i\phi}\cosh(r)\sinh(r)\langle \hat{a}^\dagger\hat{a}\rangle + \alpha^*\cosh(r)\langle \hat{a}^\dagger\rangle - e^{-i\phi}\sinh(r)\cosh(r)\langle \hat{a}\hat{a}^\dagger\rangle \\
&+ e^{-i2\phi}\sinh^2(r)\langle \hat{a}^2\rangle - \alpha^* e^{-i\phi}\sinh(r)\langle \hat{a}\rangle + \alpha^*\cosh(r)\langle \hat{a}^\dagger\rangle - \alpha^* e^{-i\phi}\sinh(r)\langle \hat{a}^\dagger\rangle + \alpha^{*2} \\
=& \alpha^{*2} - e^{-i\phi}\sinh(r)\cosh(r) \\
\langle 0|\tilde{a}^{\dagger 2}|0\rangle =& \alpha^{*2} - \frac{1}{2}e^{-i\phi}\sinh(2r).
\end{aligned}$$

$$\begin{aligned}
\langle 0|\tilde{a}\tilde{a}^\dagger|0\rangle =& \langle 0|(\cosh(r)\hat{a} - e^{i\phi}\sinh(r)\hat{a}^\dagger + \alpha)(\cosh(r)\hat{a}^\dagger - e^{-i\phi}\sinh(r)\hat{a} + \alpha^*)|0\rangle \\
\langle 0|\tilde{a}\tilde{a}^\dagger|0\rangle =& \cosh^2(r) + |\alpha|^2.
\end{aligned}$$

$$\begin{aligned}
\langle 0|\tilde{a}^\dagger\tilde{a}|0\rangle =& \langle 0|(\cosh(r)\hat{a}^\dagger - e^{-i\phi}\sinh(r)\hat{a} + \alpha^*)(\cosh(r)\hat{a} - e^{i\phi}\sinh(r)\hat{a}^\dagger + \alpha)|0\rangle \\
\langle 0|\tilde{a}^\dagger\tilde{a}|0\rangle =& \sinh^2(r) + |\alpha|^2.
\end{aligned}$$

Putting all together:

$$\begin{aligned}
\langle \hat{X}_1^2(t)\rangle &= \frac{1}{4}\left\{[\alpha^2 - \frac{1}{2}e^{i\phi}\sinh(2r)]e^{-i2\omega t} + [\alpha^{*2} - \frac{1}{2}e^{-i\phi}\sinh(2r)]e^{i2\omega t} + \cosh^2(r) + |\alpha|^2 + \sinh^2(r) + |\alpha|^2\right\} \\
&= \frac{1}{4}\left[(\alpha e^{-i\omega t} + \alpha^* e^{i\omega t})^2 + \cosh(2r) - \sinh(2r)\cos(\phi - 2\omega t)\right] \\
\langle \hat{X}_1^2(t)\rangle &= [\mathrm{Re}\left(\alpha e^{-i\omega t}\right)]^2 + \frac{1}{4}[\cosh(2r) - \sinh(2r)\cos(\phi - 2\omega t)].
\end{aligned}$$

The uncertainty is:

$$\Delta\hat{X}_1(t) = \sqrt{\langle \hat{X}_1^2(t)\rangle - \langle \hat{X}_1(t)\rangle^2} = \frac{1}{2}\sqrt{\cosh(2r) - \sinh(2r)\cos(\phi - 2\omega t)}.$$

Similarly,

$$\begin{aligned}
\langle \hat{X}_2^2(t)\rangle &= \langle 0|\hat{S}^\dagger \hat{D}^\dagger \left(\frac{\hat{a}e^{-i\omega t} - \hat{a}^\dagger e^{i\omega t}}{2i}\right)^2 \hat{D}\hat{S}|0\rangle \\
&= -\frac{1}{4}\langle 0|\hat{S}^\dagger \hat{D}^\dagger(\hat{a}e^{-i\omega t} - \hat{a}^\dagger e^{i\omega t})\hat{D}\hat{S}\hat{S}^\dagger \hat{D}^\dagger(\hat{a}e^{-i\omega t} + \hat{a}^\dagger e^{i\omega t})\hat{D}\hat{S}|0\rangle \\
&= -\frac{1}{4}\langle 0|(\tilde{a}e^{-i\omega t} - \tilde{a}^\dagger e^{i\omega t})^2|0\rangle \\
&= -\frac{1}{4}[e^{-i2\omega t}\langle 0|\tilde{a}^2|0\rangle + e^{i2\omega t}\langle 0|\tilde{a}^{\dagger 2}|0\rangle - \langle 0|\tilde{a}\tilde{a}^\dagger|0\rangle - \langle 0|\tilde{a}^\dagger\tilde{a}|0\rangle].
\end{aligned}$$

We use the previous results to get

$$\begin{aligned}
\langle \hat{X}_2^2(t)\rangle &= -\frac{1}{4}\left\{[\alpha^2 - \frac{1}{2}e^{i\phi}\sinh(2r)]e^{-i2\omega t} + [\alpha^{*2} - \frac{1}{2}e^{-i\phi}\sinh(2r)]e^{i2\omega t} - \cosh^2(r) - |\alpha|^2 - \sinh^2(r) - |\alpha|^2\right\} \\
&= -\frac{1}{4}[(\alpha e^{-i\omega t} - \alpha^* e^{i\omega t})^2 - \cosh(2r) - \sinh(2r)\cos(\phi - 2\omega t)] \\
\langle \hat{X}_2^2(t)\rangle &= [\mathrm{Im}\left(\alpha e^{-i\omega t}\right)]^2 + \frac{1}{4}[\cosh(2r) + \sinh(2r)\cos(\phi - 2\omega t)].
\end{aligned}$$

And the uncertainty is:

$$\Delta\hat{X}_2(t) = \sqrt{\langle \hat{X}_2^2(t)\rangle - \langle \hat{X}_2(t)\rangle^2} = \frac{1}{2}\sqrt{\cosh(2r) + \sinh(2r)\cos(\phi - 2\omega t)}$$

We can verify that $\Delta\hat{X}_1(0) = e^{-r}/2$ and $\Delta\hat{X}_2(0) = e^{r}/2$ when $\phi = t = 0$. The phase space representation is shown below for $t = \phi = 0$ and $r = 0.6$.

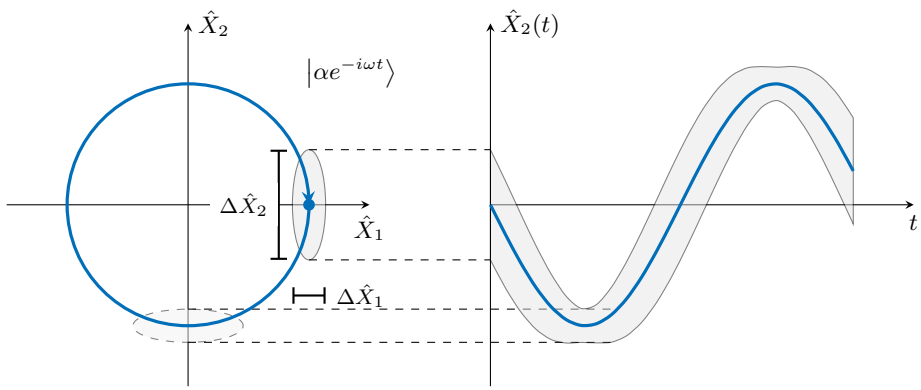


Figure 1.2 Phase space of the squeezed displaced state for $r = 0.6$.

The following table sumarizes the most improtant results for the most general case from which we can get the squeezed or displaced states.

1.6 Thermal fields

1.6.1 Thermal fields

A black body can be modeled as a cavity containing radiation at thermal equilibrium with its walls. The radiation is thus coupled to a heat bath and so is not a truly free field. Assuming the coupling is weak, we can treat the field as if it were an isolated system that can be described as a microcanonical ensemble.

Table 1.1 Summary of the displaced squeezed temporal state $|\xi,\alpha\rangle = \hat{D}(\alpha)\hat{S}(\xi)\,|0\rangle$ with $\xi = r\underline{/\phi}$.

BCH formula	$e^{L}Ae^{-L} = A + [L,A] + \frac{1}{2!}[L,[L,A]] + \cdots$
Displaced and Squeezed operator	$\hat{D}(\alpha) = e^{\alpha\hat{a}^\dagger - \alpha^*\hat{a}}$ $\quad \hat{S}(\xi) = e^{\frac{1}{2}(\xi^*\hat{a}^2 - \xi\hat{a}^{\dagger 2})}$
Quadrature operators	$\hat{X}_1(t) = \frac{\hat{a}e^{-i\omega t} + \hat{a}^\dagger e^{i\omega t}}{2}$ $\quad \hat{X}_2(t) = \frac{\hat{a}e^{-i\omega t} - \hat{a}^\dagger e^{i\omega}}{2i}$
DS Operator	$\tilde{a} = \hat{S}^\dagger\hat{D}^\dagger\hat{a}\hat{D}\hat{S} = \cosh(r)\hat{a} - e^{i\phi}\sinh(r)\hat{a}^\dagger + \alpha$
Mean value	$\langle\hat{X}_1(t)\rangle = \text{Re}\left(\alpha e^{-i\omega t}\right)$ $\quad \langle\hat{X}_2(t)\rangle = \text{Im}\left(\alpha e^{-i\omega t}\right)$
Variance	$\langle\hat{X}_1^2(t)\rangle = \langle\hat{X}_1(t)\rangle^2 + \frac{1}{4}[\cosh(2r) - \sinh(2r)\cos(\phi - 2\omega t)]$
	$\langle\hat{X}_2^2(t)\rangle = \langle\hat{X}_2(t)\rangle^2 + \frac{1}{4}[\cosh(2r) + \sinh(2r)\cos(\phi - 2\omega t)]$

Lets consider a single-mode field in thermal equilibrium with the walls of cavity at temperature T. The probability P_n that the mode is thermally xcited in the nth level is

$$P_n = \frac{\exp(-E_n/k_BT)}{\sum_n \exp(-E_n/k_BT)}, \quad k_B = 1.38\cdot 10^{-23}\ J/K.$$

We introduce the density operator for the thermal field:

$$\rho_{Th} = \frac{\exp(-\hat{H}/k_BT)}{\text{Tr}\left[\exp(-\hat{H}/k_BT)\right]}, \quad \hat{H} = \hbar\omega(\hat{a}^\dagger\hat{a} + \frac{1}{2}).$$

Ejemplo 1.1

Consider the thermal density matrix of a single-mode field

$$\rho_{th} = \sum_n P_n\,|n\rangle\langle n|\,, \quad \text{with} \quad P_n = \frac{e^{-(n+1/2)\hbar\omega/(k_BT)}}{\sum_{m=0}^{\infty} e^{-(m+1/2)\hbar\omega/(k_BT)}}.$$

a) What is the expectation value of the number of photon in a thermal state?
b) What is the average number of photons around visible wavelengths $sim500\ nm$ at room temperature $\sim 300\ K$ and temperature of the sun $\sim 6000\ K$?

Solution

a) First, the eigenequation of the number operator $\hat{n}$ and the definition of the thermal density matrix are:

$$\hat{n}\,|n\rangle = n\,|n\rangle\,, \quad \text{and} \quad \hat{\rho}_{th} = \sum_n P_n\,|n\rangle\langle n|\,.$$

The expectation value of the number of photons is:

$$\langle\hat{n}\rangle = \text{Tr}\left[\hat{n}\hat{\rho}_{th}\right] = \sum_n \langle n|\hat{n}\hat{\rho}_{th}|n\rangle = \sum_{n,n'}\langle n|\hat{n}P_{n'}|n'\rangle\langle n'|n\rangle = \sum_{n,n'} n'P_{n'}|\langle n|n'\rangle|^2 = \sum_n nP_n.$$

Now, we put the Boltzman distribution in the last summation:

$$\langle\hat{n}\rangle = \sum_n n\left[\frac{e^{-(n+1/2)\hbar\omega/(k_BT)}}{\sum_{m=0}^{\infty} e^{-(m+1/2)\hbar\omega/(k_BT)}}\right] = \sum_n n\frac{e^{-n[\hbar\omega/(k_BT)]}}{\sum_{m=0}^{\infty} e^{-m[\hbar\omega/(k_BT)]}} = \frac{1}{\sum_{m=0}^{\infty} e^{-mx}}\sum_n ne^{-nx},\ x = \frac{\hbar\omega}{k_BT}.$$

We note that the summation in n can be expressed in terms of a geometric function in the following way

$$\sum_{n=0}^{\infty} ne^{-nx} = -\frac{d}{dx}\left[\sum_{n=0}^{\infty} e^{-nx}\right] = -\frac{d}{dx}\left[\frac{1}{1-e^{-x}}\right] = \frac{e^{-x}}{(1-e^{-x})^2}.$$

Then,

$$\langle \hat{n} \rangle = \frac{1}{\sum_{m=0}^{\infty} e^{-mx}} \frac{e^{-x}}{(1-e^{-x})^2} = (1-e^{-x}) \frac{e^{-x}}{(1-e^{-x})^2} = \frac{e^{-x}}{1-e^{-x}} = \frac{1}{e^{x}-1} = \frac{1}{e^{\frac{\hbar\omega}{k_B T}}-1}.$$

which is the Planck radiation distribution.

b) By evaluating the above result, and knowing that

$$h \approx 6.6261 \cdot 10^{-34}\ m^2 kg/s, \quad \text{and} \quad k_B \approx 1.3806 \cdot 10^{-23}\ m^2 kg s^{-2} K^{-1}.$$

The average number of photon as a function of the absolute temperature is:

$$\langle \hat{n} \rangle (T) = \frac{1}{e^{\frac{[6.6261 \cdot 10^{-34}/(2\pi)][(2\pi)(3 \cdot 10^8)/(500 \cdot 10^{-9})]}{(1.3806 \cdot 10^{-23})K}} - 1} = \frac{1}{e^{28797/K}-1}.$$

At $300K$ and $6000K$ we have

$$\langle \hat{n} \rangle (300) = \frac{1}{e^{28797/300}-1} = 2.1 \cdot 10^{-42}, \quad \text{and} \quad \langle \hat{n} \rangle (6000) = \frac{1}{e^{28797/6000}-1} = 8.3 \cdot 10^{-3}.$$

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Chapter 2

Optical devices

2.1 Beam splitter 24

2.1 Beam splitter

Let us consider the action of a beam splitter (BS) on two input modes of same frequency and polarization. Clasically, the output E-field can be defined in terms of the input E-fields:

$$\begin{bmatrix} E_1' \\ E_2' \end{bmatrix} = \begin{bmatrix} t' & r \\ r' & t \end{bmatrix} \begin{bmatrix} E_1 \\ E_2 \end{bmatrix}.$$

$\{r, t\}$ and $\{r', t'\}$ are reflection/transmission coefficient associated with the two interfaces of the BS. From energy conservation, we must have

$$\text{Energy conservation in the BS} \qquad |E_1|^2 + |E_2|^2 = |E_1'|^2 + |E_2'|^2.$$

If we assume $E_2 = 0$, then

$$|E_1|^2 = |t'E_1|^2 + |r'E_1|^2 \longrightarrow |r'|^2 + |t'|^2 = 1. \tag{2.1}$$

Similarly, setting $E_1 = 0$ yields

$$|E_2|^2 = |rE_2|^2 + |tE_2|^2 \longrightarrow |r|^2 + |t|^2 = 1. \tag{2.2}$$

In general,

$$\begin{aligned}
|E_1|^2 + |E_2|^2 &= |E_1'|^2 + |E_2'|^2 \\
&= (t'E_1 + rE_2)(t'^* E_1^* + r^* E_2^*) + (r'E_1 + tE_2)(r'^* E_1^* + t^* E_2^*) \\
&= |t'|^2|E_1|^2 + |r|^2|E_2|^2 + |r'|^2|E_1|^2 + |t|^2|E_2|^2 + rt'^* E_1^* E_2 + t'r^* E_1 E_2^* + r't^* E_1 E_2^* + r'^* t E_1^* E_2 \\
|E_1|^2 + |E_2|^2 &= |E_1|^2(|r'|^2 + |t'|^2) + |E_2|^2(|r|^2 + |t|^2) + E_1^* E_2(rt'^* + r'^* t) + E_1 E_2^*(r't^* + r^* t') \\
0 &= E_1^* E_2(rt'^* + r'^* t) + c.c.
\end{aligned}$$

Thus,

$$rt'^* + r'^* t = 0. \tag{2.3}$$

For a $50:50$ BS which as $r' = 1/\sqrt{2}$, $t' = 1/\sqrt{2}$, and $r = 1/\sqrt{2}$, (2.1), (2.2) and (2.3) implies that $t = -1/\sqrt{2}$. Similarly, the quantum description of the BS is

$$\begin{bmatrix} \hat{a}_1' \\ \hat{a}_2' \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} \hat{a}_1 \\ \hat{a}_2 \end{bmatrix}. \tag{2.4}$$

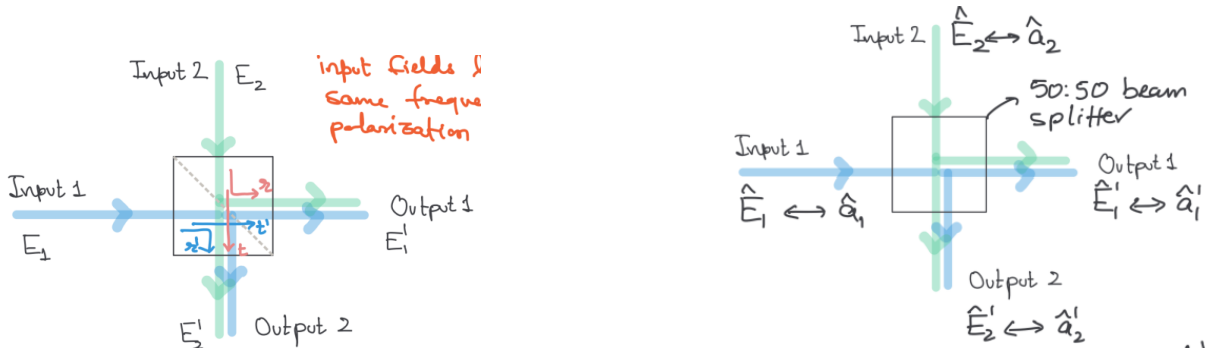
The ladder operators are equal to the E-fields. In addition, the output modes are written as a linear combination of the input modes:

$$\begin{matrix} \hat{a}_1' = \frac{1}{\sqrt{2}}(\hat{a}_1 + \hat{a}_2) \\ \hat{a}_2' = \frac{1}{\sqrt{2}}(\hat{a}_1 - \hat{a}_2) \end{matrix} \longrightarrow \begin{matrix} \hat{a}_1 = \frac{1}{\sqrt{2}}(\hat{a}_1' + \hat{a}_2') \\ \hat{a}_2 = \frac{1}{\sqrt{2}}(\hat{a}_1' - \hat{a}_2') \end{matrix}$$

Ejemplo 2.1 **Single photon incident in one input**

The input is:

$$\text{Input} = |1\rangle_1 |0\rangle_2 = (\hat{a}_1^\dagger |0\rangle_1) |0\rangle_2.$$



(a) Classical description **(b)** Quantum description

Figure 2.1 Description of a beamsplitter (BS).

Note that

$$\hat{a}_1 = \frac{1}{\sqrt{2}}(\hat{a}_1' + \hat{a}_2') \quad \text{and} \quad \hat{a}_1^\dagger = \frac{1}{\sqrt{2}}(\hat{a}_1'^\dagger + \hat{a}_2'^\dagger).$$

The output is then

$$\text{Output} = \frac{1}{\sqrt{2}}(\hat{a}_1'^\dagger + \hat{a}_2'^\dagger)\,|0\rangle_{1'}\,|0\rangle_{2'} = \frac{1}{\sqrt{2}}[|1\rangle_{1'}\,|0\rangle_{2'} + |0\rangle_{1'}\,|1\rangle_{2'}].$$

Thus, there is a $50:50$ probability of detecting the photon in one of the ouput ports.

Ejemplo 2.2 **Both input with photons**

The input is:

$$\text{Input} = |1\rangle_1\,|1\rangle_2 = \hat{a}_1^\dagger\,|0\rangle_1\,\hat{a}_2^\dagger\,|0\rangle_2\,.$$

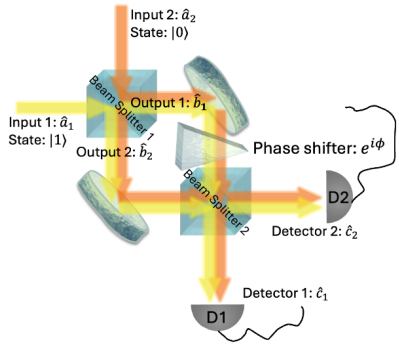
The output is:

$$\begin{aligned}
\text{Output} &= \frac{1}{2}[\hat{a}_1'^\dagger + \hat{a}_2'^\dagger][\hat{a}_1'^\dagger - \hat{a}_2'^\dagger]\,|0\rangle_{1'}\,|0\rangle_{2'} \\
&= \frac{1}{2}[\hat{a}_1'^\dagger\hat{a}_1'^\dagger - \hat{a}_2'^\dagger\hat{a}_2'^\dagger + \cancel{\hat{a}_1'^\dagger\hat{a}_2'^\dagger} - \cancel{\hat{a}_2'^\dagger\hat{a}_1'^\dagger}]\,|0\rangle_{1'}\,|0\rangle_{2'} \\
&= \frac{1}{2}[\sqrt{2}\,|2\rangle_{1'}\,|0\rangle_{2'} - \sqrt{2}\,|0\rangle_{1'}\,|2\rangle_{2'}] \\
\text{Output} &= \frac{1}{\sqrt{2}}[|2\rangle_{1'}\,|0\rangle_{2'} - |0\rangle_{1'}\,|2\rangle_{2'}].
\end{aligned}$$

Thus, when two photons are simultaneously incident on each input, both either go into output 1 or output 2 but never in both a the same time.

Ejemplo 2.3 **Mach-Zehnder**

a) Express the output modes $\{\hat{b}_1, \hat{b}_2\}$ in terms of the input modes $\{\hat{a}_1, \hat{a}_2\}$ assuming the first BS is $50:50$.
b) What is the expected number of photons $\langle \hat{b}_i^\dagger \hat{b}_i \rangle$ in each mode?
c) If $\hat{b}_1 \to \hat{b}_1 e^{i\phi}$, determine the output mode operators at the detector positions $\hat{c}_1 = \frac{1}{\sqrt{2}}(\hat{b}_1 e^{i\phi} + \hat{b}_2)$ and $\hat{b}_2 = \frac{1}{\sqrt{2}}(\hat{b}_1 e^{i\phi} - \hat{b}_2)$ in terms of the input mode operators $\{\hat{a}_1, \hat{a}_2\}$.
d) What is the number of photons detected at detector 1 as a function of ϕ?



Solution

(a) For the $50:50$ beamsplitter 1, we have:

$$\begin{bmatrix}\hat{b}_1\\ \hat{b}_2\end{bmatrix} = \frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}\begin{bmatrix}\hat{a}_1\\ \hat{a}_2\end{bmatrix}, \quad \text{with} \quad \begin{bmatrix}\hat{a}_1\\ \hat{a}_2\end{bmatrix} = \frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}\begin{bmatrix}\hat{b}_1\\ \hat{b}_2\end{bmatrix}.$$

(b) From the last part, we know that

$$\hat{a}_1 = \frac{1}{\sqrt{2}}(\hat{b}_1 + \hat{b}_2) \xrightarrow{h.c.} \hat{a}_1^\dagger = \frac{1}{\sqrt{2}}(\hat{b}_1^\dagger + \hat{b}_2^\dagger), \quad \text{and} \quad \hat{a}_2 = \frac{1}{\sqrt{2}}(\hat{b}_1 - \hat{b}_2) \xrightarrow{h.c.} \hat{a}_2^\dagger = \frac{1}{\sqrt{2}}(\hat{b}_1^\dagger - \hat{b}_2^\dagger).$$

The input is then:

$$\text{Input} = |1\rangle_1 |0\rangle_2 = \hat{a}_1^\dagger |0\rangle_1 |0\rangle_2 .$$

Output is:

$$\text{Output} = \frac{1}{\sqrt{2}}(\hat{b}_1^\dagger + \hat{b}_2^\dagger)|0\rangle_1 |0\rangle_2 = \frac{1}{\sqrt{2}}[|1\rangle_{1'} |0\rangle_{2'} + |0\rangle_{1'} |1\rangle_{2'}].$$

We expect then to have the same probability of getting one photon at each output.
The expected value at output 1 is:

$$\begin{aligned}
{}_2\langle 0|_1 \langle 1| \hat{b}_1^\dagger \hat{b}_1 |1\rangle_1 |0\rangle_2 &= \frac{1}{2}\langle 0,1|(\hat{a}_1^\dagger + \hat{a}_2^\dagger)(\hat{a}_1 + \hat{a}_2)|1,0\rangle \\
&= \frac{1}{2}\langle 0,1| (\hat{a}_1^\dagger \hat{a}_1 + \hat{a}_1^\dagger \hat{a}_2 + \hat{a}_2^\dagger \hat{a}_1 + \hat{a}_2^\dagger \hat{a}_2)|1,0\rangle \\
{}_2\langle 0|_1 \langle 1| \hat{b}_1^\dagger \hat{b}_1 |1\rangle_1 |0\rangle_2 &= \frac{1}{2}.
\end{aligned}$$

Likewise, the expected value at output 2 is:

$$\begin{aligned}
{}_2\langle 0|_1 \langle 1| \hat{b}_2^\dagger \hat{b}_2 |1\rangle_1 |0\rangle_2 &= \frac{1}{2}\langle 0,1|(\hat{a}_1^\dagger - \hat{a}_2^\dagger)(\hat{a}_1 - \hat{a}_2)|1,0\rangle \\
&= \frac{1}{2}\langle 0,1| (\hat{a}_1^\dagger \hat{a}_1 - \hat{a}_1^\dagger \hat{a}_2 - \hat{a}_2^\dagger \hat{a}_1 + \hat{a}_2^\dagger \hat{a}_2)|1,0\rangle \\
{}_2\langle 0|_1 \langle 1| \hat{b}_2^\dagger \hat{b}_2 |1\rangle_1 |0\rangle_2 &= \frac{1}{2}.
\end{aligned}$$

(c) We replace the first output as the input of the second output:

$$\begin{aligned}
\hat{c}_1 &= \frac{1}{\sqrt{2}}[\hat{b}_1 e^{i\phi} + \hat{b}_2] = \frac{1}{2}\left[(\hat{a}_1 + \hat{a}_2)e^{i\phi} + (\hat{a}_1 - \hat{a}_2)\right] = \frac{1}{2}[\hat{a}_1(1 + e^{i\phi}) + \hat{a}_2(e^{i\phi} - 1)], \\
\hat{c}_2 &= \frac{1}{\sqrt{2}}[\hat{b}_1 e^{i\phi} - \hat{b}_2] = \frac{1}{2}\left[(\hat{a}_1 + \hat{a}_2)e^{i\phi} - (\hat{a}_1 - \hat{a}_2)\right] = \frac{1}{2}[\hat{a}_1(e^{i\phi} - 1) + \hat{a}_2(1 + e^{i\phi})].
\end{aligned}$$

(d) The expected number of photons at detector 1 is:

$$\begin{aligned}
\langle 0,1|\hat{c}_1^\dagger \hat{c}_1|1,0\rangle &= \frac{1}{4}\left\{[\hat{a}_1^\dagger(1+e^{-i\phi})+\hat{a}_2^\dagger(e^{-i\phi}-1)][\hat{a}_1(1+e^{i\phi})+\hat{a}_2(e^{i\phi}-1)]\right\} \\
&= \frac{1}{4}\langle 0,1|\left\{\hat{a}_1^\dagger\hat{a}_1(\cdots)+\hat{a}_1^\dagger\hat{a}_2(\cdots)+\hat{a}_2^\dagger\hat{a}_1(\cdots)+\hat{a}_2^\dagger\hat{a}_2(\cdots)\right\}|1,0\rangle \\
&= \frac{1}{4}(1+e^{-i\phi})(1+e^{i\phi}) \\
\langle 0,1|\hat{c}_1^\dagger \hat{c}_1|1,0\rangle &= \frac{1}{2}[1+\cos\phi] = \cos^2\frac{\phi}{2}.
\end{aligned}$$

By conservation of energy (or probability), the expected number of photon at detector 2 is:

$$\langle 0,1|\hat{c}_2^\dagger \hat{c}_2|1,0\rangle = \sin^2\frac{\phi}{2}.$$

Chapter 3

Nonclassical light

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3.1 Detection of quadrature states

Modes at the beam splitter output are:

$$\hat{A}_1 = \frac{1}{\sqrt{2}}(\hat{a}_1 + \hat{a}_2), \quad \hat{A}_2 = \frac{1}{\sqrt{2}}(\hat{a}_1 - \hat{a}_2).$$

The intensity operator is the number of photons detected at each detector:

$$\text{Photocurrent} \qquad \hat{I}_k = \hat{N}_k = \hat{A}_k^\dagger \hat{A}_k \,. \tag{3.1}$$

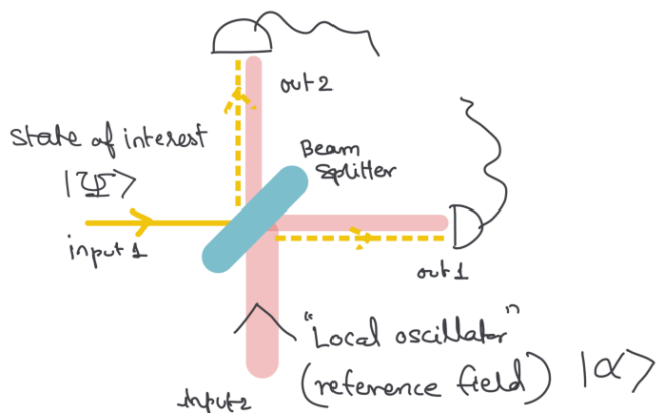


Figure 3.1 Balanced homodyne detection.

Let us consider the number operator in each output:

$$\hat{I}_1 = \hat{A}_1^\dagger \hat{A}_1 = \left(\frac{\hat{a}_1^\dagger + \hat{a}_2^\dagger}{\sqrt{2}}\right)\left(\frac{\hat{a}_1 + \hat{a}_2}{\sqrt{2}}\right) = \frac{1}{2}(\hat{a}_1^\dagger \hat{a}_1 + \hat{a}_2^\dagger \hat{a}_2 + \hat{a}_1^\dagger \hat{a}_2 + \hat{a}_2^\dagger \hat{a}_1),$$

$$\hat{I}_2 = \hat{A}_2^\dagger \hat{A}_2 = \left(\frac{\hat{a}_1^\dagger - \hat{a}_2^\dagger}{\sqrt{2}}\right)\left(\frac{\hat{a}_1 - \hat{a}_2}{\sqrt{2}}\right) = \frac{1}{2}(\hat{a}_1^\dagger \hat{a}_1 + \hat{a}_2^\dagger \hat{a}_2 - \hat{a}_1^\dagger \hat{a}_2 - \hat{a}_2^\dagger \hat{a}_1),$$

$$\hat{I}_1 - \hat{I}_2 = \hat{a}_1^\dagger \hat{a}_2 + \hat{a}_2^\dagger \hat{a}_1.$$

Let us consider that the input 2 contains a strong **coherent state field** that we refer to as *local oscillator*. Thus the total state over which the average is being taken is

$$\underbrace{|\Psi_1\rangle}_{\text{Input 1}} \quad \underbrace{|\alpha_2\rangle}_{\text{Coherent state input 2}} \quad \Longrightarrow \hat{a}_2 \, |\alpha_2\rangle = \alpha_2 \, |\alpha_2\rangle \,. \tag{3.2}$$

This produces the following expected measured intensity (epected number of photons):

$$\langle \hat{I}_1 \rangle = \langle \hat{N}_1 \rangle = \langle \Psi_1 \alpha_2 | \hat{A}_1^\dagger \hat{A}_1 | \alpha_2 \Psi_1 \rangle = \langle \Psi_1 \alpha_2 | \frac{\hat{a}_1^\dagger + \hat{a}_2^\dagger}{\sqrt{2}} \frac{\hat{a}_1 + \hat{a}_2}{\sqrt{2}} | \alpha_2 \Psi_1 \rangle = \frac{1}{2} \langle \Psi_1 \alpha_2 | (\hat{a}_1^\dagger + \alpha_2^*)(\hat{a}_1 + \alpha_2) | \alpha_2 \Psi_1 \rangle$$

$$= \frac{1}{2} \langle \Psi_1 | (\hat{a}^\dagger \hat{a}_1 + \hat{a}_1 \alpha_2^* + \hat{a}_1^\dagger \alpha_2 + |\alpha_2|^2) | \Psi_1 \rangle,$$

$$\langle \hat{I}_2 \rangle = \langle \hat{N}_2 \rangle = \langle \Psi_1 \alpha_2 | \hat{A}_2^\dagger \hat{A}_2 | \alpha_2 \Psi_1 \rangle = \langle \Psi_1 \alpha_2 | \frac{\hat{a}_1^\dagger - \hat{a}_2^\dagger}{\sqrt{2}} \frac{\hat{a}_1 - \hat{a}_2}{\sqrt{2}} | \alpha_2 \Psi_1 \rangle = \frac{1}{2} \langle \Psi_1 \alpha_2 | (\hat{a}_1^\dagger - \alpha_2^*)(\hat{a}_1 - \alpha_2) | \alpha_2 \Psi_1 \rangle$$

$$= \frac{1}{2} \langle \Psi_1 | (\hat{a}_1^\dagger \hat{a}_1 - \hat{a}_1 \alpha_2^* - \hat{a}_1^\dagger \alpha_2 + |\alpha_2|^2) | \Psi_1 \rangle.$$

By taking the difference $\hat{I}_1 - \hat{I}_2$, we can change the nature of α_2 to have different quadratures:

$$\langle \hat{I}_1 - \hat{I}_2 \rangle = \langle \hat{I}_1 \rangle - \langle \hat{I}_2 \rangle = \langle \Psi_1 | \hat{a}_1 \alpha_2^* + \hat{a}_1^\dagger \alpha_2 | \Psi_1 \rangle = \begin{cases} 2\alpha_2 \langle \Psi_1 | \hat{X}_1 | \Psi_1 \rangle, & \alpha_2 = \alpha_2^*, \\ -2i\alpha_2 \langle \Psi_1 | \hat{X}_2 | \Psi_1 \rangle, & \alpha_2 = -\alpha_2^*. \end{cases}$$

In general by changing the phase of input 2 we can measure a general quadrature. Noise in intensity corresponds to noise in specific quadratures.
Say $\alpha_2 = |\alpha_2| e^{i\varphi_\alpha}$, then:

$$\hat{N}_1 - \hat{N}_2 = |\alpha_2|(\hat{a}_1 e^{-i\varphi_\alpha} + \hat{a}_1^\dagger e^{i\varphi_\alpha}) = |\alpha_2|(\hat{a}_1 + \hat{a}_1^\dagger)\cos\varphi_\alpha + i|\alpha_2|(\hat{a}_1 - \hat{a}_1^\dagger)\sin\varphi_\alpha = 2|\alpha_2|(\cos\varphi_\alpha \hat{X}_1 - i\sin\varphi_\alpha \hat{X}_2).$$

The noise in the difference photocurrent is thus proportional to:

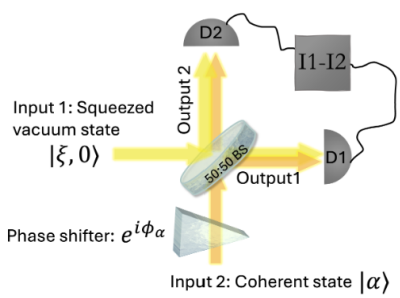
$$\langle (\hat{N}_1 - \hat{N}_2)^2 \rangle - \langle \hat{N}_1 - \hat{N}_2 \rangle^2 = \langle (\alpha_2^* \hat{a}_1 + \alpha_2 \hat{a}_1^\dagger)^2 \rangle - \langle (\alpha_2^* \hat{a}_1 + \alpha_2 \hat{a}_1^\dagger) \rangle^2.$$

The term in red can be $\hat{X}_1$ or $\hat{X}_2$ depending on the phase of α_2, as stated above.

- Noise in photocurrent difference corresponds to uncertainty in quadratures.
- If a quadrature measurement has uncertainty $\Delta X_\phi^2 < 1/4$.

Ejemplo 3.1 **Balanced homodyne deecion of squeezed light**

Consider a squeezed vacuum field mode incident on a balanced homodyne detector.



The ouput modes in the detectors are given as

$$\hat{b}_1 = \frac{1}{\sqrt{2}}(\hat{a}_1 + \hat{a}_2), \quad \hat{b}_2 = \frac{1}{\sqrt{2}}(\hat{a}_1 - \hat{a}_2).$$

What is the difference $\hat{N}_1 - \hat{N}_2$, with $\hat{N}_i = \hat{b}_i^\dagger \hat{b}_i$, and the variance $\Delta(\hat{N}_1 - \hat{N}_2)$ as a function of ϕ_α? The phase shifter changes the coherent state input so that $\alpha \to \alpha e^{i\phi_\alpha}$. Assume that $|\alpha| \gg 1$ and keep terms to linear order in $|\alpha|$.

Solution

In general, the photon number for output i is $\hat{N}_i = \hat{b}_i^\dagger \hat{b}_i$. For each output we have:

$$\hat{N}_1 = \frac{1}{2}(\hat{a}_1^\dagger + \hat{a}_2^\dagger)(\hat{a}_1 + \hat{a}_2) = \frac{1}{2}(\hat{a}_1^\dagger \hat{a}_1 + \hat{a}_1^\dagger \hat{a}_2 + \hat{a}_2^\dagger \hat{a}_1 + \hat{a}_2^\dagger \hat{a}_2),$$
$$\hat{N}_2 = \frac{1}{2}(\hat{a}_1^\dagger - \hat{a}_2^\dagger)(\hat{a}_1 - \hat{a}_2) = \frac{1}{2}(\hat{a}_1^\dagger \hat{a}_1 - \hat{a}_1^\dagger \hat{a}_2 - \hat{a}_2^\dagger \hat{a}_1 + \hat{a}_2^\dagger \hat{a}_2).$$

Substraction gives

$$\hat{N}_1 - \hat{N}_2 = \frac{1}{2}\left[(\hat{a}_1^\dagger \hat{a}_1 + \hat{a}_1^\dagger \hat{a}_2 + \hat{a}_2^\dagger \hat{a}_1 + \hat{a}_2^\dagger \hat{a}_2) - (\hat{a}_1^\dagger \hat{a}_1 - \hat{a}_1^\dagger \hat{a}_2 - \hat{a}_2^\dagger \hat{a}_1 + \hat{a}_2^\dagger \hat{a}_2)\right] = \hat{a}_1^\dagger \hat{a}_2 + \hat{a}_2^\dagger \hat{a}_1.$$

With the phase shift, the input 2 is:

$$\alpha \to \alpha e^{i\phi_\alpha}, \quad \text{and} \quad \hat{a}_2 \approx \alpha e^{i\phi_\alpha}.$$

The number photon difference is then:

$$\hat{N}_1 - \hat{N}_2 = \hat{a}_1^\dagger \alpha e^{i\phi_\alpha} + \hat{a}_1 \alpha^* e^{-i\phi_\alpha}.$$

If $\alpha = |\alpha|$, then

$$\hat{N}_1 - \hat{N}_2 = |\alpha| \left(\hat{a}_1^\dagger e^{i\phi_\alpha} + \hat{a}_1 e^{-i\phi_\alpha} \right) = 2|\alpha| \hat{X}_{\phi_\alpha}, \quad \hat{X}_{\phi_\alpha} = \frac{\hat{a}_1^\dagger e^{i\phi_\alpha} + \hat{a}_1 e^{-i\phi_\alpha}}{2},$$

where $\hat{X}_{\phi_\alpha}$ is the quadrature of mode 1 modulated by the phase shift ϕ_α.

To get the uncertainty $\Delta(\hat{N}_1 - \hat{N}_2)$, we compute the expected alue of the difference and the expected value of the difference squared. We can use the results from the previoous exercise as the input 1 is a squeezed state of the vacuum.

$$\begin{aligned}
{}_1\langle \xi, 0|_2 \langle \alpha| \left(\hat{N}_1 - \hat{N}_2\right) |\alpha\rangle_2 |\xi, 0\rangle_1 &= 2|\alpha| \langle \hat{X}_{\phi_\alpha} \rangle = 0, \\
{}_1\langle \xi, 0|_2 \langle \alpha| \left(\hat{N}_1 - \hat{N}_2\right)^2 |\alpha\rangle_2 |\xi, 0\rangle_1 &= 4|\alpha|^2 \langle \hat{X}_{\phi_\alpha}^2 \rangle = |\alpha|^2 [\cosh(2r) - \sinh(2r)\cos(\theta - 2\phi_\alpha)],
\end{aligned}$$

where $\xi = re^{i\theta}$ is the squeezing factor, and ϕ_α the phase shift imposed by the input 2. Uncertainty is therefore:

$$\Delta(\hat{N}_1 - \hat{N}_2) = |\alpha| \sqrt{\cosh(2r) - \sinh(2r)\cos(\theta - 2\phi_\alpha)}.$$

3.2 Generation of squeezed states

The following is a scheme of a nonlinear medium. Radiation from oscillating dipoles is of different frequency as the input stimulation. Nonlinear polarization induced in a medium by an E-field $\boldsymbol{E}$ is:

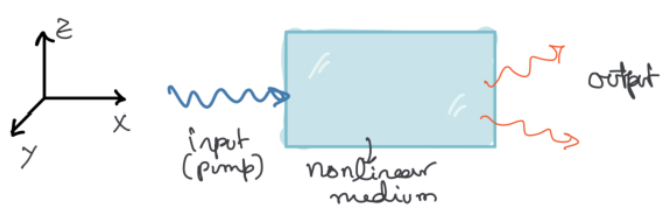


Figure 3.1 Nonlinear medium

$$\boldsymbol{P}_i = \varepsilon_0 \left[\chi_{jk}^{(1)} \boldsymbol{E}_k + \varepsilon_0 \chi_{jkl}^{(2)} \boldsymbol{E}_k \boldsymbol{E}_l + \cdots \right].$$

This polarization is for one dipole, the total would be a sum over all of them. Thses quantities are tensor. In particular, we have a linear and second-order polarization terms:

$$\begin{aligned}
\boldsymbol{P}_{Lj} &= \text{Linear polarization} = \varepsilon_0 \chi_{jk}^{(1)} \boldsymbol{E}_k \\
\boldsymbol{P}_{NLj} &= \text{Second-order polarization} = \varepsilon_0 \chi_{jkl}^{(2)} \boldsymbol{E}_k \boldsymbol{E}_l.
\end{aligned}$$

The interaction (dipole-field) energy density associated with the field becomes:

$$
\begin{aligned}
U = -\boldsymbol{D}\cdot\boldsymbol{E} &= \underbrace{-\varepsilon_0\boldsymbol{E}^2 - \boldsymbol{P}_L\cdot\boldsymbol{E}}_{\alpha\boldsymbol{E}^2(\text{linear})} - \boldsymbol{P}_{NL}^{(2)}\cdot\boldsymbol{E} \\
&= -\varepsilon_0[\chi_{jk}^{(1)}\boldsymbol{E}_k + \chi_{jkl}^{(2)}\boldsymbol{E}_j\boldsymbol{E}_k]\boldsymbol{E}_l = \underbrace{-\varepsilon_0\chi_{jk}^{(1)}\boldsymbol{E}_k\boldsymbol{E}_l}_{U_L} + \underbrace{-\varepsilon_0\chi_{jkl}^{(2)}\boldsymbol{E}_j\boldsymbol{E}_k\boldsymbol{E}_l}_{U_{NL}^{(2)}\ (\text{Squeezed output})} .
\end{aligned}
$$

We will analyze the above nonlinear Hamiltonian for the specific interaction where we have an incident blue photon corresponding spontaneously down-converted to two red photons (nonlinear process). The total E-field present is:

$$
\hat{\boldsymbol{E}}_{tot} = \boldsymbol{E}_B + \hat{\boldsymbol{E}}, \qquad \begin{array}{l} \boldsymbol{E}_B = \text{Incidence classical blue field} \\ \hat{\boldsymbol{E}} = \text{Quantized E-field} \end{array} .
$$

Hamiltonian becomes

$$
\hat{H}_{NL}^{(2)} = -\int dV U_{NL}^{(2)} = -\varepsilon_0\int dV\chi_{jkl}^{(2)}(\boldsymbol{E}_B + \hat{\boldsymbol{E}})_j(\boldsymbol{E}_B + \hat{\boldsymbol{E}})_k(\boldsymbol{E}_B + \hat{\boldsymbol{E}})_l
$$

For further simplification let us assume that only $\chi_{zzz}^{(2)} \neq 0 = \chi^{(2)}$. This is a simplifying assumption, used only for making the calculations easier. The E-field and the E-field operators to be used are of the form:

$$
\boldsymbol{E}_B = \hat{\boldsymbol{z}}\left[\mathcal{E}_B e^{i(k_B z - \omega_B t)} + \mathcal{E}_B^* e^{-i(k_B z - \omega_B t)}\right], \quad \hat{\boldsymbol{E}}_z = i\sum_{\boldsymbol{k}}\hat{\boldsymbol{z}}\sqrt{\frac{\hbar\omega}{2\varepsilon_0 V}}\left[\hat{a}_{\boldsymbol{k},z}e^{i(\boldsymbol{k}\cdot\boldsymbol{r}-\omega t)} - \hat{a}_{\boldsymbol{k},z}^\dagger e^{-i(\boldsymbol{k}\cdot\boldsymbol{r}-\omega t)}\right]
$$

We are only studyng the simplest case where squeezed light is generated. There will be 4 different kinds of terms resulting from the above combination of E-fields:

$$
\underbrace{(E_{Bz} + \hat{E}_z)}_{(1)}\underbrace{(E_{Bz} + \hat{E}_z)}_{(2)}\underbrace{(E_{Bz} + \hat{E}_z)}_{(3)} .
$$

(1) $E_{Bz}E_{Bz}E_{Bz}$ This is constant and doest no contain any operator: only adds global phase.

(2) $E_{Bz}E_{Bz}\hat{E}_z$ Such a term can be simplified as follows

$$
\begin{aligned}
& iE_{Bz}^2\sum_{\boldsymbol{k}}\sqrt{\frac{\hbar\omega}{2\varepsilon_0 V}}\left[\hat{a}_{\boldsymbol{k},z}e^{i(\boldsymbol{k}\cdot\boldsymbol{r}-\omega t)} - \hat{a}_{kz}^\dagger e^{-i(\boldsymbol{k}\cdot\boldsymbol{r}-\omega t)}\right] \\
=& i\left[\mathcal{E}_B^2 e^{i2(k_B z-\omega_B t)} + \mathcal{E}_B^{*2}e^{-i2(k_B z-\omega_B t)} + |\mathcal{E}_B|^2\right]\sum_{\boldsymbol{k}}\sqrt{\frac{\hbar\omega}{2\varepsilon_0 V}}\left[\hat{a}_{\boldsymbol{k},z}e^{i(\boldsymbol{k}\cdot\boldsymbol{r}-\omega t)} - \hat{a}_{\boldsymbol{k},z}^\dagger e^{-i(\boldsymbol{k}\cdot\boldsymbol{r}-\omega t)}\right] .
\end{aligned}
$$

The above product has time dependent terms of the type $e^{i(2\omega_B\pm\omega)t},\ e^{\pm i\omega t}$. For optical frequencies $\{\omega_B, \omega\}$ both of these terms are fast oscillating and they average to zero over typical timescales of observations $t \sim 1\ ns$.

(3) $E_{Bz}\hat{E}_z\hat{E}_z$ This term contributes to the nonlinear Hamiltonian as follows:

$$
\begin{aligned}
\hat{H}_{NL}^{(2)} &= -3\varepsilon_0\chi^{(2)}\int dV E_{Bz}\hat{E}_z\hat{E}_z \\
E_{Bz} &= \mathcal{E}_B e^{i(\boldsymbol{k}_B\cdot\boldsymbol{r}-\omega_B t)} + \mathcal{E}_B^* e^{-i(\boldsymbol{k}_B\cdot\boldsymbol{r}-\omega_B t)}, \\
\hat{E}_z &= i\sum_{\boldsymbol{k}}\sqrt{\frac{\hbar\omega}{2\varepsilon_0 V}}\left[\hat{a}_{\boldsymbol{k},z}e^{i(\boldsymbol{k}\cdot\boldsymbol{r}-\omega t)} - \hat{a}_{\boldsymbol{k},z}^\dagger e^{-i(\boldsymbol{k}\cdot\boldsymbol{r}-\omega t)}\right]
\end{aligned}
$$

Substituting the E-fields in the nonlinear Hamiltonian:

$$\hat{H}_{NL}^{(2)} = -\frac{3\varepsilon_0}{2V}\chi^{(2)}\hbar \int dV \left[\mathcal{E}_B e^{i(\boldsymbol{k}_B\cdot\boldsymbol{r}-\omega_B t)} + \mathcal{E}_B^* e^{-i(\boldsymbol{k}_B\cdot\boldsymbol{r}-\omega t)}\right]$$
$$\sum_{k_1}\sqrt{\omega_1}\left[\hat{a}_{\boldsymbol{k}_1} e^{i(\boldsymbol{k}_1\cdot\boldsymbol{r}-\omega_1 t)} - \hat{a}_{\boldsymbol{k}_1}^\dagger e^{-i(\boldsymbol{k}_1\cdot\boldsymbol{r}-\omega_1 t)}\right]\sum_{k_2}\sqrt{\omega_2}\left[\hat{a}_{k_2} e^{i(\boldsymbol{k}_2\cdot\boldsymbol{r}-\omega_2 t)} - \hat{a}_{k_2}^\dagger e^{-i(\boldsymbol{k}_2\cdot\boldsymbol{r}-\omega_2 t)}\right]$$

Considering the time-dependent terms:

$$e^{i(\omega_B-\omega_1-\omega_2)t},\ e^{i(\omega_B+\omega_1+\omega_2)t},\ e^{i(\omega_B-\omega_1+\omega_2)t},\ e^{i(\omega_B+\omega_1-\omega_2)t}.$$

These are all fast oscillating terms, which averages to zero. To get a non-zero contribution we choose

$$\omega_B = \omega_1 + \omega_2 \Longrightarrow \text{Energy conservation}\,. \tag{3.3}$$

Similarly,

$$\int dV e^{i(\boldsymbol{k}_B-\boldsymbol{k}_1-\boldsymbol{k}_2)\cdot\boldsymbol{r}}.$$

For this integral to not vanish

$$\boldsymbol{k}_B = \boldsymbol{k}_1 + \boldsymbol{k}_2 \Longrightarrow \text{Momentum conservation}\,. \tag{3.4}$$

This is the **spontaneous parametric down-conversion** SPDC process.

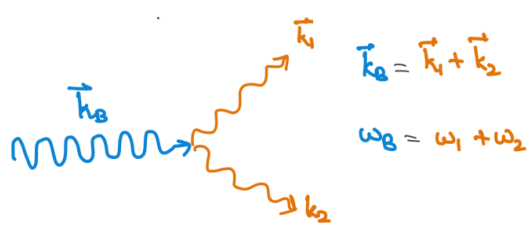


Figure 3.2 Energy and momentum conservation.

Thus, only keeping the terms that conserve energy and momentum:

$$\hat{H}_{NL}^{(2)} = -\frac{3}{2}\hbar\chi^{(2)}\sqrt{\omega_1\omega_2}\left[\mathcal{E}_B\hat{a}_{k_1}^\dagger\hat{a}_{k_2}^\dagger + \mathcal{E}_B^*\hat{a}_{k_1}\hat{a}_{k_2}\right].$$

If the two modes are the same:

$$\hat{H}_{NL}^{(2)} = -\frac{1}{2}\left[\xi^*\hat{a}^2 + \xi\hat{a}^{\dagger 2}\right], \quad \xi = 3\hbar\chi^{(2)}\omega\mathcal{E}_B, \quad \text{and} \quad \begin{array}{l}\omega_B = 2\omega \Longrightarrow \omega = \omega_B/2\\ \boldsymbol{k}_B = 2\boldsymbol{k} \Longrightarrow \boldsymbol{k} = \boldsymbol{k}_B/2\end{array}.$$

The corresponding evolution operator becomes

$$U(t,0) = e^{-i\hat{H}_{NL}^{(2)}t/\hbar} = e^{\frac{1}{2}[\xi^*\hat{a}^2 - \xi\hat{a}^{\dagger 2}]}, \quad \xi = 3i\chi^{(2)}\mathcal{E}_B\omega t. \tag{3.5}$$

Thus starting with an initial vacuum state in the red modes the system one obtains a squeezed vacuum output.

3.3 Generation of entangled photons

Lets consider a nonlinear medium with $\chi^{(2)}_{zzz}, \chi^{(2)}_{zyy} \neq 0$ and the following Hamiltonian

$$\hat{H}^{(2)}_{NL} = -\varepsilon_0 \int dV [\chi^{(2)}_{zzz} E_z E_z E_z + \chi^{(2)}_{zyy} E_z E_y E_y].$$

E-field along z and y is:

$$E_z = E_{BZ}\hat{E}_z, \quad E_y = \hat{E}_y, \quad \text{where} \quad \begin{matrix} \text{Input classical field} \\ \text{Quantized E-field} \end{matrix} .$$

The first term leads to the Hamiltonian:

$$\hat{H}^{(2)}_{NL,1} = -\frac{3\varepsilon_0}{2}\chi^{(2)}_{zzz}\sqrt{\omega_1\omega_2}(\mathcal{E}^*_B \hat{a}_{1z}\hat{a}^\dagger_{2z} + \mathcal{E}_B \hat{a}^\dagger_{1z}\hat{a}^\dagger_{2z}), \quad \begin{matrix} \omega_1 + \omega_2 = \omega_B \\ \boldsymbol{k}_1 + \boldsymbol{k}_2 = \boldsymbol{k}_B \end{matrix} . \tag{3.6}$$

The second term similarly yields

$$\hat{H}^{(2)}_{NL,2} = -\varepsilon_0 \int dV \chi^{(2)}_{zyy}(E_{Bz} + \hat{E}_z)\hat{E}_y\hat{E}_y = -\frac{\varepsilon_0}{2}\chi^{(2)}_{zyy}\sqrt{\omega_1\omega_2}(\mathcal{E}^\dagger_B \hat{a}_{1y}\hat{a}_{2y} + \mathcal{E}_B \hat{a}^\dagger_{1y}\hat{a}^\dagger_{2y}), \quad \begin{matrix} \omega_1 + \omega_2 = \omega_B \\ \boldsymbol{k}_1 + \boldsymbol{k}_2 = \boldsymbol{k}_B \end{matrix} .$$

Applying the total evolution operator to the vacuum (of all modes):

$$e^{-i(\hat{H}^{(2)}_{NL,1} + \hat{H}^{(2)}_{NL,2})t/\hbar} |\{0\}\rangle = |\psi_{out}\rangle .$$

Assuming that the nonlinearity is weak, we can expand the exponential up to first-order Taylor series:

$$\begin{aligned} |\psi_{out}\rangle \approx & \left[\mathbb{1} - \frac{it}{\hbar}(\hat{H}^{(2)}_{NL,1} + \hat{H}^{(2)}_{NL,2})\right] |\{0\}\rangle \\ & \left[\mathbb{1} + (\xi_z \hat{a}^\dagger_{1z}\hat{a}^\dagger_{2z} + \xi^*_z \hat{a}_{1z}\hat{a}_{2y}) + (\xi_y \hat{a}^\dagger_{1y}\hat{a}^\dagger_{2y} + \xi^*_y \hat{a}_{1y}\hat{a}_{2y})\right] |\{0\}\rangle , \end{aligned}$$

where ξ_z, ξ_y are the squeezing parameters:

$$\xi_z = -\frac{3i\varepsilon_0 t}{2\hbar}\chi^{(2)}_{zzz}\sqrt{\omega_1\omega_2}, \quad \xi_y = -\frac{i\varepsilon_0 t}{2\hbar}\chi^{(2)}_{zyy}\sqrt{\omega_1\omega_2}.$$

Therefore,

$$|\psi_{out}\rangle \approx |\{0\}\rangle + \xi_z |1\rangle_{1z} |1\rangle_{2z} + \xi_y |1\rangle_{1y} |1\rangle_{2y} .$$

Normalizing the above state, assuming that $\xi_z = \xi_y$ and assuming measurement when there is a single photon present in the output:

$$\text{Bell state} \qquad |\psi_{out}\rangle \approx \frac{1}{\sqrt{2}}(|1\rangle_{1z} |1\rangle_{2z} + |1\rangle_{1y} |1\rangle_{2y}) \Longleftrightarrow \frac{1}{\sqrt{2}}(|z\rangle_\uparrow |z\rangle_\downarrow + |y\rangle_\uparrow |y\rangle_\downarrow). \tag{3.7}$$

Ejemplo 3.2 **Four-wave mixing**

Consider a third-order nonlinear medium with induced nonlinear polarization $\hat{P}^{NL}_i = \varepsilon_0 \chi^{(3)}_{ijkl}\hat{E}_j\hat{E}_k\hat{E}_l$. Consider that $\chi_{zzzz} \neq 0$.

a) Consider a green laser incident on such a medium. The nonlinearity of the medium can lead to a process where two green photons can combine to generate a pair of entangled red and blue photons. Simplify the following nonlinear Hamiltonian to obtain the energy and momentum conservation conditions

$$\hat{H}^{(3)}_{NL} = -\int_V dV \varepsilon_0 \chi^{(3)}_{zzzz} \hat{E}_z \hat{E}_z \hat{E}_z \hat{E}_z.$$

The E-field of the green laser is $\hat{E}_z = \hat{\boldsymbol{z}}[\mathcal{E}_G e^{i(kx-\omega_G t)} + \mathcal{E}^*_G e^{-i(kx-\omega_G t)}]$.

b) What is the output state for the red and blue modes, considering that the ini- tial state of the field in all modes other than the pump is in vacuum and the nonlinearity is weak?

c) onsidering a nonlinear medium with nonzero $(\chi^{(3)}_{zzzz}, \chi^{(3)}_{zzyy} \neq 0)$,using the simplified Hamiltonian from part (a), show that four-wave mixing can generate two polarization entangled photons.

Solution

a) Assuming a green laser that combine to generate a pair of entangled red and blue photons. The E-field and operators are:

$$\boldsymbol{E}_z = \hat{\boldsymbol{z}} \left[\mathcal{E}_G e^{i(kx-\omega_G t)} + \mathcal{E}^*_G e^{-i(kx-\omega_G t)}\right],$$

$$\hat{\boldsymbol{E}}_z = i \sum_{\boldsymbol{k}_R} \sqrt{\frac{\hbar\omega_R}{2\varepsilon_0 V}} \left[\hat{a}_{\boldsymbol{k}_R,z} e^{i(\boldsymbol{k}_R\cdot\boldsymbol{r}-\omega_R t)} - \hat{a}^\dagger_{\boldsymbol{k}_R,z} e^{-i(\boldsymbol{k}_R\cdot\boldsymbol{r}-\omega_R t)}\right],$$

$$\hat{\boldsymbol{E}}_z = i \sum_{\boldsymbol{k}_B} \sqrt{\frac{\hbar\omega_B}{2\varepsilon_0 V}} \left[\hat{a}_{\boldsymbol{k}_B,z} e^{i(\boldsymbol{k}_B\cdot\boldsymbol{r}-\omega_B t)} - \hat{a}^\dagger_{\boldsymbol{k}_B,z} e^{-i(\boldsymbol{k}_B\cdot\boldsymbol{r}-\omega_B t)}\right].$$

Using these definitions in the product integrand of the Hamiltonian yields

$$\boldsymbol{E}_z \boldsymbol{E}_z \hat{\boldsymbol{E}}_z \hat{\boldsymbol{E}}_z = -\frac{\hbar}{2\varepsilon_0 V} \hat{\boldsymbol{z}}\cdot\hat{\boldsymbol{z}} \sum_{\boldsymbol{k}_B,\boldsymbol{k}_R} \sqrt{\omega_R\omega_B} \left[\mathcal{E}_G e^{i(kx-\omega_G t)} + \mathcal{E}^*_G e^{-i(kx-\omega_G t)}\right] \left[\mathcal{E}_G e^{i(kx-\omega_G t)} + \mathcal{E}^*_G e^{-i(kx-\omega_G t)}\right]$$
$$\left[\hat{a}_{\boldsymbol{k}_R,z} e^{i(\boldsymbol{k}_R\cdot\boldsymbol{r}-\omega_R t)} - \hat{a}^\dagger_{\boldsymbol{k}_R,z} e^{-i(\boldsymbol{k}_R\cdot\boldsymbol{r}-\omega_R t)}\right] \left[\hat{a}_{\boldsymbol{k}_B,z} e^{i(\boldsymbol{k}_B\cdot\boldsymbol{r}-\omega_B t)} - \hat{a}^\dagger_{\boldsymbol{k}_B,z} e^{-i(\boldsymbol{k}_B\cdot\boldsymbol{r}-\omega_B t)}\right].$$

By distributing all the terms above, we will have different combinations of frequency and wavevectors. In particular,

$$\omega: \ e^{i(\pm 2\omega_G \pm \omega_R \pm \omega_B)t} \quad \wedge \quad \boldsymbol{k}: \ e^{i[\pm 2kx + (\pm\boldsymbol{k}_R \pm \boldsymbol{k}_B)\cdot\boldsymbol{r}]}.$$

The only terms that are slow oscillating are the ones that are nearly zero, obtaining the energy and momentum conservation:

$$\pm 2\omega_G \mp (\omega_R + \omega_B) = 0 \Longrightarrow \text{Energy conservation}: 2\omega_G = \omega_R + \omega_B,$$
$$\pm 2kx \mp (\boldsymbol{k}_R + \boldsymbol{k}_B) \Longrightarrow \text{Momentum conservation}: 2kx = \boldsymbol{k}_R + \boldsymbol{k}_B.$$

The difference of blue and red frequencies is still in the optical frequency. The Hamiltonian then reduces to

$$\hat{H}^{(3)}_{NL,1} = \frac{\hbar\sqrt{\omega_R\omega_B}}{2} \chi^{(3)}_{zzzz} \left[\mathcal{E}^2_G \hat{a}^\dagger_{\boldsymbol{k}_R,z} \hat{a}^\dagger_{\boldsymbol{k}_B,z} + \mathcal{E}^{*2}_G \hat{a}_{\boldsymbol{k}_R,z} \hat{a}_{\boldsymbol{k}_B,z}\right].$$

b) The evolution operator applied to the vaccum state of R and B light $|0\rangle_R |0\rangle_B = |\{0\}\rangle$ is:

$$\hat{U}(t,0) = e^{-i\hat{H}^{(3)}_{NL,1} t/\hbar} |\{0\}\rangle = e^{-i\left\{\frac{\hbar\sqrt{\omega_R\omega_B}}{2}\chi^{(3)}_{zzzz}\left[\mathcal{E}^2_G \hat{a}^\dagger_{\boldsymbol{k}_R,z}\hat{a}^\dagger_{\boldsymbol{k}_B,z} + \mathcal{E}^{*2}_G \hat{a}_{\boldsymbol{k}_R,z}\hat{a}_{\boldsymbol{k}_B,z}\right]\right\} t/\hbar} = |\psi_{out}\rangle$$

Expanding the exponential to the first-order Taylor series yields:

$$\left\{\mathbb{1} - i\frac{\sqrt{\omega_R\omega_B}\chi^{(3)}_{zzzz}}{2}\left[\mathcal{E}^2_G \hat{a}^\dagger_{\boldsymbol{k}_R,z}\hat{a}^\dagger_{\boldsymbol{k}_B,z} + \mathcal{E}^{*2}_G \hat{a}_{\boldsymbol{k}_R,z}\hat{a}_{\boldsymbol{k}_B,z}\right] t\right\} |\{0\}\rangle.$$

By defining the squeezing parameter ξ_z as:

$$\xi_z = -i\frac{\sqrt{\omega_R\omega_B}\chi^{(3)}_{zzzz}}{2}\mathcal{E}_G^2 t,$$

we have:

$$\left[\mathbb{1} + \xi_z \hat{a}^\dagger_{\boldsymbol{k}_R,z}\hat{a}^\dagger_{\boldsymbol{k}_B,z} + \xi_z^* \hat{a}_{\boldsymbol{k}_R,z}\hat{a}_{\boldsymbol{k}_B,z}\right]|\{0\}\rangle = |\{0\}\rangle + \xi_z |1\rangle_{R_z} |1\rangle_{B_z}.$$

c) The process for the new term is esentially the same, with the change of the last two indices to y polarization. The Hamiltonian for this term is:

$$\hat{H}^{(3)}_{NL,2} = \frac{\hbar\sqrt{\omega_R\omega_B}}{2}\chi^{(3)}_{zzyy}\left[\mathcal{E}_G^2\hat{a}^\dagger_{\boldsymbol{k}_R,y}\hat{a}^\dagger_{\boldsymbol{k}_B,y} + \mathcal{E}_G^{*2}\hat{a}_{\boldsymbol{k}_R,y}\hat{a}_{\boldsymbol{k}_B,y}\right]$$

The total Hamiltonian is the sum of the individual Hamiltonian:

$$\begin{aligned}
\hat{H}^{(3)}_{NL} &= \hat{H}^{(3)}_{NL,1} + \hat{H}^{(3)}_{NL,2} \\
\hat{H}^{(3)}_{NL} &= \frac{\hbar\sqrt{\omega_R\omega_B}}{2}\left[\chi^{(3)}_{zzzz}\mathcal{E}_G^2\hat{a}^\dagger_{\boldsymbol{k}_R,z}\hat{a}^\dagger_{\boldsymbol{k}_B,z} + \mathcal{E}_G^{*2}\hat{a}_{\boldsymbol{k}_R,z}\hat{a}_{\boldsymbol{k}_B,z} + \chi^{(3)}_{zzyy}\mathcal{E}_G^2\hat{a}^\dagger_{\boldsymbol{k}_R,y}\hat{a}^\dagger_{\boldsymbol{k}_B,y} + \mathcal{E}_G^{*2}\hat{a}_{\boldsymbol{k}_R,y}\hat{a}_{\boldsymbol{k}_B,y}\right].
\end{aligned}$$

The first-order Taylor expansion of the evolution operator is:

$$U(t,0) = e^{-i\hat{N}^{(3)}_{NL}t/\hbar} \approx \mathbb{1} + \xi_z\hat{a}^\dagger_{\boldsymbol{k}_R,z}\hat{a}^\dagger_{\boldsymbol{k}_B,z} + \xi_z^*\hat{a}_{\boldsymbol{k}_R,z}\hat{a}_{\boldsymbol{k}_B,z} + \xi_y\hat{a}^\dagger_{\boldsymbol{k}_R,y}\hat{a}^\dagger_{\boldsymbol{k}_B,y} + \xi_y^*\hat{a}_{\boldsymbol{k}_R,y}\hat{a}_{\boldsymbol{k}_B,y},$$

where the squeezing parameters are:

$$\xi_z = -i\frac{\sqrt{\omega_R\omega_B}\chi^{(3)}_{zzzz}}{2}\mathcal{E}_G^2 t, \quad \text{and} \quad \xi_y = -i\frac{\sqrt{\omega_R\omega_B}\chi^{(3)}_{zzyy}}{2}\mathcal{E}_G^2 t.$$

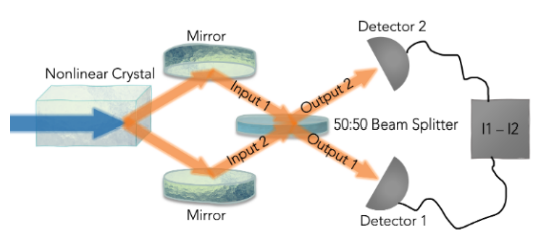
Application of the evolution operator in the vaccum state produces:

$$\left[\mathbb{1} + \xi_z\hat{a}^\dagger_{\boldsymbol{k}_R,z}\hat{a}^\dagger_{\boldsymbol{k}_B,z} + \xi_z^*\hat{a}_{\boldsymbol{k}_R,z}\hat{a}_{\boldsymbol{k}_B,z} + \xi_y\hat{a}^\dagger_{\boldsymbol{k}_R,y}\hat{a}^\dagger_{\boldsymbol{k}_B,y} + \xi_y^*\hat{a}_{\boldsymbol{k}_R,y}\hat{a}_{\boldsymbol{k}_B,y}\right]|\{0\}\rangle = |\{0\}\rangle + \xi_z|1\rangle_{R_z}|1\rangle_{B_z} + \xi_y|1\rangle_{R_y}|1\rangle_{B_y}.$$

Generating in that way, two polarization entangled photons.

Ejemplo 3.3 **Entangled photons through a BS**

Consider the following setup.



The state of the two photons exiting the nonlinear crystal is: i) $|1\rangle_1|1\rangle_2$, ii) $\frac{1}{\sqrt{2}}(|1\rangle_1|0\rangle_2 + |0\rangle_1|1\rangle_2)$, and iii) $\frac{1}{\sqrt{2}}(|1\rangle_1|0\rangle_2 - |0\rangle_1|1\rangle_2)$. Find $\langle\hat{I}_k\rangle$ and $\langle\hat{I}_1 - \hat{I}_2\rangle$.

Solution

The intensity operator for each detector and the difference is:

$$\begin{aligned}
\hat{I}_1 &= \hat{b}_1^\dagger \hat{b}_1 = \frac{1}{2}(\hat{a}_1^\dagger + \hat{a}_2^\dagger)(\hat{a}_1 + \hat{a}_2) = \frac{1}{2}(\hat{a}_1^\dagger \hat{a}_1 + \hat{a}_1^\dagger \hat{a}_2 + \hat{a}_2^\dagger \hat{a}_1 + \hat{a}_2^\dagger \hat{a}_2), \\
\hat{I}_2 &= \hat{b}_2^\dagger \hat{b}_2 = \frac{1}{2}(\hat{a}_1^\dagger - \hat{a}_2^\dagger)(\hat{a}_1 - \hat{a}_2) = \frac{1}{2}(\hat{a}_1^\dagger \hat{a}_1 - \hat{a}_1^\dagger \hat{a}_2 - \hat{a}_2^\dagger \hat{a}_1 + \hat{a}_2^\dagger \hat{a}_2), \\
\hat{I}_1 - \hat{I}_2 &= \hat{a}_1^\dagger \hat{a}_2 + \hat{a}_2^\dagger \hat{a}_1.
\end{aligned}$$

We use these to compute the expected number of photons $\langle \hat{I}_k \rangle$ at individual detectors and the difference $\langle \hat{I}_1 - \hat{I}_2 \rangle$.

i)

$$\begin{aligned}
\langle \hat{I}_1 \rangle &= \frac{1}{2}\langle 1,1|(\hat{a}_1^\dagger \hat{a}_1 + \hat{a}_1^\dagger \hat{a}_2 + \hat{a}_2^\dagger \hat{a}_1 + \hat{a}_2^\dagger \hat{a}_2)|1,1\rangle = 1, \\
\langle \hat{I}_2 \rangle &= \frac{1}{2}\langle 1,1|(\hat{a}_1^\dagger \hat{a}_1 - \hat{a}_1^\dagger \hat{a}_2 - \hat{a}_2^\dagger \hat{a}_1 + \hat{a}_2^\dagger \hat{a}_2)|1,1\rangle = 1, \\
\langle \hat{I}_1 - \hat{I}_2 \rangle &= 0.
\end{aligned}$$

ii) In this case of entangled photons, we follow the definition of the expected value and we use distribution property:

$$\langle \hat{I}_1 \rangle = \left[\frac{1}{\sqrt{2}}(\langle 0,1| + \langle 1,0|)\right] |\hat{b}_1^\dagger \hat{b}_1| \left[\frac{1}{\sqrt{2}}(|1,0\rangle + |0,1\rangle)\right] = \frac{1}{4}(\langle 0,1| + \langle 1,0|)|(\hat{a}_1^\dagger \hat{a}_1 + \hat{a}_1^\dagger \hat{a}_2 + \hat{a}_2^\dagger \hat{a}_1 + \hat{a}_2^\dagger \hat{a}_2)|(|1,0\rangle + |0,1\rangle).$$

I should distribute and solve the 16 terms. However, I can express the input in the following way:

$$|\psi\rangle = \frac{1}{\sqrt{2}}(\hat{a}_1^\dagger + \hat{a}_2^\dagger)\,|0,0\rangle = \hat{b}_1^\dagger\,|0,0\rangle = |1',0'\rangle\,.$$

which is simpler to use, because:

$$\langle \hat{I}_1 \rangle = \langle 0',1'|\hat{b}_1^\dagger \hat{b}_1|1',0'\rangle = \langle 0',1'|1',0'\rangle = 1.$$

For the detector 2, the expected number of photon is:

$$\langle \hat{I}_2 \rangle = \langle 0',1'|\hat{b}_2^\dagger \hat{b}_2|1',0'\rangle = 0.$$

Therefore,

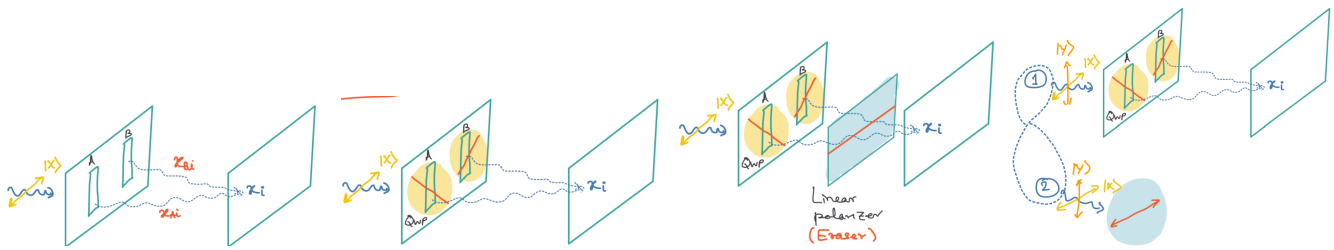
$$\langle \hat{I}_1 - \hat{I}_2 \rangle = \langle \hat{I}_1 \rangle - \langle \hat{I}_2 \rangle = 1.$$

iii) Using the form from previous part, the entangled state is:

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|1,0\rangle - |0,1\rangle) = \frac{1}{\sqrt{2}}(\hat{a}_1^\dagger - \hat{a}_2^\dagger)\,|0,0\rangle = \hat{b}_2^\dagger\,|0,0\rangle = |0',1'\rangle\,.$$

Now,

$$\begin{aligned}
\langle \hat{I}_1 \rangle &= \langle 0',1'|\hat{b}_1^\dagger \hat{b}_2'|1',0'\rangle = 0, \\
\langle \hat{I}_2 \rangle &= \langle 0',1'|\hat{b}_2^\dagger \hat{b}_2|0',1'\rangle = 1, \\
\langle \hat{I}_1 - \hat{I}_2 \rangle &= -1.
\end{aligned}$$



(a) Case 1 **(b)** Case 2 **(c)** Case 3 **(d)** Case 4

Figure 3.1 Double-slit quantum eraser with different cases.

3.4 Double-slit quantum eraser

3.4.1 Case 1: Single photon incident on a double slit apparatus

The state of the photon after it passes through the slit is:

$$|\psi\rangle = \frac{1}{\sqrt{2}} \underbrace{(|A\rangle + |B\rangle)}_{\text{position}} \otimes \underbrace{|X\rangle}_{\text{polarization}} .$$

As the photon propagates from the slit to screen

$$|A\rangle \to \sum_i N e^{ikx_{Ai}} |x_i\rangle , \quad |B\rangle \to \sum_i N e^{ikx_{bi}} |x_i\rangle .$$

State of the photon at screen is

$$|\psi\rangle = \frac{1}{\sqrt{2}} \sum_i N[e^{ikx_{Ai}} + e^{ikx_{Bi}}] |x_i\rangle \otimes |X\rangle = \frac{1}{\sqrt{2}} \sum_i N e^{ikx_{Ai}} [1 + e^{ik(x_{Bi} - x_{Ai})}] |x_i\rangle \otimes |X\rangle .$$

The probability of finding the photon at psition x_j is:

$$|\langle x_j|\psi\rangle|^2 = \frac{1}{2}|N e^{ikx_{Ai}}|^2 |1 + e^{ik(x_{Bj} - x_{Aj})}|^2 = 2|N|^2 \cos^2[\frac{k}{2}(x_{Bj} - x_{Ai})]. \tag{3.8}$$

3.4.2 Case 2: Encoding which path information

State of the photon after it passes through the two slits is:

$$|\psi\rangle = \frac{1}{\sqrt{2}}[|A\rangle |+\rangle + |B\rangle |-\rangle], \quad |\pm\rangle = \frac{1}{\sqrt{2}}(|x\rangle \pm i |y\rangle),$$

where $|\pm\rangle$ is left and right circularly polarized, respectively. At the screen:

$$|\psi\rangle = \frac{1}{\sqrt{2}} \left[\sum_i N e^{ikx_{Ai}} |+\rangle + \sum_i N e^{ikx_{Bi}} |-\rangle \right] |x_i\rangle .$$

The probability of detecting a photon at x_j is:

$$|\langle x_j|\psi\rangle|^2 = \frac{1}{\sqrt{2}}|N|^2 |e^{ikx_{Ai}} |+\rangle + e^{ikx_{Bj}} |-\rangle |^2 = \frac{2}{\sqrt{2}}|N|^2, \quad \langle \pm|\mp\rangle = 0.$$

No interference!

3.4.3 Case 3: Erasing which path information

State after the slit is:

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|A\rangle\,|+\rangle + |B\rangle\,|-\rangle).$$

After the eraser:

$$|\psi\rangle = \frac{1}{\sqrt{2}}\,|x\rangle\,\langle x|\,(|A\rangle\,|+\rangle + |B\rangle\,|-\rangle) = \frac{1}{2}(|A\rangle + |B\rangle) \otimes |x\rangle\,.$$

Same as case 1 (reduced norm) No which path information. Interference is back!

3.4.4 Case 4: Quantum erasure via an entangled photon

Photon 1 and 2 are entangled in polarization:

$$|\psi\rangle_{12} = \frac{1}{\sqrt{2}}(|x\rangle_1\,|x\rangle_2 + |y\rangle_1\,|y\rangle_2).$$

After passing through the slits:

Table 3.1 Action of the QWP.

	A	B
$\|x\rangle_1$	$\|+\rangle_1$	$\|-\rangle_1$
$\|y\rangle_1$	$\|-\rangle_1$	$\|+\rangle_1$

The total state of the photons becomes:

$$|\psi_{12}\rangle = \frac{1}{2}\left[(|A\rangle_1\,|+\rangle_1 + |B\rangle_1\,|-\rangle_1) \otimes |x\rangle_2 + (|A\rangle_1\,|-\rangle_1 + |B\rangle_1\,|+\rangle_1) \otimes |y\rangle_2\right].$$

Case 4a

Applying a linear polarizer to photon 2 oriented along x:

$$|x\rangle_2\;{}_2\langle x|\psi_{12}\rangle = \frac{1}{2}(|A\rangle_1\,|+\rangle_1 + |B\rangle_1\,|-\rangle_1).$$

This corresponds to having which-path information (see case 2). No interference. The same if obtained when polarized is oriented along $|y\rangle$.

Case 4b

Say the polarizer in photon 2's path is oriented along $|+\pi/4\rangle$:

$$\begin{aligned}
&\frac{1}{2}(|x\rangle_2 + |y\rangle_2)({}_2\langle x| + {}_2\langle y|)\,|\psi_{12}\rangle \\
=&\frac{1}{4}(|x\rangle_2 + |y\rangle_2)({}_2\langle x| + {}_2\langle y|)[(|A\rangle_1\,|+\rangle_2 + |B\rangle_1\,|-\rangle_1) \otimes |x\rangle_2 + (|A\rangle_1\,|-\rangle_1 + |B\rangle_1\,|+\rangle_1) \otimes |y\rangle_2] \\
=&\frac{1}{2}(|x\rangle_2 + |y\rangle_2)[|A\rangle_1\,|x\rangle_1 + |B\rangle_1\,|x\rangle_1] \\
=&\frac{1}{2}\underbrace{[|A\rangle_1 + |B\rangle_1]}_{\text{No which path!}}|x\rangle_1 \otimes (|x\rangle_2 + |y\rangle_2).
\end{aligned}$$

Table 3.2 Some generalization of the quantum interferometry idea.

Initial state	$\|\psi^-\rangle = [\cos\theta\,\|X\rangle + \sin\theta e^{i\phi}\,\|Y\rangle]$	Slits	$\|\psi^+\rangle = \frac{1}{\sqrt{2}}(\|A\rangle + \|B\rangle) \otimes \|\psi^-\rangle$
Polarizer	$\hat{P} = \|p\rangle\langle p\|$	$\|\psi^{++}\rangle = \hat{P}\,\|\psi^+\rangle$	$\|p\rangle =$ same form as $\|\psi^-\rangle$
Evolution	$\|A\rangle \to \sum_i N\exp(ikx_{Ai})\,\|x_i\rangle$	Entangled	$\|\psi\rangle_{12} = \frac{1}{\sqrt{2}}(\|x\rangle_1\,\|x\rangle_2 + \|y\rangle_1\,\|y\rangle_2)$

Interference is back. Quantum erasure via an entangled photon!

Ejemplo 3.4 **Quantum interferometry**

The incoming photon has a polarization given by $|Y\rangle$. There are two LPs in fron of each slit oriented along $\theta_A = -\theta_B = \theta$ such that their operators are $\hat{P}_A = |\theta_A\rangle\langle\theta_A|$ and $\hat{P}_B = |\theta_B\rangle\langle\theta_B|$, with $|\theta_A\rangle = \cos\theta\,|X\rangle + \sin\theta\,|Y\rangle$ and $|\theta_B\rangle = \cos\theta\,|X\rangle - \sin\theta\,|Y\rangle$.

a) What is the state $|\Psi\rangle$ once it passes through the double slit?
b) What is the probability $P_k = |\langle x_k|\Psi\rangle|^2$ of finding the photon at x_k on the screen?
c) How does the probability change as function of θ? Consider $\theta = 0, \pi/4$. We now place an eraser as shown in the right figure.
d) What is the state of the photon once it passes through the eraser?
e) Consider $\theta = \pi/4$ and $\theta_E = \pi/4$, is there an interference pattern on the screen?
f) For $\theta = \pi/4$ and $\theta_E = 0$, is there an interference pattern on the screen?

Solution

a) State just after the slits, and just after the linear polarizers are:

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|A\rangle + |B\rangle) \otimes |Y\rangle\,,$$
$$|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|A\rangle\,\hat{P}_A + |B\rangle\,\hat{P}_B) \otimes |Y\rangle = \frac{\sin\theta}{\sqrt{2}}(|A\rangle\,|\theta_A\rangle - |B\rangle\,|\theta_B\rangle) \longrightarrow \left|\Psi^{+'}\right\rangle = \frac{|\Psi^+\rangle}{\sin\theta}.$$

b) At the screen, $|A\rangle$ and $|B\rangle$ should evolve as we seen in the lectures. However, I will leave them as they are and make the projection.

$$\langle x_k|\Psi^{+'}\rangle = \frac{1}{\sqrt{2}}(\langle x_k|A\rangle\,|\theta_A\rangle - \langle x_k|B\rangle\,|\theta_B\rangle) = \frac{1}{\sqrt{2}}[\psi_A(x_k)\,|\theta_A\rangle + \psi_B(x_k)\,|\theta_B\rangle].$$

Then, the magnitude is:

$$\begin{aligned}|\langle x_k|\Psi^{+'}\rangle|^2 &= \frac{1}{2}[\psi_A(x_k)\,|\theta_A\rangle + \psi_B(x_k)\,|\theta_B\rangle][\Psi_A^*(x_k)\,\langle\theta_A| + \Psi_B^*(x_k)\,\langle\theta_B|] \\ &= \frac{1}{2}\left[|\psi_A(x_k)|^2 + |\psi_B(x_k)|^2 - \psi_A(x_k)\psi_B\langle\theta_B|\theta_A\rangle - \psi_B(x_k)\psi_A^*\langle\theta_A|\theta_B\rangle\right].\end{aligned}$$

We need to find the projection of the polarizers:

$$\langle\theta_A|\theta_B\rangle = [\cos\theta\,\langle X| + \sin\theta\,\langle Y|][\cos\theta\,|X\rangle - \sin\theta\,|Y\rangle] = \cos^2\theta - \sin^2\theta = \cos 2\theta,$$
$$\langle\theta_B|\theta_A\rangle = [\cos\theta\,\langle X| - \sin\theta\,\langle Y|][\cos\theta\,|X\rangle + \sin\theta\,|Y\rangle] = \cos^2\theta - \sin^2\theta = \cos 2\theta.$$

If we assume that

$$\psi_A(x_k) = |\psi_A|\angle\theta_A, \quad \text{and} \quad \psi_B(x_k) = |\psi_B|\angle\theta_B,$$

then usinging it with the projection computed we finally have:

$$P_k = |\langle x_k|\Psi^{+'}\rangle|^2 = \frac{1}{2}\left[|\psi_A|^2 + |\psi_B|^2 - 2\cos 2\theta|\psi_A||\psi_B|\cos(\Delta\phi)\right], \quad \Delta\phi = \theta_B - \theta_A.$$

c) For each angle, we have:

$$\theta = 0: \qquad P_k = \frac{1}{2}\left[|\psi_A|^2 + |\psi_B|^2 - 2|\psi_A||\psi_B|\cos(\Delta\phi)\right] \Longrightarrow \text{Interference term maximized},$$
$$\theta = \pi/4: \qquad P_k = \frac{1}{2}\left[|\psi_A|^2 + |\psi_B|^2\right] \Longrightarrow \text{No interference term.}$$

As we can see, when $\theta = \pi/4$ the interference term vanished and therefore there is no interference.

d) The eraser projects onto its basis:

$$\begin{aligned}
\left|\Psi^{++}\right\rangle &= |\theta_E\rangle\langle\theta_E|\left|\Psi^{+'}\right\rangle \\
&= \frac{1}{\sqrt{2}}|\theta_E\rangle\langle\theta_E|\left(|A\rangle|\theta_A\rangle - |B\rangle|\theta_B\rangle\right) \\
&= \frac{1}{\sqrt{2}}\left[(\cos\theta\cos\theta_E + \sin\theta\sin\theta_E)|A\rangle - (\cos\theta\cos\theta_E - \sin\theta\sin\theta_E)|B\rangle\right]\otimes|\theta_E\rangle \\
\left|\Psi^{++}\right\rangle &= \frac{1}{\sqrt{2}}\left[\cos(\theta-\theta_E)|A\rangle - \cos(\theta+\theta_E)|B\rangle\right]\otimes|\theta_E\rangle.
\end{aligned}$$

e) For $\theta = \pi/4$ and $\theta_E = \pi/4$ the state vector is:

$$\left|\Psi^{++}\right\rangle = \frac{1}{\sqrt{2}}|A\rangle\otimes|\theta_E\rangle \longrightarrow \left|\Psi^{++'}\right\rangle = |A\rangle\otimes|\theta_E\rangle.$$

Because there is not interaction with the slit B, there is no interference pattern detected on the screen.

f) For $\theta = \pi/4$ and $\theta_E = 0$ we have:

$$\left|\Psi^{++}\right\rangle = \frac{\cos(\pi/4)}{\sqrt{2}}[|A\rangle - |B\rangle]\otimes|X\rangle \longrightarrow \left|\Psi^{++'}\right\rangle = \frac{1}{\sqrt{2}}[|A\rangle - |B\rangle]\otimes|X\rangle.$$

As both slits are present in the state vector, when computed the probability $|\langle x_j|\Psi^{++'}\rangle$ there will be a cross-term that will implies interfence.

Part II

Atom-field interaction

Chapter 4

Atom field interaction

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4.1 Atom-field interactions

4.1.1 Introduction

The electric-dipole Hamiltonian is:

$$H_{int} = -\hat{\boldsymbol{d}} \cdot \hat{\boldsymbol{E}}(\boldsymbol{r}_A), \qquad \begin{array}{l} \boldsymbol{d} = \text{Atomic dipole moment} \\ \boldsymbol{E} = \text{E-field at the position of the atom} \end{array}. \tag{4.1}$$

- Interaction between a classical dipole and classical E-field: models of atom polarizability.
- Interaction between a quantized atom structure (two-level) and classical EM-field: Raby cycle, quantum logic states, 3-level systems.
- Interaction between a single atom and quantuzed EM-field (free space): spontaneous emission, Lamb shift, caivty WED, Purcell effect, etc.
- Multiple atoms and quantized EM field: dipole-dipole interactcions, collective spontaneous emission.
- Single atom, classical field, quantized motion: atom cooling and trapping, atom interferometry.
- Single atom (polarizable particle), quantized field, etc.

4.1.2 Atom-matter interaction

At the most fundamental level light-matter interaction reduces to the interaction between EM fields and single atoms, which are composed of charges.

Let us consider the atom to be a distribution of charges $\{q_n, m_n\}$ located at $\boldsymbol{r}_n$ relative to the atomic center of mmass $\boldsymbol{r}_A$.

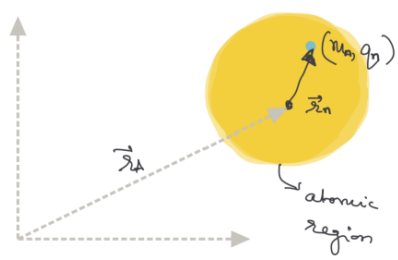


Figure 4.1 Atom distribution and distribution of charges.

The Lorentz force on a charge is:

$$\boldsymbol{F}(\boldsymbol{r}) = q\boldsymbol{E}(\boldsymbol{r}) + q\boldsymbol{v} \times \boldsymbol{B}(\boldsymbol{r}). \tag{4.2}$$

The corresponding electric and magnetic potentials are:

$$V_e = \text{Electric potential} = -\int d\boldsymbol{s} \cdot \sum_n q_n \delta(\boldsymbol{s} - \boldsymbol{r}_n) \boldsymbol{E}(\boldsymbol{r}_A + \boldsymbol{s}), \tag{4.3}$$

$$V_m = \text{Magnetic potential} = -\int d\boldsymbol{s} \cdot \sum_n q_n \delta(\boldsymbol{s} - \boldsymbol{r}_n) \boldsymbol{v}_n \times \boldsymbol{B}(\boldsymbol{r}_A + \boldsymbol{s}). \tag{4.4}$$

We note that the variation of the EM-fields over the size of th atom is small.

Review of Bohr model

Centrifugal force=electrostatic force ($z = 1$) $$\frac{m_e v_n^2}{r_n} = \frac{1}{4\pi\varepsilon_0}\frac{e^2}{r_n^2} \longrightarrow m_e v_n^2 r_n = \frac{e^2}{4\pi\varepsilon_0}.$$

Quantization of angular momentum $$m_e v_n r_n = n\hbar$$

Dividing the above equations:

$$v_n = \frac{e^2}{4\pi\varepsilon_0 n\hbar}.$$

Assuming $n = 1$:

$$v = \frac{e^2}{4\pi\varepsilon_0\hbar},$$

Substituting in the first expression:

$$r = \frac{\hbar}{m_e v} = \frac{4\pi\varepsilon_0\hbar^2}{m_e e^2},$$

Transition wavelength:

$$\lambda = \frac{2\pi\hbar c}{E} = \frac{4\pi\hbar c}{m_e v^2}, \quad E = \frac{1}{2}m_e v^2.$$

$$\frac{r}{\lambda} \sim \frac{\hbar}{m_e v}\frac{m_e v^2}{2\pi\hbar c} = \frac{v}{2\pi c} = \frac{e^2}{4\pi\varepsilon_0\hbar c} = \alpha.$$

We note that the ratio of atomic radius to transition wavelength r/λ ad the ratio of speed of e^- in orbit to speed of light v/c are both of the order of α, the fine-structure constant.

From this we conclude that the variation of the EM-fields over the size of the atom is small. Let us compare the size of the magnetic to electric potential:

$$\left|\frac{V_m}{V_e}\right| \sim \frac{|\boldsymbol{v}||\boldsymbol{B}|}{|\boldsymbol{E}|} \overset{\text{Ampere's law}}{\sim} \frac{|\boldsymbol{v}|}{c} \sim \frac{e^2}{4\pi\varepsilon_0\hbar c} \equiv \alpha \approx \frac{1}{137}. \tag{4.5}$$

Magnetic interactions between atoms and EM fields are waker by a factor of α compared to electric interactions.

Let us look at the electric potential:

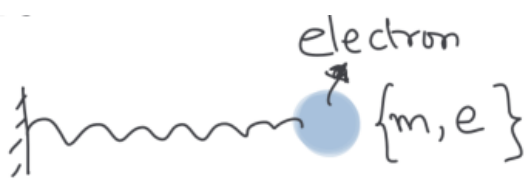
$$
\begin{aligned}
V_e &= -\int d\boldsymbol{s} \cdot \sum_n q_n \delta(\boldsymbol{s} - \boldsymbol{r}_n) \boldsymbol{E}(\boldsymbol{r}_A + \boldsymbol{s}) \\
&= -\sum_n q_n \underbrace{\Phi(\boldsymbol{r}_A + \boldsymbol{r}_n)}_{\text{scalar potential}} \quad \text{(Sifting property of delta)} \\
&= -\sum_n q_n \left[\Phi(\boldsymbol{r}_A) + \boldsymbol{r}_n \cdot \nabla\Phi(\boldsymbol{r}_A) + \text{higher order terms}\right] \\
V_e &= -\underbrace{\sum_n q_n}_{=0(\text{neutral atom})} \Phi(\boldsymbol{r}_A) - \underbrace{\sum_n q_n \boldsymbol{r}_n}_{=\boldsymbol{d}(\text{dipole moment})} \cdot \boldsymbol{E}(\boldsymbol{r}_A) + \text{multipolar terms.}
\end{aligned}
$$

Thus, the atom-field interaction under the electric-dipole approximation is given by:

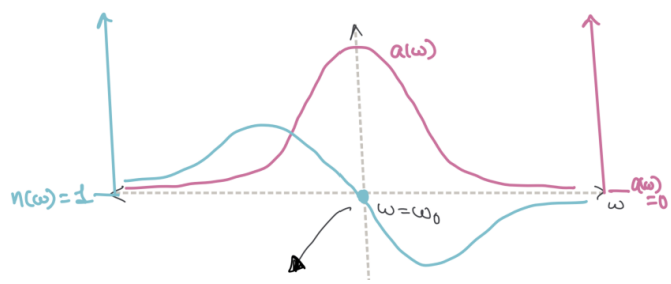
$$
V_e = -\boldsymbol{d} \cdot \boldsymbol{E}(\boldsymbol{r}_A) \xrightarrow{\text{Quantization}} \hat{H}_{int} = -\hat{\boldsymbol{d}} \cdot \hat{\boldsymbol{E}}(\boldsymbol{r}_A) \,. \tag{4.6}
$$

4.2 The Lorentz atom

A Lorentz atom refers to the toy model of an atom where a single classical electron is harmonically bound to the nucleus, like a mass-spring.



(a) Lorentz atom



(b) Refractive index behavior

Equation of motion for the electron is:

$$
m\frac{d^2\boldsymbol{r}}{dt^2} + m\omega_0^2\boldsymbol{r} + 2m\gamma\frac{d\boldsymbol{r}}{dt} = -e\boldsymbol{E}(t), \tag{4.7}
$$

where γ is the frictional decay rate of oscilattory motion. The E-field and the induced position are:

$$
\boldsymbol{E}(t) = \sum_n \mathcal{E}_n e^{i\omega_n t} + \mathcal{E}_n^* e^{-i\omega_n t} \xrightarrow{\text{induced motion}} \boldsymbol{r} = \sum \left[\boldsymbol{r}_n e^{i\omega_n t} + \boldsymbol{r}_n^* e^{-i\omega_n t}\right]. \tag{4.8}
$$

By substituting both in the equation of motion and solving for $\boldsymbol{r}_n$ we find that:

$$
\boldsymbol{r}_n = \left[-\frac{e}{m}\frac{1}{\omega_0^2 - \omega_n^2 + 2i\gamma\omega_n}\right]\mathcal{E}_n. \tag{4.9}
$$

The dipole moment of the atom induced by the E-field is:

$$
\boldsymbol{d}_n = -e\boldsymbol{r}_n = \alpha(\omega_n)\mathcal{E}_n = \frac{e^2}{m}\frac{1}{\omega_0^2 - \omega_n^2 + 2i\gamma\omega_n}\mathcal{E}_n, \quad \alpha(\omega_n) = \text{polarizability of the atom.} \tag{4.10}
$$

Therefore, the polarizability of the atom is:

$$\text{polarizability of the atom} \qquad \alpha(\omega_n) = \frac{e^2/m}{\omega_0^2 - \omega_n^2 + 2i\gamma\omega_n}. \tag{4.11}$$

If the induced dipole moment for a single atom is $\boldsymbol{d}$, for N atoms per uint volume we have:

$$\boldsymbol{P}_n = N\boldsymbol{d}_n = N\alpha(\omega_n)\mathcal{E}_n = \varepsilon_0\chi(\omega_n)\boldsymbol{E} \longrightarrow \chi(\omega_n) = \frac{N\alpha(\omega_n)}{\varepsilon_0}. \tag{4.12}$$

Thus, the complex refractive index is:

$$\tilde{n}^2(\omega_n) = 1 + \chi(\omega_n) = 1 + \frac{N\alpha(\omega_n)}{\varepsilon_0} \longrightarrow \boxed{\tilde{n}(\omega_n) = \sqrt{1 + \frac{N\alpha(\omega_n)}{\varepsilon_0}}}. \tag{4.13}$$

As an EM-field passes through a medium it experiences:

- Dispersion, defined by Re $(n(\omega))$.
- Absorption, described by Im $(n(\omega))$.

If $\chi(\omega_n)$ is small, we approximate above by the first-order of the square root:

$$\chi(\omega_n) \approx 1 + \frac{N\alpha(\omega_n)}{2\varepsilon_0} \Longrightarrow \begin{aligned} n(\omega) &= \text{Re}\,(\tilde{n}(\omega_n)) = 1 + \frac{Ne^2/m\varepsilon_0(\omega_0^2 - \omega_n^2)}{2[(\omega_0^2 - \omega_n^2) + 4\omega_n^2\gamma^2]} \\ a(\omega_n) &= \text{Im}\,(\tilde{n}(\omega_n)) = \frac{-i\omega\gamma Ne^2/m\varepsilon_0}{[(\omega_0 - \omega_n^2)^2 + 4\omega_n^2\gamma^2]}. \end{aligned} \tag{4.14}$$

In the curve above both parts are plotted againd the frequency. At $\omega = \omega_0$ (resonance), the absorption is maximum whereas the dispersion is minimum (vanishes).

4.3 Quantized energy level of atom

So far we have trated the atom semi-clasically. We will now describe the atomic degree of freedom-electrons as quantum mechanics objects. The Hamiltonian and eigenequation for a single e^- in the atom are:

$$\hat{H}_A = \frac{\hat{p}^2}{2m} + V(\hat{\boldsymbol{r}}), \quad \hat{H}_A\,|\psi_n\rangle = E_n\,|\psi_n\rangle\,. \tag{4.15}$$

Not that $|\psi_n\rangle$ is not a position eigenstate, since $\hat{H}_0$ contains both $\hat{\boldsymbol{r}}$ and $\hat{\boldsymbol{p}}$ operators.

The electronic wavefunction fo an e^- in the state $|\psi_n\rangle$ is given by $\langle \boldsymbol{r}|\psi_n\rangle = \psi_n(\boldsymbol{r})$.

Let us consider two-energy eigenstates:

$$\underbrace{|\psi_g\rangle = |g\rangle}_{\text{ground}} \quad \text{and} \quad \underbrace{|\psi_e\rangle = |e\rangle}_{\text{excited}}. \tag{4.16}$$

We restrict out attention to two-energy levels since they might have a stronger posibility of transition between them.

We would like to express the dipole operator in the energy basis:

$$\langle g|\hat{\boldsymbol{d}}|e\rangle = -e\langle g|\hat{\boldsymbol{r}}|e\rangle = -e\int d^3r \int d^3r' \, \langle g|\boldsymbol{r}\rangle\langle \boldsymbol{r}|\hat{\boldsymbol{r}}'|\boldsymbol{r}'\rangle\langle \boldsymbol{r}'|e\rangle = -e\int d^3r \, \boldsymbol{r}\psi_g^*(\boldsymbol{r})\psi_e(\boldsymbol{r}) = \boldsymbol{d}_{ge}.$$

Thus, the dipole transition matrix corresponds to the overlap between the electronic wavefunction in two energy levels. Similarly,

$$\langle e|\hat{\boldsymbol{d}}|g\rangle = -e\int d^3r \, \boldsymbol{r}\psi_g(\boldsymbol{r})\psi_e^*(\boldsymbol{r}) = \boldsymbol{d}_{eg} = \boldsymbol{d}_{ge}^*, \quad \langle g|\hat{\boldsymbol{d}}|g\rangle = \langle e|\hat{\boldsymbol{d}}|e\rangle = 0$$

The dipole matrix is symmetric. We can thus express the dipole operator as:

$$\hat{\boldsymbol{d}} = \boldsymbol{d}_{eg}\,|e\rangle\,\langle g| + \boldsymbol{d}_{ge}\,|g\rangle\,\langle e| = \boldsymbol{d}\hat{\sigma}_+ + \boldsymbol{d}^*\hat{\sigma}_-, \tag{4.17}$$

where $\hat{\sigma}$ operators are Pauli matrices and corresponds to atomic raising and lowering operators, respectively.

The free Hamiltonian of the atom $\hat{H}_0$ can be expressed in terms of its energy eigenstates:

$$\hat{H}_0 = \sum_n E_n\,|n\rangle\,\langle n| = E_g\,|g\rangle\,\langle g| + E_e\,|e\rangle\,\langle e| = \hbar\omega_0\,|e\rangle\,\langle e| \quad (E_g = 0 \wedge E_e = \hbar\omega_0) = \hbar\omega_0\hat{\sigma}_+\hat{\sigma}_-.$$

The total Hamiltonian of an atom interacting with quantized EM-field is:

$$\hat{H}_{tot} = \underbrace{\hat{H}_A}_{\text{atom}} + \underbrace{\hat{H}_F}_{\text{field}} + \underbrace{\hat{H}_{int}}_{\text{interaction}}, \tag{4.18}$$

where

$$\hat{H}_A = \hbar\omega_0\,|e\rangle\,\langle e| = \hbar\omega_0\hat{\sigma}_+\hat{\sigma}_0, \tag{4.19}$$

$$\hat{H}_F = \sum_{\boldsymbol{k},\lambda}\hbar\omega_{\boldsymbol{k}}\left(\hat{a}^\dagger_{\boldsymbol{k},\lambda}\hat{a}_{\boldsymbol{k},\lambda} + \frac{1}{2}\right), \quad [\hat{a}_{\boldsymbol{k},\lambda}, \hat{a}^\dagger_{\boldsymbol{k},\lambda}] = 1, \tag{4.20}$$

$$\hat{H}_{int} = -\hat{\boldsymbol{d}}\cdot\hat{\boldsymbol{E}}(\boldsymbol{r}_A), \quad \begin{array}{l} \hat{\boldsymbol{d}} = \boldsymbol{d}\,|e\rangle\,\langle g| + \boldsymbol{d}^*\,|g\rangle\,\langle e| = \boldsymbol{d}\hat{\sigma}_+ + \boldsymbol{d}^*\hat{\sigma}_-, \\ \hat{\boldsymbol{E}}(\boldsymbol{r}_A) = i\sum_{\boldsymbol{k},\lambda}\hat{\boldsymbol{e}}_{\boldsymbol{k},\lambda}\sqrt{\frac{\hbar\omega_{\boldsymbol{k}}}{2\varepsilon_0 V}}[\hat{a}_{\boldsymbol{k},\lambda}e^{i\boldsymbol{k}\cdot\boldsymbol{r}_A} - \hat{a}^\dagger_{\boldsymbol{k},\lambda}e^{-i\boldsymbol{k}\cdot\boldsymbol{r}_A}]. \end{array} \tag{4.21}$$

We now move to the interaction picture where we define $H_{tot} = H_0 + H_{int}$. In Schrodinger picture,

$$i\hbar\partial_t\,|\psi\rangle_s = (H_0 + H_{int})\,|\psi\rangle_s$$

In the interaction picture we define

$$|\psi\rangle_I = e^{iH_0t/\hbar}\,|\psi\rangle_s\,,$$

and

$$\begin{aligned} i\hbar\partial_t\,|\psi\rangle_I &= i\hbar\partial_t[e^{iH_0t/\hbar}\,|\psi\rangle_s] = i\hbar\frac{iH_0}{\hbar}e^{iH_0t/\hbar}\,|\psi\rangle_s + i\hbar e^{iH_0t/\hbar}\partial_t\,|\psi\rangle_s \\ &= -H_0\cancel{e^{iH_0t/\hbar}}\,|\psi\rangle_s + e^{iH_0t/\hbar}(\cancel{H_0} + H_{int})\,|\psi\rangle_s = \underbrace{e^{iH_0t/\hbar}H_{int}e^{-iH_0t/\hbar}}_{H_{int,T}\text{ or }\tilde{H}_{int}}\underbrace{e^{iH_0t/\hbar}\,|\psi\rangle_s}_{|\psi\rangle_I} \\ i\hbar\partial_t\,|\psi\rangle_I &= \tilde{H}_{int}\,|\psi\rangle_I\,. \end{aligned}$$

Let us consider the interaction Hamiltonian in interaction picture:

$$\tilde{H}_{int} = e^{iH_0t/\hbar}\hat{H}_{int}e^{-iH_0t/\hbar} = e^{iH_0t/\hbar}[-\hat{\boldsymbol{d}}\cdot\hat{\boldsymbol{E}}(\boldsymbol{r}_A)]e^{-iH_0t/\hbar}$$

$$\tilde{H}_{int} = -e^{i(H_A+H_F)t/\hbar}(\boldsymbol{d}\hat{\sigma}_+ + \boldsymbol{d}^*\hat{\sigma}_-)\cdot i\sum_{\boldsymbol{k},\lambda}\hat{\boldsymbol{e}}_{\boldsymbol{k},\lambda}\sqrt{\frac{\hbar\omega_{\boldsymbol{k}}}{2\varepsilon_0 V}}[\hat{a}_{\boldsymbol{k},\lambda}e^{i\boldsymbol{k}\cdot\boldsymbol{r}} - \hat{a}^\dagger_{\boldsymbol{k},\lambda}e^{-i\boldsymbol{k}\cdot\boldsymbol{r}}]e^{-i(H_A+H_F)t/\hbar}.$$

In solving the above, we encounter

$$e^{iH_At/\hbar}\hat{\sigma}_+e^{-iH_At/\hbar} = \hat{\sigma}_+e^{i\omega_0t}, \quad e^{iH_At/\hbar}\hat{\sigma}_-e^{-iH_At/\hbar} = \hat{\sigma}_-e^{-i\omega_0t},$$
$$e^{iH_Ft/\hbar}\hat{a}_{\boldsymbol{k},\lambda}e^{-iH_Ft/\hbar} = \hat{a}_{\boldsymbol{k},\lambda}e^{-i\omega_{\boldsymbol{k}}t}, \quad e^{iH_Ft/\hbar}\hat{a}^\dagger_{\boldsymbol{k},\lambda}e^{-iH_Ft/\hbar} = \hat{a}^\dagger_{\boldsymbol{k},\lambda}e^{i\omega_{\boldsymbol{k}}t}.$$

We then have:

$$\tilde{H}_{int} = -(\boldsymbol{d}\hat{\sigma}_+e^{i\omega_0t} + \boldsymbol{d}^*\hat{\sigma}_-e^{-i\omega_0t})\cdot i\sum_{\boldsymbol{k},\lambda}\hat{\boldsymbol{e}}_{\boldsymbol{k},\lambda}\sqrt{\frac{\hbar\omega_{\boldsymbol{k}}}{2\varepsilon_0 V}}\left[\hat{a}_{\boldsymbol{k},\lambda}e^{i(\boldsymbol{k}\cdot\boldsymbol{r}-\omega_{\boldsymbol{k}}t)} - \hat{a}^\dagger_{\boldsymbol{k},\lambda}e^{-i(\boldsymbol{k}\cdot\boldsymbol{r}-\omega_{\boldsymbol{k}}t)}\right].$$

Rotating wave approximation (RWA): Terms $e^{\pm i(\omega_0+\omega_n)t}$ oscillate on timescales of $\sim 10^{-15}\ s$. On typical observation timescales $\sim 1\ ns$ and it contains about $(\omega_0+\omega_{\boldsymbol{k}})t \sim 10^{-15}\cdot 10^{-9} \sim 10^8$ oscillations. Thus, these fast-oscilating terms averages to zero.

Under the RWA, the interaction Hamiltonian becomes:

$$\tilde{H}_{int} \approx \sum_{\boldsymbol{k},\lambda} -i\boldsymbol{d}\cdot\hat{\boldsymbol{e}}_{\boldsymbol{k},\lambda}\sqrt{\frac{\hbar\omega_{\boldsymbol{k}}}{2\varepsilon_0 V}}\left[\hat{\sigma}_+\hat{a}_{\boldsymbol{k},\lambda}e^{i\boldsymbol{k}\cdot\boldsymbol{r}}e^{-i(\omega_{\boldsymbol{k}}-\omega_0)t} - \hat{\sigma}_-\hat{a}^\dagger_{\boldsymbol{k},\lambda}e^{-i\boldsymbol{k}\cdot\boldsymbol{r}}e^{i(\omega_{\boldsymbol{k}}-\omega_0)t}\right].$$

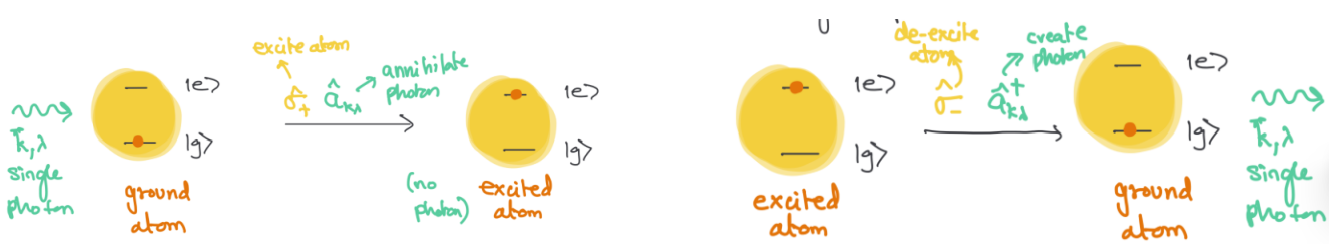
Defining

$$\hbar g_{\boldsymbol{k},\lambda} = -i\boldsymbol{d}\cdot\hat{\boldsymbol{e}}_{\boldsymbol{k},\lambda}\sqrt{\frac{\hbar\omega_{\boldsymbol{k}}}{2\varepsilon_0 V}}e^{i\boldsymbol{k}\cdot\boldsymbol{r}},$$

we finally have:

Interaction Hamiltonian
$$\tilde{H}_{int} \approx \hbar\sum_{\boldsymbol{k},\lambda} g_{\boldsymbol{k},\lambda}\hat{\sigma}_+\hat{a}_{\boldsymbol{k},\lambda}e^{-i(\omega_{\boldsymbol{k}}-\omega_0)t} + g^*_{\boldsymbol{k},\lambda}\hat{\sigma}_-\hat{a}^\dagger_{\boldsymbol{k},\lambda}e^{i(\omega_{\boldsymbol{k}}-\omega_0)t}. \tag{4.22}$$

There the first term of the type $\hat{\sigma}_+\hat{a}_{\boldsymbol{k},\lambda}$ represents the process where $\hat{\sigma}_+ = |e\rangle\langle g|$ takes a ground state atom to its excited state and $\hat{a}_{\boldsymbol{k},\lambda}$ annihilates a photon. Similarly, the second term represents the fundamental process of emission of a single photon from an excited atom.



(a) First term **(b)** Second term

Figure 4.1 Process of interaction light-matter.

4.4 Spontaneous emission

The initial state of the system in the figure is:

$$|\psi\rangle = |e\rangle \otimes |\{0\}\rangle\,, \quad |\{0\}\rangle = \text{All field modes in vacuum state.} \tag{4.23}$$

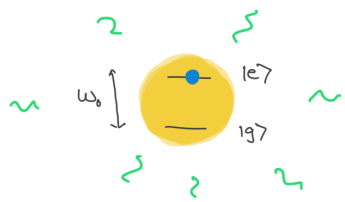


Figure 4.1 Initial state

We observe that the total Hamilonian in the SP

$$H_{tot} = \hbar\omega_0\hat{\sigma}_+\hat{\sigma}_- + \sum_{\boldsymbol{k},\lambda}\hbar\omega_{\boldsymbol{k}}\hat{a}^\dagger_{\boldsymbol{k},\lambda}\hat{a}_{\boldsymbol{k},\lambda} + \hbar\sum_{\boldsymbol{k},\lambda}(g_{\boldsymbol{k},\lambda}\hat{\sigma}_+\hat{a}_{\boldsymbol{k},\lambda} + g^*_{\boldsymbol{k},\lambda}\hat{\sigma}_-\hat{a}^\dagger_{\boldsymbol{k},\lambda}).$$

preserves the total number of exctiations in atom + field supbspaces

$$\hat{N}_{AF} = \hat{\sigma}_+\hat{\sigma}_- + \sum_{\boldsymbol{k},\lambda}\hat{a}^\dagger_{\boldsymbol{k},\lambda}\hat{a}_{\boldsymbol{k},\lambda}$$

$$[\hat{N}_{AF}, \hat{H}_{tot}] = 0 \Longrightarrow \begin{array}{c}\text{Total number of atoms+field exctiations is}\\ \text{preserved by the dynamics under } H_{tot}\end{array}$$

We can thus maeke the following onsate for th eatom+field stat at any time t:

$$|\psi(t)\rangle = c_e(t)\,|\{0\}\rangle + \sum_{\boldsymbol{k},\lambda} c_{g,\boldsymbol{k},\lambda}(t)\,|g\rangle\,|1_{\boldsymbol{k},\lambda}\rangle \tag{4.24}$$

Let us qrite the dynamics of the quantum state in the interaction picture:

$$i\hbar\partial_t\,|\psi(t)\rangle = \tilde{H}_{int}\,|\psi(t)\rangle$$

$$i\hbar\partial_t[c_e(t)\,|\{0\}\rangle + \sum_{\boldsymbol{k},\lambda} c_{g,\boldsymbol{k},\lambda}(t)\,|g\rangle\,|1_{\boldsymbol{k},\lambda}\rangle] = \left[\sum_{\boldsymbol{k},\lambda} g_{\boldsymbol{k},\lambda}\hat{\sigma}_+\hat{a}_{\boldsymbol{k},\lambda}e^{-i(\omega_{\boldsymbol{k}}-\omega_0)t} + g^*_{\boldsymbol{k},\lambda}\hat{\sigma}_-\hat{a}^\dagger_{\boldsymbol{k},\lambda}e^{i(\omega_{\boldsymbol{k}}-\omega_0)t}\right]$$
$$\left[c_e(t)\,|e\rangle\,|\{0\}\rangle + \sum_{\boldsymbol{k}',\lambda'} c_{g,\boldsymbol{k}',\lambda'}(t)\,|g\rangle\,\left|1_{\boldsymbol{k}',\lambda'}\right\rangle\right]$$

$$i\hbar\partial_t c_e\,|e\rangle\,|\{0\}\rangle + i\hbar\sum_{\boldsymbol{k},\lambda}\partial_t c_{g,\boldsymbol{k},\lambda}\,|g\rangle\,|1_{\boldsymbol{k},\lambda}\rangle \overset{(a)}{=} \hbar\sum_{\boldsymbol{k},\lambda}\left(g_{\boldsymbol{k},\lambda}c_{g,\boldsymbol{k},\lambda}e^{-i(\omega_{\boldsymbol{k}}-\omega_0)t}\,|e\rangle\,|\{0\}\rangle + g^*_{\boldsymbol{k},\lambda}c_e e^{i(\omega_{\boldsymbol{k}}-\omega_0)t}\,|g\rangle\,|1_{\boldsymbol{k},\lambda}\rangle\right)$$

In (a), we used the fact that $\hat{a}_{\boldsymbol{k},\lambda}\left|1_{\boldsymbol{k}',\lambda'}\right\rangle = \delta^3(\boldsymbol{k}-\boldsymbol{k}')\delta_{\lambda\lambda'}$. Comparing coefficients for $|e\rangle\,|\{0\}\rangle$ and $|g\rangle\,|1_{\boldsymbol{k},\lambda}\rangle$:

$$i\partial_t c_e = \sum_{\boldsymbol{k},\lambda} g_{\boldsymbol{k},\lambda}c_{g,\boldsymbol{k},\lambda}e^{-i(\omega_{\boldsymbol{k}}-\omega_0)t}$$

$$i\partial_t c_{g,\boldsymbol{k},\lambda} = g^*_{\boldsymbol{k},\lambda}c_e e^{i(\omega_{\boldsymbol{k}}-\omega_0)t}$$

We are interested in $c_e(t)$, so we will eliminate $c_{g,\boldsymbol{k},\lambda}(t)$ from the equation of motion for the atom:

$$i[c_{g,\boldsymbol{k},\lambda}(t) - \cancelto{0}{c_{g,\boldsymbol{k},\lambda}(0)}] = g^*_{\boldsymbol{k},\lambda} \int_0^t d\tau \; c_e(\tau) e^{i(\omega_{\boldsymbol{k}}-\omega_0)\tau}, \tag{4.25}$$

where $c_{g,\boldsymbol{k},\lambda}(0) = 0$ as there are no excitation in the field in the beginning. Substituting this coeficient in the equation of motion of the atom:

$$\begin{aligned} i\partial_t c_e &= -i \sum_{\boldsymbol{k},\lambda} |g_{\boldsymbol{k},\lambda}|^2 \int_0^t d\tau \; c_e(\tau) e^{-i(\omega_{\boldsymbol{k}}-\omega_0)(t-\tau)} \\ \partial_t c_e &= -\sum_{\boldsymbol{k},\lambda} |g_{\boldsymbol{k},\lambda}|^2 \int_0^t d\tau' \; e^{-i(\omega_{\boldsymbol{k}}-\omega_0)\tau'} c_e(t-\tau') \quad (\tau' = t-\tau) \end{aligned}$$

At this point we make the *Markov approximation* which means that the system evolution at any time is determined by its present state $c_e(t)$ (memoryless system). Thus, aproximating $c_e(t-\tau') \to c_e(t)$ in the integral, we obtain:

$$\partial_t c_e \approx -c_e(t) \sum_{\boldsymbol{k},\lambda} |g_{\boldsymbol{k},\lambda}|^2 \int_0^t d\tau \; e^{-i(\omega_{\boldsymbol{k}}-\omega_0)(t-\tau)} = -\frac{\gamma_0}{2} c_e(t).$$

And finally,

$$c_e(t) = e^{-\gamma_0 t/2} c_e(0) \Longrightarrow |c_e(t)|^2 = e^{-\gamma_0 t}, \tag{4.26}$$

which is an exponential decay of an excited atom probability at a rate γ_0.

4.4.1 Interaction Hamiltonian in the interaction picture

We have:

$$\tilde{H}_{int} = \sum_{\boldsymbol{k},\lambda} \hbar g_{\boldsymbol{k}\lambda} \left[\hat{\sigma}_+ \hat{a}_{\boldsymbol{k}\lambda} e^{-i(\omega-\omega_0)t} + \hat{\sigma}_- \hat{a}^\dagger_{\boldsymbol{k}\lambda} e^{i(\omega-\omega_0)t} \right]. \tag{4.27}$$

From Schrodinger equation:

$$\frac{d}{dt} |\Psi(t)\rangle = -\frac{i}{\hbar} \tilde{H}_{int} |\Psi(t)\rangle .$$

The state of the system at any time t is:

$$|\Psi(t)\rangle = c_e(t) |e\rangle |\{0\}\rangle + \sum_{\boldsymbol{k},\lambda} c_{\boldsymbol{k}\lambda}(t) |g\rangle |1_{\boldsymbol{k}\lambda}\rangle .$$

There is at most one excitation in the total atom+field system since the Hamiltonian preserves the total number of excitations. Substituting $|\Psi(t)\rangle$ in the Schrodinger equation yields

$$\frac{dc_e}{dt} = -i \sum_{\boldsymbol{k},\lambda} g_{\boldsymbol{k}\lambda} c_{\boldsymbol{k}\lambda}(t) e^{-i(\omega_{\boldsymbol{k}}-\omega_0)t}, \quad \frac{dc_{\boldsymbol{k}\lambda}}{dt} = -i g^*_{\boldsymbol{k}\lambda}(t) c_e(t) e^{i(\omega_k-\omega_0)t}.$$

From substituting

$$c_{\boldsymbol{k}\lambda}(t) = -i g^*_{\boldsymbol{k}\lambda} \int_0^t c_e(t) e^{i(\omega_{\boldsymbol{k}}-\omega_0)\tau} \, d\tau$$

in the equation of motion for c_e:

$$\frac{dc_e}{dt} = -\int_0^t d\tau \sum_{\boldsymbol{k},\lambda} |g_{\boldsymbol{k}\lambda}|^2 c_e(\tau) e^{-i(\omega_{\boldsymbol{k}}-\omega_0)(t-\tau)}, \quad \frac{dc_e}{dt} = -c_e(t) \sum_{\boldsymbol{k},\lambda} |g_{\boldsymbol{k}\lambda}|^2 \int_0^\infty d\tau \; e^{-i(\omega_{\boldsymbol{k}}-\omega_0)(t-\tau)}.$$

We now sum over the continuum of modes in free space

$$\sum_{\boldsymbol{k},\lambda} \to \sum_{\lambda} \int d^2k \; \frac{V}{(2\pi)^3} \tag{4.28}$$

$$\frac{\gamma_0}{2} = -\frac{1}{\hbar^2} \sum_{\lambda} \int d^3k \; \frac{V}{(2\pi)^3} |\boldsymbol{d} \cdot \hat{\boldsymbol{e}}_{\boldsymbol{k}\lambda}|^2 \frac{\hbar\omega_{\boldsymbol{k}}}{2\varepsilon_0 V} \left[\pi\delta(\omega-\omega_0) - iP\frac{i}{\omega-\omega_0} \right].$$

In free space $\omega_{\boldsymbol{k}} = kc$. Also, the orientation of the atomic dipole $\boldsymbol{d}$ can be chosen arbitrarily. We choose the atomic dipole to be oriented along $\hat{\boldsymbol{z}}$, i.e., $\boldsymbol{d} = d\hat{\boldsymbol{z}}$.

$$\begin{aligned}\frac{\gamma_0}{2} &= -\frac{1}{16\pi^2\hbar\varepsilon_0} \sum_{\lambda} \int_0^\infty dk \; k^2 \int_0^\pi d\theta \; \sin\theta \int_0^{2\pi} d\phi \; d^2 |\hat{\boldsymbol{z}} \cdot \hat{\boldsymbol{e}}_{\boldsymbol{k}\lambda}|^2 kc\delta(\omega_{\boldsymbol{k}}-\omega_0) \\ &= -\frac{d^2c}{16\pi^2\hbar\varepsilon_0} \int_0^\infty dk \; k^3 \int_0^\pi d\theta \; \sin\theta \int_0^{2\pi} d\phi \left(\sum_{\lambda} |\hat{\boldsymbol{z}} \cdot \hat{\boldsymbol{e}}_{\boldsymbol{k}\lambda}|^2 \right) \delta(\omega_{\boldsymbol{k}}-\omega_0).\end{aligned}$$

Geometric calculation of $\hat{z} \cdot \hat{e}_{k\lambda}$

$$\boldsymbol{d} = d\{0,0,1\}, \quad \boldsymbol{k} = k\{\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta\}, \quad \begin{aligned}\hat{\boldsymbol{e}}_{\boldsymbol{k}\lambda_1} &= \{\cos\theta\cos\phi, \cos\theta\sin\phi, -\sin\theta\} \\ \hat{\boldsymbol{e}}_{\boldsymbol{k}\lambda_2} &= \{-\sin\phi, \cos\phi, 0\}\end{aligned}$$

Therefore,

$$\hat{\boldsymbol{z}} \cdot \hat{\boldsymbol{e}}_{\boldsymbol{k}\lambda} = \begin{cases} \hat{\boldsymbol{z}} \cdot \hat{\boldsymbol{e}}_{\boldsymbol{k}\lambda_1} = -\sin\theta \\ \hat{\boldsymbol{z}} \cdot \hat{\boldsymbol{e}}_{\boldsymbol{k}\lambda_2} = 0 \end{cases}.$$

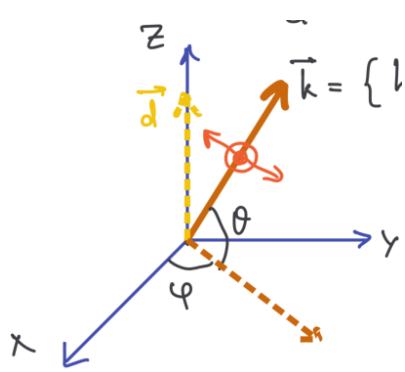


Figure 4.2 Geometric calculation of $\hat{\boldsymbol{z}} \cdot \hat{\boldsymbol{e}}_{\boldsymbol{k}\lambda}$.

Then,

$$\begin{aligned}\frac{\gamma_0}{2} &= -\frac{d^2c}{16\pi^2\hbar\varepsilon_0}\int_0^\infty dk\ k^2\int_0^\pi d\theta\ \sin^3\theta\int_0^{2\pi}d\phi\ \delta(\omega_{\boldsymbol{k}}-\omega_0)\\ &= -\frac{d^2c}{16\pi^2\hbar\varepsilon_0}2\pi\frac{4}{3}\frac{\omega_0^3}{c^4} = -\frac{d^2k_0^3}{6\pi^2\hbar\varepsilon_0}.\end{aligned}$$

$$\frac{dce}{dt} = -\frac{\gamma_0}{2}c_e(t) \longrightarrow c_e(t) = c_e(0)e^{-i\gamma_0 t/2}.$$

Spontaneous emitted radiation

Substituting in the field coefficient expression:

$$\begin{aligned}c_{\boldsymbol{k}\lambda}(t) &= -i\int_0^t d\tau\ g^*_{\boldsymbol{k}\lambda}c_e(\tau)e^{i(\omega-\omega_0)\tau} = -i\int_0^t d\tau\ g^*_{\boldsymbol{k}\lambda}e^{-\gamma\tau/2}e^{i(\omega-\omega_0)\tau}\\ &= -ig^*_{\boldsymbol{k}\lambda}\frac{e^{[-\gamma/2+i(\omega-\omega_0)]\tau}}{-\gamma/2+i(\omega-\omega_0)}\bigg|_0^t = -ig^*_{\boldsymbol{k}\lambda}\frac{e^{[-\gamma/2+i(\omega-\omega_0)]t}-1}{-\gamma/2+i(\omega-\omega_0)}.\end{aligned}$$

The value sqaured is:

$$|c_{\boldsymbol{k}\lambda}(t)|^2 = |g_{\boldsymbol{k}\lambda}|^2\frac{e^{-\gamma t}-2e^{-\gamma t/2}\cos[(\omega-\omega_0)t]+1}{(\gamma/2)^2+(\omega-\omega_0)^2}.$$

Then,

$$\begin{aligned}t=0: &\qquad |c_{\boldsymbol{k}\lambda}(0)|^2 = 0\\ t\to\infty: &\qquad |c_{\boldsymbol{k}\lambda}|^2 \to \frac{|g_{\boldsymbol{k}\lambda}|^2}{(\gamma/2)^2+(\omega-\omega_0)^2} = \text{Lorentzian with width } \gamma/2\end{aligned}$$

Finally, the emitted E-field is:

$$\hat{E}(\boldsymbol{r},t) = i\sum_{\boldsymbol{k},\lambda}\mathcal{E}_0\left[\hat{a}_{\boldsymbol{k}\lambda}e^{-i(\boldsymbol{k}\cdot\boldsymbol{r}-\omega t)} - \hat{a}^\dagger_{\boldsymbol{k}\lambda}e^{i(\boldsymbol{k}\cdot\boldsymbol{r}-\omega t)}\right].$$

The average value of the E-field over the $\Psi(t)$ is:

$$\langle\Psi(t)|\hat{E}(\boldsymbol{r},t)|\Psi(t)\rangle\sum\langle e,0_{\boldsymbol{k}\lambda}|\hat{a}_{\boldsymbol{k}\lambda}|g,1_{\boldsymbol{k}\lambda}\rangle = 0.$$

4.5 Driving the atom with a laser

Se consider how a laser drives a two-level atom. Note that for simplicity we will ignore the interaction of the atom with the quantized EM field modes- this is for us to be able to focus on the atom-laser interaction (even though the quantized field and its effects are always present).

The Hamiltonian for the total system is:

$$H = H_A + H_{AL}, \qquad \begin{aligned}H_A &= \hbar\omega_0\hat{\sigma}_+\hat{\sigma}_-\\ H_{AL} &= -\hat{\boldsymbol{d}}\cdot\boldsymbol{E}_L(\boldsymbol{r}_A,t)\end{aligned} \tag{4.29}$$

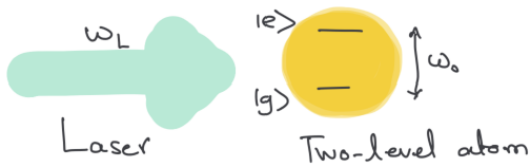


Figure 4.1 Driving the atom (two-level) with a laser.

The E-field is:

$$\hat{E}_L(\boldsymbol{r}_A, t) = \mathcal{E}_L e^{-i(\omega_L t - \boldsymbol{k}_L \cdot \boldsymbol{r}_A)} + \mathcal{E}_L^* e^{i(\omega_L t - \boldsymbol{k}_L \cdot \boldsymbol{r}_A)}$$

$$H_{AL} = -\left[\boldsymbol{d}\hat{\sigma}_+ + \boldsymbol{d}^*\hat{\sigma}_-\right] \cdot \left[\mathcal{E}_L e^{-i(\omega_L t - \boldsymbol{k}_L \cdot \boldsymbol{r}_A)} + \mathcal{E}_L^* e^{i(\omega_L t - \boldsymbol{k}_L \cdot \boldsymbol{r}_A)}\right].$$

Moving to the interaction picture with respect to H_A, assuming $\boldsymbol{r}_A = 0$

$$\tilde{\sigma}_\pm = e^{iH_A t/\hbar}\hat{\sigma}_\pm e^{-iH_A t/\hbar} = \hat{\sigma}_\pm e^{\pm i\omega_0 t}. \tag{4.30}$$

The Hamiltonian in the interaction picture is:

$$\tilde{H}_{AL} = e^{iH_A t/\hbar} H_{AL} e^{-iH_A t/\hbar} = -\left[\boldsymbol{d}\hat{\sigma}_+ e^{i\omega_0 t} + \boldsymbol{d}^*\hat{\sigma}_- e^{-i\omega_0 t}\right] \cdot \left[\mathcal{E}_L e^{-i\omega_L t} + \mathcal{E}_L^* e^{i\omega_L t}\right].$$

Making the RWA:

$$\tilde{H}_{AL} = -\left[\boldsymbol{d} \cdot \mathcal{E}_L \hat{\sigma}_+ e^{i(\omega_0 - \omega_L)t} + \boldsymbol{d}^* \cdot \mathcal{E}_l^* \hat{\sigma}_- e^{-i(\omega_0 - \omega_L)t}\right] \approx \frac{\hbar}{2}(\Omega\hat{\sigma}_+ e^{-i\Delta_L t} + \Omega^*\hat{\sigma}_- e^{i\Delta_L t}).$$

We identify two things:

$$\text{Raby frequency}: \quad \frac{1}{2}\hbar\Omega = -\boldsymbol{d} \cdot \mathcal{E}_l, \quad \text{and} \quad \text{Detuning}: \quad \Delta_L = \omega_L - \omega_0. \tag{4.31}$$

Let us consider the evolution of the atomic state via the Schrodinger equation:

$$\begin{aligned}
i\hbar\partial_t \left|\Psi(t)\right\rangle_I &= \tilde{H}_{AL}\left|\Psi(t)\right\rangle_I \\
i\hbar\frac{d}{dt}\left[c_g(t)\left|g\right\rangle + c_e(t)\left|e\right\rangle\right] &= \frac{\hbar}{2}\left[\Omega\hat{\sigma}_+ e^{-i\Delta_L t} + \Omega^*\hat{\sigma}_- e^{i\Delta_L t}\right] \cdot \left[c_g(t)\left|g\right\rangle + c_e(t)\left|e\right\rangle\right] \\
i\left[\frac{dc_g}{dt}\left|g\right\rangle + \frac{dc_e}{dt}\left|e\right\rangle\right] &= \frac{\Omega}{2}c_g e^{-i\Delta_L t}\left|e\right\rangle + \frac{\Omega^*}{2}c_e e^{i\Delta_L t}\left|g\right\rangle, \quad (\hat{\sigma}_+\left|g\right\rangle = \left|e\right\rangle, \hat{\sigma}_-\left|e\right\rangle = \left|g\right\rangle).
\end{aligned}$$

Equating coefficients of $|g\rangle$ and $|e\rangle$ yields:

$$\text{Equation of motions} \quad \boxed{\begin{aligned}\frac{dc_g}{dt} &= -i\frac{\Omega^*}{2}c_e e^{i\Delta_L t} \\ \frac{dc_e}{dt} &= -i\frac{\Omega}{2}c_g e^{-i\Delta_L t}\end{aligned}} \tag{4.32}$$

Let us first consider the case when the laser is at resonance with the atomic transition frequency, that is, $\Delta_L = 0$.

$$\frac{dc_g}{dt} = -i\frac{\Omega^*}{2}c_e, \ \frac{dc_e}{dt} = -i\frac{\Omega}{2}c_g \Longrightarrow \frac{d^2c_e}{dt^2} = -\frac{|\Omega|^2}{4}c_e.$$

The solution of this expression is:

$$c_e(t) = A\cos\left(\frac{|\Omega|t}{2}\right) + B\sin\left(\frac{|\Omega|t}{2}\right). \tag{4.33}$$

The coefficients are found as:

$$\begin{aligned} t = 0: &\quad c_e(0) = A \\ t = 0': &\quad \frac{dc_e}{dt}(0) = -i\frac{\Omega}{2}c_g(0) = \frac{B|\Omega|}{2} \longrightarrow B = -i\frac{\Omega}{|\Omega|}c_g(0). \end{aligned}$$

The probability amplitude for being in the excited and ground states are:

$$c_e(t) = c_e(0)\cos\left(\frac{|\Omega|t}{2}\right) - \frac{i\Omega}{|\Omega|}c_g(0)\sin\left(\frac{|\Omega|t}{2}\right), \tag{4.34}$$

$$c_g(t) = c_g(0)\cos\left(\frac{|\Omega|t}{2}\right) - \frac{i\Omega^*}{|\Omega|}c_e(0)\sin\left(\frac{|\Omega|t}{2}\right) \tag{4.35}$$

This is the general dynamics of a laser-driven two-level atom.

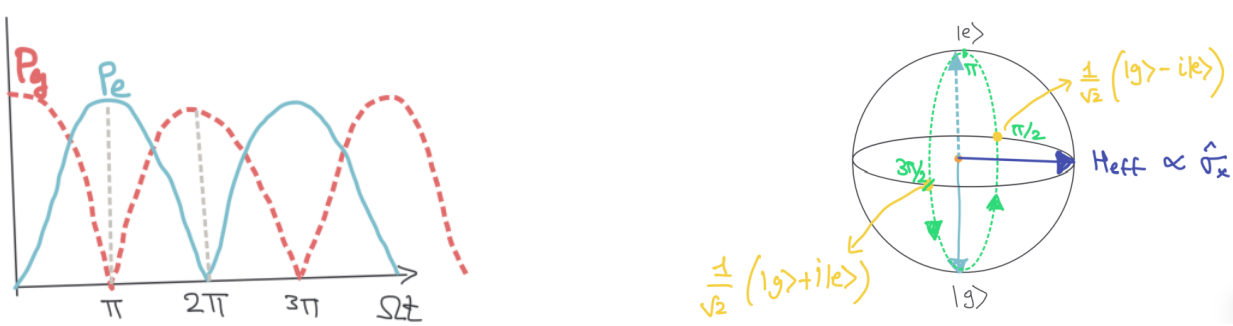
Let us now considee that the atom is initially in the ground state and $\Omega = \Omega^* = |\Omega|$:

$$c_g(t) = \cos\left(\frac{\Omega t}{2}\right), \quad c_e(t) = -i\sin\left(\frac{\Omega t}{2}\right).$$

Probability of finding the atom in the ground or excited states are:

$$P_g = |c_g(t)|^2 = \cos^2\left(\frac{|\Omega|t}{2}\right), \quad P_e = |c_e(t)|^2 = \sin^2\left(\frac{|\Omega|t}{2}\right).$$

The following figure illustrates the evolution of ground and excited states. We see that population of the atom oscilatess between those levels at a frequency Ω, called **Rabi frequency**. This phenomenon is called **Rabi flopping**.



(a) Time evolution of states

(b) Bloch sphere

Figure 4.2 Rabi flopping from initial ground state.

Inceease in excited state population corresponds to laser absorption and decrease in excited state implies stimulatred emission. In particular,

- $\Omega t = \pi$ The atom in the ground state ends up in the excited state: π-pulse.
- $\Omega t = \pi/2$ The atom ends up in a superposition of $(1/\sqrt{2})(|g\rangle - i\,|e\rangle)$: $\pi/2$-pulse.

We can depict this evolution of the atomic state in the BLoch sphere. The state preceses around the Hamiltonian vector H_{eff} with a frequency $|\Omega|/2$.

4.5.1 Dressed states

Let us look at the equations of motion:

$$\begin{aligned}\frac{dc_g}{dt} &= -i\frac{\Omega^*}{2}c_e e^{i\Delta_L t} \\ \frac{dc_e}{dt} &= -i\frac{\Omega}{2}c_g e^{-i\Delta_L t}\end{aligned} \longrightarrow \begin{aligned}\frac{dc_g}{dt}e^{-i\Delta_L t/2} &= -i\frac{\Omega^*}{2}c_e e^{i\Delta_L t/2} \\ \frac{dc_e}{dt}e^{i\Delta_L t/2} &= -i\frac{\Omega}{2}c_g e^{-i\Delta_L t/2}\end{aligned}.$$

For a finite detuning Δ_L, let us define

$$\tilde{c}_e = c_e e^{i\Delta_L t/2}, \quad \tilde{c}_g = c_g e^{-i\Delta_L t/2}. \tag{4.36}$$

This allows us to rewrite the coupled differential equations without any explicit time dependence:

$$\begin{aligned}\frac{d\tilde{c}_e}{dt} &= \frac{dc_e}{dt}e^{i\Delta_L t/2} + \frac{i\Delta_L}{2}c_e e^{i\Delta_L t/2} = -\frac{i\Omega}{2}c_g e^{i\Delta_L t/2} + \frac{i\Delta_L}{2}\tilde{c}_e = -\frac{i\Omega}{2}\tilde{c}_g + \frac{i\Delta_L}{2}\tilde{c}_e \\ \frac{d\tilde{c}_g}{dt} &= \frac{dc_g}{dt}e^{-i\Delta_L t/2} - \frac{i\Delta_L}{2}c_g e^{-i\Delta_L t/2} = -\frac{i\Omega}{2}\tilde{c}_e - \frac{i\Delta_L}{2}\tilde{c}_g.\end{aligned}$$

Then,

$$\frac{d}{dt}\begin{bmatrix}\tilde{c}_e \\ \tilde{c}_g\end{bmatrix} = -\frac{i}{\hbar}\begin{bmatrix}-\hbar\Delta_L/2 & \hbar\Omega/2 \\ \hbar\Omega/2 & \hbar\Delta_L/2\end{bmatrix}\begin{bmatrix}\tilde{c}_e \\ \tilde{c}_g\end{bmatrix} \Longrightarrow \frac{d}{dt}\boldsymbol{c} = -\frac{i}{\hbar}H_{\text{eff}}\boldsymbol{c}. \tag{4.37}$$

One can diagonalize the above Hamiltonian such that

$$V\frac{d\boldsymbol{c}}{dt} = -\frac{i}{\hbar}\underbrace{VH_{eff}V^{-1}}_{H_{eff,D}}\underbrace{V\boldsymbol{c}}_{\text{New eigenstates}}$$

$$\frac{d}{dt}\underbrace{\begin{bmatrix}C_+ \\ C_-\end{bmatrix}}_{V\boldsymbol{c}} = -\frac{i}{\hbar}\begin{bmatrix}E_+ & 0 \\ 0 & E_-\end{bmatrix}\underbrace{\begin{bmatrix}C_+ \\ C_-\end{bmatrix}}_{V\boldsymbol{c}}.$$

Then, the dressed eigenstates are:

$$\text{Dessed eigenstates} \qquad \begin{aligned}|+\rangle &= \cos\theta\,|e\rangle + \sin\theta\,|g\rangle \\ |-\rangle &= -\sin\theta\,|e\rangle + \cos\theta\,|g\rangle\end{aligned}, \quad \tan 2\theta = -\frac{\Omega}{\Delta_L}. \tag{4.38}$$

The atomic state expressed in dresses basis:

$$|\Psi(t)\rangle = c_+|+\rangle + c_-|-\rangle, \quad c_\pm(t) = c_\pm(0)e^{-iE_\pm t/\hbar}, \quad E_\pm = \pm\frac{\hbar}{2}\sqrt{\Delta_L^2 + \Omega^2}.$$

A qualitative evolution is given in the following figure.

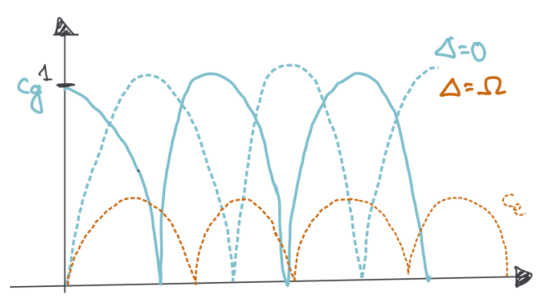
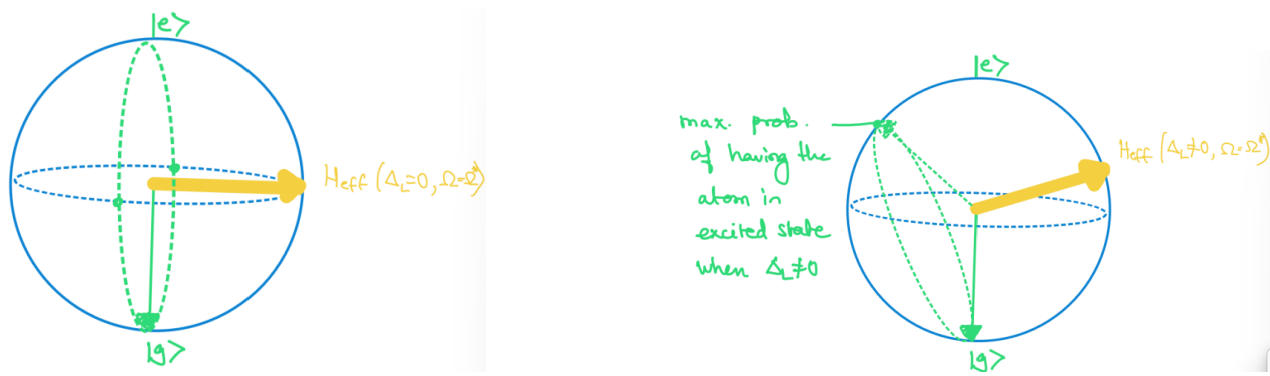


Figure 4.3 Evolution of c_g and c_e for different detuning values Δ_L.

For a detuned laser the atom does not get fully excited. To see this qualitative dynamics we look at the effective Hamiltonian:

$$H_{eff} = \begin{bmatrix} -\Delta_L/2 & \Omega/2 \\ \Omega^*/2 & \Delta_L/2 \end{bmatrix} = -\frac{\Delta_L}{2}\hat{\sigma}_z + \frac{\mathrm{Re}\,(\Omega)}{2}\hat{\sigma}_x - \frac{\mathrm{Im}\,(\Omega)}{2}\hat{\sigma}_y. \tag{4.39}$$

Let us depict this Hamiltonian and the initial state on the Bloch sphere.



4.5.2 Dressed states and energies variation with Δ_L

The angle θ is:

$$\tan 2\theta = -\frac{\Omega}{\Delta_L} \longrightarrow 2\theta = -\underbrace{\tan^{-1}\left(\frac{\Omega}{\Delta_L}\right)}_{[0,\pi]}$$

Then, for different ragions of the detuning we have:

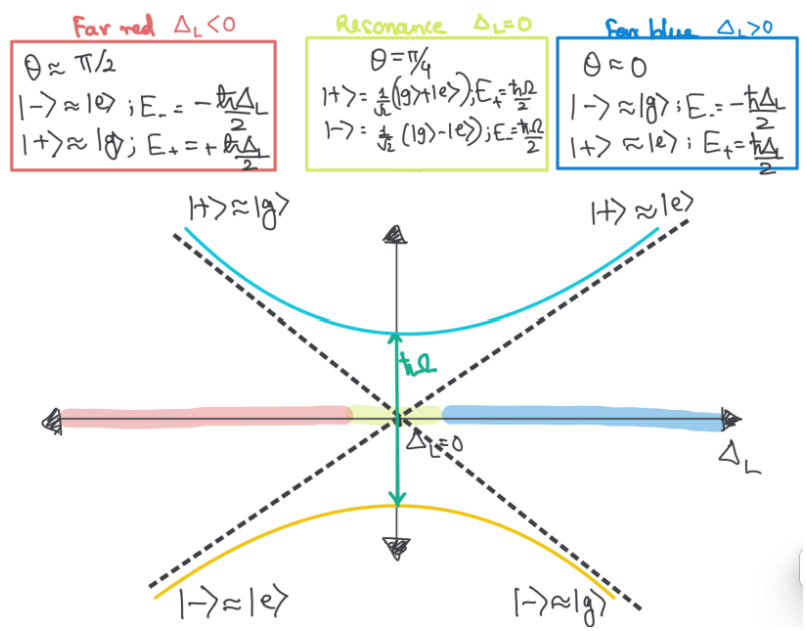


Figure 4.5 Chirping the external field slowly enough such that the atomic state remains in an eigenstate of the Hamiltonian.

It can be shown that the probability of graing from the ground to excited state is:

$$P_e(t \to \infty) = 1 - \exp\left(-\frac{\pi\Omega^2}{2|\partial_t \Delta_L|}\right). \tag{4.40}$$

This effect is referred to as **adiabatic passage**.

4.6 Ramsey interferometry

Mach-Zehnder interferometer for atomic internal states. Consider that pulse 1 is a $\pi/2$ pulse such that:

$$|g\rangle \to \frac{1}{\sqrt{2}}(|e\rangle + |g\rangle). \tag{4.41}$$

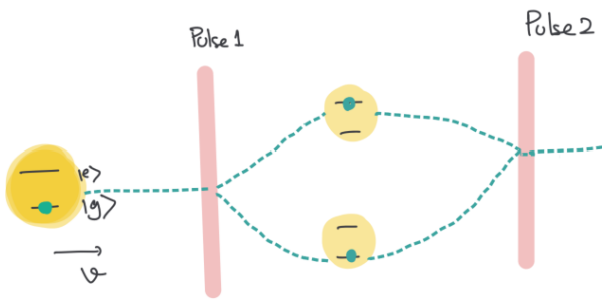
Say there is a general state-dependent phase shift applied to the states $|g\rangle, |e\rangle$ such that

$$\frac{1}{\sqrt{2}}(|e\rangle + |g\rangle) \to \frac{1}{\sqrt{2}}(|e\rangle + e^{i\phi}|g\rangle).$$

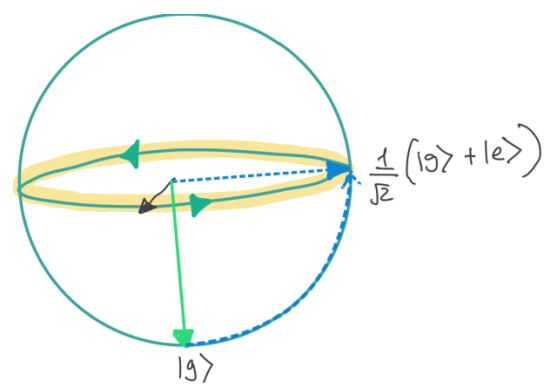
Pulse 2 then:

$$\frac{1}{\sqrt{2}}(|e\rangle + e^{i\phi}|g\rangle) \to \frac{1}{\sqrt{2}}[\frac{1}{\sqrt{2}}(|e\rangle - |g\rangle) + \frac{e^{i\phi}}{\sqrt{2}}(|e\rangle + |g\rangle)] = \frac{1}{2}[|e\rangle \underbrace{(1+e^{i\phi})}_{P_e = \frac{1+\cos\phi}{2}} + |g\rangle \underbrace{(1-e^{i\phi})}_{\frac{1-\cos\phi}{2}}]$$

For $\phi = 0$, $P_e = 1$, $P_g = 0$ $\{\pi/2 + \pi/2 = \pi\text{pulse}\}$. The phase ϕ depends on possible external field and can be detected by the final excitation probability of the atom.



(a) Ramsey interferometer



(b) Bloch sphere after the π pulse

4.6.1 Analogy with MZI

We have:

$$\begin{aligned}|1\rangle_1 &= \hat{a}_1^\dagger|0\rangle \\ &= \frac{\hat{a}_2^\dagger + \hat{a}_3^\dagger}{\sqrt{2}}|0\rangle\end{aligned}$$

$$|1\rangle_1 = \frac{1}{\sqrt{2}}(|1\rangle_2 + |1\rangle_3) \xrightarrow{\pi-\text{pulse}} \frac{1}{\sqrt{2}}(|1\rangle_2 + e^{i\phi}|1\rangle_3) \xrightarrow{BS} \frac{1}{\sqrt{2}}\left[\frac{\hat{a}_4^\dagger + \hat{a}_5^\dagger}{\sqrt{2}}|0\rangle + e^{i\phi}\frac{\hat{a}_4^\dagger - \hat{a}_5^\dagger}{\sqrt{2}}|0\rangle\right] = \frac{1}{2}\left[(1+e^{i\phi})|1\rangle_4 + (1-e^{i\phi})|1\rangle_5\right].$$

Probability of detecting a photon at detector D4 and D5 changes with the phase ϕ.

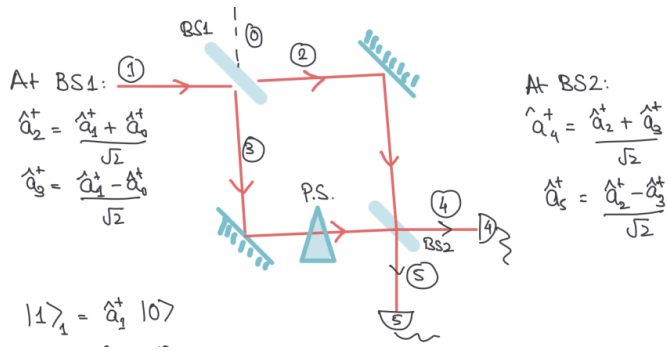
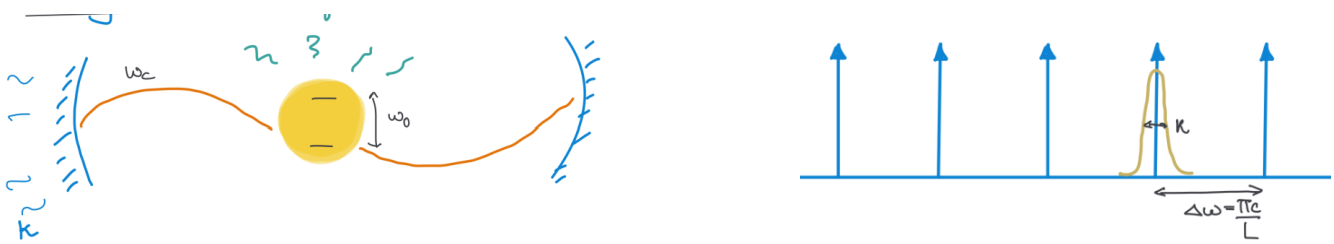


Figure 4.2 Analogy with MZI.

4.7 Cavity QED

Let us consider an atom inside an optical cavity. The EM field modes in the cavity correspond to standing wave modes with a frequency spacing $\Delta\omega = \pi c/L$ between adjacent modes.



(a) Cavity **(b)** Modes spectrum

Figure 4.1 Cacity QED and modes.

In the presence of lossy mirrors the cavity has a decay such that there is a Lorentzian linewidth κ to each cavity resonance.

Consider a cavity of length $L \ll \lambda_0$, the fundamental mode of the cavity starts at

$$\underbrace{\omega_{c0} = \frac{\pi c}{L}}_{\text{cavity fundamental mode}} \quad \gg \quad \underbrace{\omega_0 = \frac{\pi c}{\lambda_0}}_{\text{Atomic resonance}} \quad . \tag{4.42}$$

To such a case the atom does not see any modes to emit into and its spontaneous emission is inhibited. More generally cavity modifies the EM field mode structure seen by the atom. A typical Fabry-Perot cavity is $\sim 100\ \mu m$ long with $\Delta\omega \sim 10^{13}\ Hz$.

In such a regime the atom typically only sees one of the modes of the EM field inside the cavity. The total atom+cavity system is described by the Hamiltonian:

$$H = H_A + H_C + H_{AC}, \qquad \begin{aligned} H_A &= \hbar\omega_0 \hat{\sigma}_+ \hat{\sigma}_- \\ H_C &= \hbar\omega_c \hat{a}^\dagger \hat{a} \\ H_{AC} &= -\hat{\boldsymbol{d}} \cdot \hat{\boldsymbol{E}}(\boldsymbol{r}_A) = -(\boldsymbol{d}\hat{\sigma}_+ + \boldsymbol{d}^*\hat{\sigma}_-) \cdot \sqrt{\tfrac{\hbar\omega_c}{2\varepsilon_0 V}}\hat{\boldsymbol{e}}_c(\hat{a} + \hat{a}^\dagger)\sin(k_c r_a). \end{aligned}$$

Making RWA, we have

$$H_{AC} = \underbrace{-\boldsymbol{d}\cdot\hat{\boldsymbol{e}}_c\sqrt{\frac{\hbar\omega_c}{2\varepsilon_0 V}}\sin(k_c r_a)}_{i\hbar g}[\hat{\sigma}_+\hat{a} + \hat{\sigma}_x\hat{a}^\dagger].$$

The total Hamiltonian is therefore:

$$\text{Joynes-Cumming Hamiltonian} \qquad H = \hbar\omega_0\hat{\sigma}_+\hat{\sigma}_- + \hbar\omega_c\hat{a}^\dagger\hat{a} + i\hbar g[\hat{\sigma}_+\hat{a} + \hat{\sigma}_x\hat{a}^\dagger]. \tag{4.43}$$

The above Hamiltonian conserves the total number of excitations such that the eigenstates of the Hamiltonian can be expressed in terms of linea combinartions of $\{|g,n\rangle, |e,n-1\rangle\}$ in the n-excitation manifold.

$$|\Psi_n\rangle = c_{gn}|g,n\rangle + c_{e,n-1}|e,n-1\rangle.$$

From Schrodinger equation:

$$\begin{aligned} i\hbar\partial_t|\Psi_n\rangle &= H|\Psi_n\rangle \\ \partial_t\begin{bmatrix} c_{e,n-1} \\ c_{g,n}\end{bmatrix} &= -\frac{i}{\hbar}\begin{bmatrix}(n-1)\hbar\omega_c + \hbar\omega_0 & i\hbar g\sqrt{n} \\ -i\hbar g\sqrt{n} & n\hbar\omega_c\end{bmatrix}\begin{bmatrix} c_{e,n-1} \\ c_{g,n}\end{bmatrix}. \end{aligned}$$

From diagonalizing the Hamiltonian in the n-ecitation manifold the eigenstates of the atom+field system are:

$$\begin{aligned} |+,n\rangle &= i\sin\theta_n|g,n\rangle + \cos\theta_n|e,n-1\rangle \\ |-,n\rangle &= -\cos\theta_n|g,n\rangle + \sin\theta_n|e,n-1\rangle, \quad \tan 2\theta_n = \frac{2g\sqrt{n}}{\Delta_c},\ \Delta_c = \underbrace{\omega_0 - \omega_c}_{\text{cavity detuning}}. \end{aligned}$$

The eigenenergies are:

$$E_n^\pm = \frac{\hbar}{2}\left[\omega_0 + (2n-1)\omega_c \pm \sqrt{4g^2 n + (\omega_0-\omega_c)^2}\right].$$

- Uncoupled case $g \to 0$

$$\begin{aligned} E_n^\pm &= \frac{\hbar}{2}[\omega_0 + (2n-1)\omega_c \pm (\omega_0-\omega_c)] \\ |e,n-1\rangle:\ E_n^+ &= \frac{\hbar}{2}[2\omega_0 + 2(n-1)\omega_c] = \hbar(\omega_0 + (n-1)\omega_c) \\ |g,n\rangle:\ E_n^- &= \frac{\hbar}{2}[2n\omega_c] = \hbar n\omega_c. \end{aligned}$$

- **Resonant case** $\Delta_c \to 0$

$$E_n^\pm = \frac{\hbar}{2}[\omega_0 + (2n-1)\omega_c \pm 2h\sqrt{n}] = n\hbar\omega_c \pm \hbar g\sqrt{n}.$$

  Eigenstates are:

$$\begin{aligned} |+,n\rangle &= \frac{1}{\sqrt{2}}[i|g,n\rangle + |e,n-1\rangle] \\ |-,n\rangle &= \frac{1}{\sqrt{2}}[-i|g,n\rangle + |e,n-1\rangle] \end{aligned}$$

Example: How does an excited atom evolve in an empty resonant single-mode cavity? The initial state of the system $|e,0\rangle$.

$$\text{states in the single subspace} \quad |e,0\rangle \to |g,1\rangle\,.$$

The eigenstates are:

$$|+,1\rangle = \frac{1}{\sqrt{2}}[i\,|g,1\rangle + |e,0\rangle],\ E_+ = n\hbar\omega_c + \hbar g$$
$$|-,1\rangle = \frac{1}{\sqrt{2}}[-i\,|g,1\rangle + |e,0\rangle],\ E_- = \hbar\omega_c - \hbar g.$$

The initial state can be written in terms of the eigenstates as:

$$|e,0\rangle = \frac{1}{\sqrt{2}}[|+,1\rangle + |-,1\rangle]$$

each eigenstate evolves with the corresponding eigenenergies

$$\frac{1}{\sqrt{2}}[|+,1\rangle\, e^{-iE_+t/\hbar} + |-,1\rangle\, e^{-iE_-t/\hbar}]$$
$$= \frac{1}{2}[\{i\,|g,1\rangle + |e,0\rangle\}e^{-i\omega_c t}e^{-igt} + \{-i\,|g,1\rangle + |e,0\rangle\}e^{-i\omega_c t}e^{igt}]$$
$$= \frac{1}{2}e^{-i\omega_c t}[i\,|g,1\rangle\,\{e^{-igt} - e^{igt}\} + |e,0\rangle\,\{e^{igt} + e^{-igt}\}]$$
$$= e^{-i\omega_c t}[\sin(gt)\,|g,1\rangle + \cos(gt)\,|e,0\rangle].$$

Vacuum Rabi oscilattions of an atom in a single mode cavity.

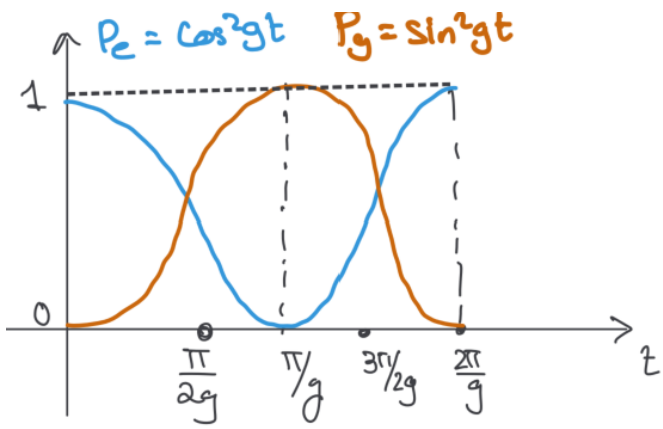


Figure 4.2 Rabi oscilaltion in a single-mode cavity.

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Chapter A

Density operator

A.1 Density operator

A.1.1 Definition

State vectors $|\psi\rangle$ convey the maximal amount of information about a system allowed by the laws of quantum mechanics.

If $|\psi_1\rangle$ and $|\psi_2\rangle$ are two possible quantum states then so is their coherent superporition

$$|\psi\rangle = c_1 |\psi_1\rangle + c_2 |\psi_2\rangle , \quad |c_1|^2 + |c_2|^2 = 1.$$

However, there are situations where the state vector is not precisely known. It may be possible to write state vectors for the multicomponent system but not for the subsystem of interest.

For instance, a system of two spin-1/2 with eigenstates $\{|\uparrow\rangle , |\downarrow\rangle\}$. A possible state of the combined system is

$$|\psi\rangle = \frac{1}{\sqrt{2}} \left[|\uparrow\rangle_1 |\downarrow\rangle_2 - |\downarrow\rangle_1 |\uparrow\rangle_2 \right] .$$

This equation is an entangled state. An entangled state cannot be factored, in any basis, into a product of states of the two subsystems, that is,

$$\text{Entangled state} \qquad |\psi\rangle \neq |\text{spin } 1\rangle |\text{spin } 2\rangle .$$

Quantum states described by state vectors are said to be *pure states*. States that cannot be described by state vectors are said to be in *mixed states*. Mixed states are described by the density operator

$$\text{Mixed state} \qquad \hat{\rho} = \sum_i p_i |\psi_i\rangle \langle\psi_i| , \tag{A.1}$$

where the sym is over an ensemble and p_i is the probability of the system being in the ith state of the ensemble $|\psi_i\rangle$. The probabilities satisfy the obvious relations

$$0 \leq p_i \leq 1, \qquad \sum_i p_i = 1, \qquad \sum_i p_i^2 \leq 1.$$

For the pure state, all the p_i vanish except the jth with $p_i \delta_{ij}$. Then,

$$\text{Pure state} \qquad \hat{\rho} = |\psi_j\rangle \langle\psi_j| .$$

The density operator for the pure state $|\psi_j\rangle$ is just the projection onto the state $|\psi_j\rangle$, while the general case (A.1) is a sum of the projection operators over the ensemble, weighted with the probabilities of each member of the ensemble.

In an orthonormal basis $\{|\varphi_n\rangle\}$, the ith member of the ensemble can be projected as

$$|\psi_i\rangle = \sum_n |\varphi_n\rangle \langle\varphi_n|\psi_i\rangle = \sum_n c_n^{(i)} |\varphi_n\rangle, \quad c_n^{(i)} = \langle\varphi_n|\psi_i\rangle.$$

The matrix element of $\hat{\rho}$ between the n and n' eigenstates is

$$\langle\varphi_n|\hat{\rho}|\varphi_{n'}\rangle = \sum_i \langle\varphi_n|\psi_i\rangle p_i \langle\psi_i|\varphi_{n'}\rangle = \sum_i p_i c_n^{(i)} c_{n'}^{(i)*}.$$

The quantities $\langle\varphi_n|\hat{\rho}|\varphi_{n'}\rangle$ form the elements of the density matrix. The trace of this matrix is

$$\mathrm{Tr}\,[\hat{\rho}] = \sum_n \langle\varphi_n|\hat{\rho}|\varphi_n\rangle = \sum_{i,n} \langle\varphi_n|\psi_i\rangle p_i \langle\psi_i|\varphi_n\rangle = \sum_{i,n} p_i \langle\psi_i|\varphi_n\rangle\langle\varphi_n|\psi_i\rangle = \sum_i p_i = 1.$$

Since $\hat{\rho}$ is Hermitian, the diagonal elements $0 \leq \langle\varphi_n|\hat{\rho}|\varphi_n\rangle \leq 1$.
Let us now consider the square $\hat{\rho}^2 = \hat{\rho} \cdot \hat{\rho}$. For a pure state, we have

$$\hat{\rho}^2 = |\psi\rangle \langle\psi|\psi\rangle \langle\psi| = |\psi\rangle \langle\psi| = \hat{\rho} \Longrightarrow \mathrm{Tr}\left[\hat{\rho}^2\right] = \mathrm{Tr}\,[\hat{\rho}] = 1.$$

For a statistical mixture

$$\hat{\rho}^2 = \sum_{i,j} p_i p_j |\psi_i\rangle \langle\psi_i|\psi_j\rangle \langle\psi_j|.$$

Its trace is

$$\mathrm{Tr}\left[\hat{\rho}^2\right] = \sum_n \langle\varphi_n|\hat{\rho}^2|\varphi_n\rangle = \sum_{n,i,j} p_i p_j \langle\varphi_n|\psi_i\rangle\langle\psi_i|\psi_j\rangle\langle\psi_j|\varphi_n\rangle = \sum_{i,j} p_i p_j |\langle\psi_i|\psi_j\rangle|^2 \leq \left[\sum_i p_i\right]^2 = 1.$$

Thus, we have

$$\mathrm{Tr}\left[\hat{\rho}^2\right] \begin{cases} = 1, & \text{pure state} \\ < 1, & \text{mixed state} \end{cases}. \tag{A.2}$$

The ensemble average (or expectation value for pure state) of some operator $\hat{O}$ is

$$\text{Expectation} \qquad \langle\hat{O}\rangle = \mathrm{Tr}\left[\hat{\rho}\hat{O}\right] = \begin{cases} \langle\psi_i|\hat{O}|\psi_i\rangle, & \text{pure state} \\ \sum_i p_i \langle\psi_i|\hat{O}|\psi_i\rangle, & \text{mixed state} \end{cases}. \tag{A.3}$$

Formally we can use the density operator to express the expectation as:

$$\langle\hat{O}\rangle = \mathrm{Tr}\left[\hat{\rho}\hat{O}\right] = \sum_n \langle\varphi_n|\hat{\rho}\hat{O}|\varphi_n\rangle = \sum_{n,i} p_i \langle\varphi_n|\psi_i\rangle\langle\psi_i|\hat{O}|\varphi_n\rangle = \sum_{i,n} p_i \langle\psi_i|\hat{O}|\varphi_n\rangle\langle\varphi_n|\psi_i\rangle = \sum_i p_i \langle\psi_i|\hat{O}|\varphi_i\rangle.$$

A.1.2 Two-state system

For a two-state system, there always exists a description in terms of the Pauli operators $\hat{\sigma}_1, \hat{\sigma}_2, \hat{\sigma}_3$ satisfying the commutation relations $[\hat{\sigma}_i, \hat{\sigma}_j] = 2i\varepsilon_{ijk}\hat{\sigma}_k$.

In a basis where $\hat{\sigma}_3$ and $\hat{\sigma}^2 = \hat{\sigma}_1^2 + \hat{\sigma}_2^2 + \hat{\sigma}_3^2$ are diagonal, these operators can be written in matric form:

$$\sigma_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \sigma_2 = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad \sigma_3 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}. \tag{A.4}$$

Any Hermitian 2×2 matrix can be expressed in terms of the Pauli matrices and the identity matriz $\hat{I}_2$, and of course, $\hat{\rho}$. We can then write:

$$\hat{\rho} = \begin{bmatrix} \rho_{11} & \rho_{12} \\ \rho_{21} & \rho_{22} \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 - s_3 & s_1 + is_2 \\ s_1 - is_2 & 1 - s_3 \end{bmatrix} = \frac{1}{2}(\hat{I}_2 + \boldsymbol{s} \cdot \sigma), \tag{A.5}$$

where $\boldsymbol{s} = \{s_1, s_2, s_3\}$ is the *Bloch vector*. The associated quantum state can be represented as:

$$|\psi\rangle = \cos\frac{\theta}{2} e^{-i\phi/2} |\uparrow\rangle + \sin\frac{\theta}{2} e^{i\phi/2} |\downarrow\rangle . \tag{A.6}$$

The density operator above has two eigenvalues:

$$g_{\frac{1}{2}} = \frac{1}{2}\left[1 \pm \sqrt{s_1^2 + s_2^2 + s_3^2}\right] = \frac{1}{2}[1 \pm |\boldsymbol{s}|] , \tag{A.7}$$

and its eigenvectors are determined by the two vectors $\pm\boldsymbol{u}$ shown in the Bloch sphere. For pure states where $|\boldsymbol{s}| = 1$, $\pm\boldsymbol{u} = \pm\boldsymbol{s}$ and $\boldsymbol{s}$ lies on the surface of the Bloch sphere. For mixed states where $|\boldsymbol{s}| < 1$, $\boldsymbol{s}$ lies inside the Bloch sphere and $\pm\boldsymbol{u} = \pm\frac{\boldsymbol{s}}{|\boldsymbol{s}|}$.

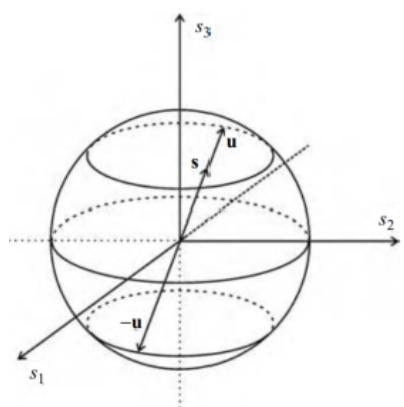

Figure A.1 Representation of the density matrix of a two-state ssystem in terms of the Bloch sphere and Bloch vector.

A.1.3 Entangled states

Lets consider a two-particle system and that each particle can be in either of two one-particle states $|\psi_1\rangle$ or $|\psi_2\rangle$. Using the notation $\left|\psi_i^{(j)}\right\rangle$=particle j in state i, we consider a pure two-particle superposition state

$$|\psi\rangle = C_1 \left|\psi_1^{(1)}\right\rangle \otimes \left|\psi_2^{(2)}\right\rangle + C_2 \left|\psi_2^{(1)}\right\rangle \otimes \left|\psi_1^{(2)}\right\rangle .$$

This can be extended for multiparticle systems, where we can define the *reduced density operators* for each of the subsystems by tracing the density operator over the states of all the other systems. With

$\phi = |\psi\rangle\langle\psi|$, the reduced density operator for particle 1 is

$$\hat{\rho}^{(1)} = \mathrm{Tr}_2[\hat{\rho}] = \langle\psi_1^{(2)}|\hat{\rho}|\psi_1^{(2)}\rangle + \langle\psi_2^{(2)}|\hat{\rho}|\psi_2^{(2)}\rangle = |C_1|^2 \left|\psi_1^{(1)}\right\rangle \left\langle\psi_1^{(1)}\right| + |C_2|^2 \left|\psi_2^{(1)}\right\rangle \left\langle\psi_2^{(1)}\right|.$$

This has the form of a mixed state for particle 1 as long as $C_i \neq 0,\ i = 1, 2$. Simiarly, for particle 2,

$$\hat{\rho}^{(2)} = \mathrm{Tr}_1[\hat{\rho}] = |C_1|^2 \left|\psi_1^{(2)}\right\rangle \left\langle\psi_1^{(2)}\right| + |C_2|^2 \left|\psi_2^{(2)}\right\rangle \left\langle\psi_2^{(2)}\right|.$$

When one of the particles is considered without regards to the other, it is generally in a mixed state. Thus one may characterized the degree of entanglement according to the degree of purity of either of the subsystems.

- $\mathrm{Tr}\left[\hat{\rho}^{(2)}\right]^2 = 1$, the state $|\psi\rangle$ is not an entangled state.
- $\mathrm{Tr}\left[\hat{\rho}^{(2)}\right]^2 < 1$, then $|\psi\rangle$ describes an entanglement between subsystems 1 and 2.

A.1.4 Dynamics of the density operator

The density operator evolves unitarily according to

$$\text{von Neumann eq 1} \qquad \partial_t \hat{\rho} = \frac{i}{\hbar}[\hat{\rho}, \hat{H}], \quad \text{with} \quad i\hbar\partial_t |\psi_i\rangle = \hat{H}|\psi_i\rangle. \tag{A.8}$$

Alternatively, we may write

$$\text{Heisenberg picture of } \hat{\rho} \qquad \hat{\rho}(t) = \hat{U}(t,0)\hat{\rho}(0)\hat{U}^\dagger(t,0),$$

where the unitary evolution operator $\hat{U}(t,0)$ satisfies the equation

$$i\hbar\partial_t \hat{U} = \hat{H}\hat{U}. \tag{A.9}$$

The equation is sometimes written as

$$\text{von Neumann eq 2} \qquad i\hbar\partial_t\hat{\rho}(t) = [\hat{H}, \hat{\rho}(t)] = \mathcal{L}\hat{\rho}(t), \tag{A.10}$$

where $\mathcal{L}$ is known as the Liouvilian superoperator.

Chapter B

Shcrodinger and Heisenberg pictures

B.0.1 Time-dependent reference frames

The state of a system can be evaluated in various **reference frames** that evolve in time.

The state $|\psi(t)\rangle$ in one reference frame evolves in time according to the Schrodinger equation

$$i\hbar\partial_t |\psi(t)\rangle = H(t) |\psi(t)\rangle .$$

A second reference frame that may evolve in time relative to the first one if we assume the existence of a time-dependent unitary operator that operates over the first frame provifing an effective state in the second frame:

$$\text{Effective state of the second frame} \qquad |\psi_E(t)\rangle = F(t) |\psi(t)\rangle , \quad F(t_0) = \mathbb{1} . \tag{B.1}$$

The effective state obeys the **effective Schrodinger equation** obtained by inserting $|\psi(t)\rangle = F^\dagger(t) |\psi_E(t)\rangle$ in the Schrodinger equation of the first frame:

$$\text{Effective Schrodinger equation and Hamiltonian} \qquad \begin{aligned} & i\hbar\partial_t |\psi_E(t)\rangle = H_E(t) |\psi_E(t)\rangle \\ & H_E(t) = F(t)H(t)F^\dagger(t) - i\hbar F(t) \left(\partial_t F^\dagger(t)\right) \end{aligned} \tag{B.2}$$

Frame transformations are generally used to simplify calculations and time dependence of the Schrodinger equation.

B.0.2 Schrodinger, Heisenberg, and interaction pictures

They are different frames of reference, and are distinguished by the specfic time-dependent unitary transformations.

- **Schrodinger picture** State vectors $|\psi_S(t)\rangle$ evolve in time under the action of the Hamiltonian $|\psi_S(t)\rangle = U(t, t_0) |\psi_S(t)\rangle$. Position and momentum operators have no time dependence.
- **Heisenberg picture** Defined by the adjoint of the evolution operator of the S picture, so that its application on the Schrodinger-picture state vector $|\psi_H\rangle = U^\dagger(t, t_0) |\psi_S(t)\rangle = |\psi_S(t_0)\rangle$ vanishes the time dependence. On the other hand, operators that have no time dependence in the S picture may now depend on time.
- **Interaction picture** Used when the S picture Hamiltonian is time dependent. In this picture, operators and state vectors generally evolve in time.

B.0.3 Schrodinger picture

Position and momentum operators have no explicit tiem dependence in this picture. However, other operators with time dependence may be constructed.

The expectation value of an operator $A_S(t)$ who generally have time dependence that results from both the time dependence of $|\psi_S(t)\rangle$ and from the operator itself:

$$\langle A_S(t)\rangle(t) = \langle\psi_S(t)|A_S(t)|\psi_S(t)\rangle \longrightarrow \boxed{\frac{d}{dt}\langle A_S\rangle = \frac{1}{i\hbar}\langle[A_S, H_S]\rangle + \langle\frac{\partial A_S}{\partial t}\rangle}\,. \tag{B.3}$$

Ehrenfest's equations are obtained by replacing $A_S(t)$ with the position and momentum operator $\boldsymbol{R} = (X, Y, Z)$ and $\boldsymbol{P} = (P_x, P_y, P_z)$, and noting that $\partial_t\boldsymbol{R} = \partial_t\boldsymbol{P} = 0$:

$$\text{Ehrenfest's equations} \qquad \boxed{\begin{aligned} &\frac{d}{dt}\langle\boldsymbol{R}\rangle = \frac{1}{m}\langle\boldsymbol{P}\rangle \\ &\frac{d}{dt}\langle\boldsymbol{P}\rangle = -\langle\nabla V(\boldsymbol{R})\rangle \end{aligned}} \quad , \quad \langle V(\boldsymbol{R})\rangle = \big[\langle\partial_X V(\boldsymbol{R})\rangle, \langle\partial_Y V(\boldsymbol{R})\rangle, \langle\partial_Z V(\boldsymbol{R})\rangle\big]\,. \tag{B.4}$$

B.0.4 Heisenberg picture

This picture vanishes the time dependence of S picture state vector by applying the adjoint of the time evolution oeprator $U^\dagger(t, t_0)$, which defines a unitary transformation. In addition, because expectation values of oeprators must be **conserved** regardless the interaction used, nn arbitrary operator $A_S(t)$ in the S picture is transformed to the H picture as

$$\text{Transformed state vector and operator} \qquad \boxed{\begin{aligned} &|\psi_H\rangle = U^\dagger(t, t_0)\,|\psi_S(t)\rangle = |\psi_S(t_0)\rangle \\ &A_H(t) = U^\dagger(t, t_0)A_S(t)U(t, t_0) \end{aligned}}\,.$$

As,

$$\langle A\rangle = \langle\psi_S(t)|U(t, t_0)|U^\dagger(t, 0)AU(t, 0)U^\dagger(t, 0)|\psi_S(t)\rangle.$$

The evolution of the operator $A_H(t)$ is then given by:

$$\begin{aligned} \frac{d}{dt}A_H(t) =& -\frac{1}{i\hbar}U^\dagger(t, t_0)H_S(t)A_S(t)U(t, t_0) + U^\dagger(t, t_0)\frac{dA_S(t)}{dt}U(t, t_0) + \frac{1}{i\hbar}U^\dagger(t, t_0)A_S(t)H_S(t)U(t, t_0) \\ =& -\frac{1}{i\hbar}U^\dagger(t, t_0)H_S(t)U(t, t_0)U^\dagger(t, t_0)A_S(t)U(t, t_0) + U^\dagger(t, t_0)\frac{dA_S(t)}{dt}U(t, t_0) \\ &+ \frac{1}{i\hbar}U^\dagger(t, t_0)A_S(t)U(t, t_0)U^\dagger(t, t_0)H_S(t)U(t, t_0) \\ i\hbar\partial_t A_H(t) =& [A_H(t), H_H(t)] + i\hbar U^\dagger(t, t_0)(\partial_t A_S(t))U(t, t_0). \end{aligned}$$

In the table, H_S and $|\psi_S(t)\rangle$ are the S picture Hamiltonian and state vector.

The effective Hamiltonian of the H picture is $H_E = 0$. Therefore, the effect Schrodinger equation $i\hbar\partial_t|\psi_H\rangle = 0$ is solved by $|\psi_H\rangle = |\psi_S(t_0)\rangle$. In the H picture, only operators evolve in time following the ODE in the last line of the table.

An advantage of Heisenberg picture is that it leads to equations formally similar to those of classical mechanics.

Heisenberg picture quantities and dynamics

$\lvert\psi_H\rangle$	$\equiv U^\dagger(t,t_0)\lvert\psi_S(t)\rangle = \lvert\psi_S(t_0)\rangle$
$A_H(t)$	$\equiv U^\dagger(t,t_0)A_S(t)U(t,t_0)$
H_H	$= H_S,$ for time-independent H_S
$H_H(t)$	$= H_S(t),$ for $[H_S(t), H_S(t')] = 0$
$\langle A_H(t)\rangle(t)$	$= \langle\psi_H\lvert A_H(t)\rvert\psi_H\rangle$
	$= \langle\psi_S(t_0)\lvert U^\dagger(t,t_0)A_S(t)U(t,t_0)\rvert\psi_S(t_0)\rangle$
	$= \langle\psi_S\lvert A_S(t)\rvert\psi_S(t)\rangle = \langle A_S(t)\rangle(t)$
$i\hbar\partial_t A_H(t)$	$= [A_H(t), H_H(t)] + i\hbar U^\dagger(t,t_0)\left(\partial_t A_S(t)\right)U(t,t_0)$

B.0.5 Interaction picture

Obtained with a unitary transformation of state vectors and operators of the S picture. This picture removes some of the time dependence of the S picture state vectors, while also altering the time dependence of operators. The interaction picture is typically used with a time-dependent S picture Hamiltonian

$$\text{Time-dependent S picture Hamiltonian} \qquad H_S(t) = H_0 + W(t)\,, \tag{B.5}$$

where the eigenstates of H_0 are knwon, and $W(t)$ is a time-dependent **perturbation** which induces time-dependent dynamics and **transitions** between the eigenstates of H_0. To transform into the interaction picture, an evolutino operator $U_0(t,t_0) = e^{-iH_0(t-t_0)/\hbar}$ is associated with H_0. The transformed state vector and arbitrary operator expressed in the interaction picture are

$$\text{Transformed state vector and operator} \qquad \begin{aligned} \lvert\psi_I(t)\rangle &= U_0^\dagger(t,t_0)\lvert\psi_S(t)\rangle \\ A_I(t) &= U_0^\dagger(t,t_0)A_S(t)U_0(t,t_0) \end{aligned} \tag{B.6}$$

The effective Schrodinger equation in the interaction picture is

$$i\hbar\partial_t\lvert\psi_I(t)\rangle = H_E(t)\lvert\psi_I(t)\rangle\,, \quad \text{where} \quad H_E(t) = U_0^\dagger(t,t_0)W(t)U(t,t_0) \tag{B.7}$$

is the effective Hamiltonian. If $W(t) = 0$, then the interaction picture reduces to the H picture: $H_E(t) = 0$ and $\lvert\psi_I(t)\rangle = \lvert\psi_S(t_0)\rangle$.

Meaning of the interaction picture

In the I picture (E-picture), we are isolating the evolutino of H_0 while only allow evolve H_1.